

Project Data Page:





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DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\LTS Reduction System KO Drum (173-D)Rev0-7%.pvdb

Date of Analysis : Sep 16,2026 2:07pm

PV Elite 28 U1, March 2026

Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 2450.000 mm. ID	
Vessel Design Length, Tangent to Tangent	2825.00 mm.
Distance of Bottom Tangent above Grade	3300.00 mm.
Distance of Base above Grade	300.00 mm.
Specified Datum Line Distance	3000.00 mm.
Internal Design Temperature	82 °C
Internal Design Pressure	10.799 bars
External Design Temperature	0 °C
Maximum Allowable Working Pressure	10.8 bars
Shop Test Pressure	19.521 bars
Field Test Pressure	14.039 bars
Required Minimum Design Metal Temperature	-12.0 °C
Warmest Computed Minimum Design Metal Temperature	-28.0 °C
Wind Design Code	ASCE/SEI 7-16
Earthquake Design Code	ASCE/SEI 7-16

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	
Head	SA-516 70	...	...	K02700	
Skirt	SA-36	...	...	K02600	
Skirt	SA-516 70	...	...	K02700	
Nozzle	SA-106 B	...	...	K03006	
Nozzle	SA-105	...	...	K03504	
Re-Pad	SA-516 70	...	...	K02700	
Nozzle Flg	SA-105	...	...	K03504	
Basering	SA-516 70	...	...	K02700	
Base Bolting	SA-193 B7	...	<= 64	G41400	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (bars & mm.):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
SKIRT	...	...	...	3.0000	N/A	No
SKIRT	...	...	...	3.0000	N/A	No
BOTTOM HEAD	10.980	0.00	12.300	3.0000	No	No
SHELL	10.915	0.00	12.200	3.0000	N/A	No
TOP HEAD	10.806	0.00	12.400	3.0000	No	No

Liquid Level: 1787.50 mm. Dens.: 0.001 kg./cm³ Sp. Gr.: 0.989

Element Types and Properties:

Element Type	"To" Elev mm.	Element Length mm.	Nominal Thickness mm.	Finished Thickness mm.	Reqd Thk Internal mm.	Reqd Thk External mm.	Long Eff	Circ Eff
Skirt	-1000.0	2000.0	14.0	14.0	...	...	0.70	0.70
Skirt	0.0	1000.0	14.0	14.0	...	...	0.70	0.50

Ellipse	50.0	50.0	20.0	16.0	14.5	...	0.85	0.85
Cylinder	2775.0	2725.0	16.0	16.0	14.5	...	0.85	0.85
Ellipse	2825.0	50.0	20.0	16.0	14.3	...	0.85	0.85

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
D1-2"	WRC-107/537	0.672	n/a	Passed
N1-16"	WRC-297	0.920	Shell	Passed
N2-14"	WRC-107/537	0.672	n/a	Passed

Wind/Earthquake Shear, Bending:

From	To	Distance to Support mm.	Cumulative Wind Shear Kg-f	Earthquake Shear Kg-f	Wind Bending Kg-m.	Earthquake Bending Kg-m.
10	20	1000	1502.97	1289.82	6089.03	5866
20	30	2500	1257.85	1256.84	3328.16	3319.29
30	40	3025	1135.29	1229.62	2131.56	2076.03
40	50	4412.5	873.307	973.046	2081.35	2020.97
50	60	5800	539.469	420.69	156.401	121.965

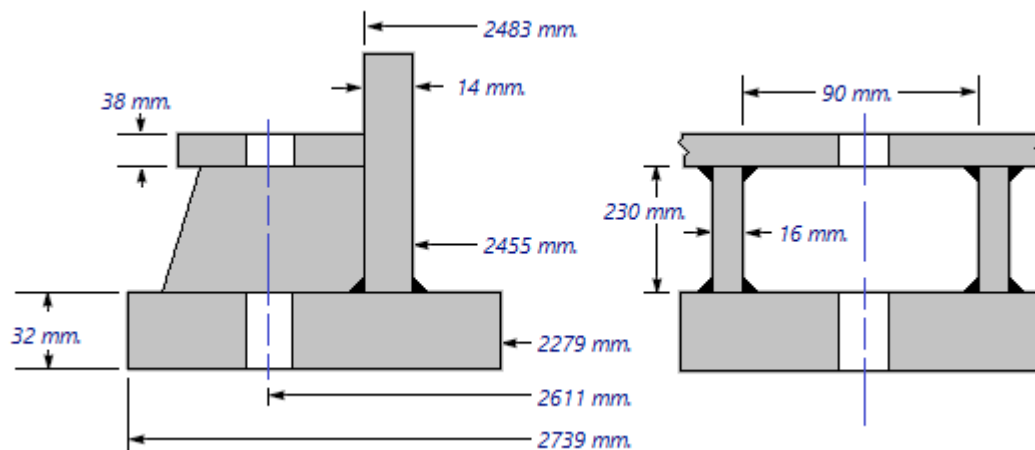
Abs Max of the all of the Stress Ratio's : 0.4377

Basing Data : Continuous Top Ring W/Gussets

Thickness of Basing	32.0000	mm.
Inside Diameter of Basing	2279.0000	mm.
Outside Diameter of Basing	2739.0000	mm.
Nominal Diameter of Bolts	42.0000	mm.
Diameter of Bolt Circle	2611.0000	mm.
Number of Bolts	16	

Thickness of Gusset Plates	16.0000	mm.
Average Width of Gusset Plates	118.0000	mm.
Height of Gussets	230.0000	mm.
Distance between Gussets	90.0000	mm.
Thickness of Top Plate or Ring	38.0000	mm.
Radial Width of the Top Plate	128.0000	mm.
Number of Gussets	32	

Basing Sketch for Cylindrical Skirts:



Basing Cross Section View



Loads for Foundation/Support Design:

Total Wind Shear on Support	4027.	Kgf
Total Earthquake Shear on Support	3814.	Kgf
Wind Moment on Support	24809.	Kg-m.
Earthquake Moment on Support	24586.	Kg-m.
Maximum Initial Bolt Load	1149.	Kgf

Wind and Earthquake moments include the effects of user defined forces and moments if any exist in the job and were specified to act (compute loads and stresses) during these cases. Also included are moment effects due to eccentric weights if any are present in the input.

Weights:

Fabricated - Bare w/o removable internals	11322.1	kg.
Shop Test - Fabricated + water (full)	28532.8	kg.
Shipping - Fab. + rem. intls.+ shipping app.	11322.1	kg.
Erected - Fab. + rem. intls.+ insul. (etc)	19618.8	kg.
Empty - Fab. + intls. + details + wghts.	19618.8	kg.
Operating - Empty + operating liquid (no ca)	26997.8	kg.
Field Test - Empty weight + test medium (full)	36669.5	kg.

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Nozzle Calculation Summary (bars & mm.):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
D1-2"	10.8	...	...	OK	6.42	OK	Passed
N1-16"	10.8	...	...	OK	11.33	OK	Passed
K1-2"	10.8	...	...	OK	7.80	OK	Passed
K2-2"	10.8	...	...	OK	7.80	OK	Passed
K3-2"	10.8	...	...	OK	7.80	OK	Passed
K4-2"	10.8	...	...	OK	7.80	OK	Passed
M1-24"	10.8	...	...	...		OK	Passed
N2-14"	10.8	...	...	OK	11.33	OK	Passed
K5-2"	10.8	...	...	OK	7.80	OK	Passed
K6-2"	10.8	...	...	OK	7.80	OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 10.8 Nozzle : K6-2"
 Minimum MAWP Shells/Flanges : 10.8 Element : SHELL
 Minimum MAPnc Shells/Flanges : 15.2 Element : SHELL

Computed Vessel M.A.W.P. : 10.8 bars

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	Y Coordinate mm.	Layout Angle deg	Dia. Limit mm.
30	D1-2"	0.000	0.000	88.352
40	N1-16"	1675.000	90.000	760.800
40	K1-2"	1125.000	135.000	113.600
40	K2-2"	599.999	90.000	113.600
40	K3-2"	1325.000	270.000	113.600
40	K4-2"	325.000	270.000	113.600
40	M1-24"	799.999	0.000	1175.200
50	N2-14"	0.000	0.000	647.000
50	K5-2"	0.000	270.000	113.600
50	K6-2"	0.000	90.000	113.600

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.

lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Pad Diameter	Pad Thk mm	Cut Length mm	Flg Class
D1-2"	2.000 in	XXS	WNF	2.375	11.074	170.00	16.00	176.21	300
K1-2"	2.000 in	Actual	LWN	3.320	16.764	...	...	176.73	300
K2-2"	2.000 in	Actual	LWN	3.320	16.764	...	...	176.73	300
K3-2"	2.000 in	Actual	LWN	3.320	16.764	...	...	176.73	300
K4-2"	2.000 in	Actual	LWN	3.320	16.764	...	...	176.73	300
K5-2"	2.000 in	Actual	LWN	3.320	16.764	...	...	510.12	300
K6-2"	2.000 in	Actual	LWN	3.320	16.764	...	...	510.12	300
N2-14"	14.000 in	80	WNF	14.000	19.050	550.00	16.00	273.14	150
N1-16"	16.000 in	Actual	WNF	16.000	16.000	850.00	16.00	2466.00	150
M1-24"	24.000 in	Actual	WNF	24.000	14.000	1050.00	14.00	404.53	150

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm.):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
D1-2"	SA-106 B	14.000	10.000	12.000	16.000	...
K1-2"	SA-105	14.000	10.000	...	...	...
K2-2"	SA-105	14.000	10.000	...	...	...
K3-2"	SA-105	14.000	10.000	...	...	...
K4-2"	SA-105	14.000	10.000	...	...	...
K5-2"	SA-105	14.000	10.000	...	...	...
K6-2"	SA-105	14.000	10.000	...	...	...
N2-14"	SA-106 B	14.000	10.000	12.000	16.000	...
N1-16"	SA-106 B	14.000	10.000	12.000	16.000	14.000
M1-24"	SA-106 B	14.000	10.000	12.000	14.000	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm.	Layout Angle deg	Proj Outside mm.	Proj Inside mm.	Installed in Component
D1-2"	...	0.0	160.00	0.00	BOTTOM HEAD
K1-2"	1125.000	135.0	160.00	0.00	SHELL
K2-2"	599.999	90.0	160.00	0.00	SHELL
K3-2"	1325.000	270.0	160.00	0.00	SHELL
K4-2"	325.000	270.0	160.00	0.00	SHELL
K5-2"	...	270.0	480.00	0.00	TOP HEAD
K6-2"	...	90.0	480.00	0.00	TOP HEAD
N2-14"	...	0.0	250.00	0.00	TOP HEAD
N1-16"	1674.999	90.0	250.00	2200.00	SHELL
M1-24"	799.999	0.0	350.00	0.00	SHELL



Bill of Materials:

QTY	DESCRIPTION	MATERIAL
2	ELLIPTICAL HEAD: 2.0 X 1, 20.0mm. THK X 2450.0mm. ID X 50.0mm.	SA-516 70
1	PLATFORM: 1000mm X 1500mm WIDE	
1	LADDER: 3000.0mm	...
1	PLATFORM: 1000mm X 5000mm WIDE	
1	LADDER: 3440.0mm	...
1	CLASS 150 GR 1.1, 24.0" BLIND FLANGE(S)	SA-105
1	BASERING: 2739.0mm. OD X 2279.0 ID X 32.0mm.	SA-516 70
16	BASERING BOLTS: 42.0mm. DIA	SA-193 B7
16	BASERING BOLT NUTS: 42.0mm. DIA	...
1	BASERING TOP RING: 2739.0mm. OD X 2483.0 ID X 38.0mm.	...
32	BASERING GUSSET PLATES: 118.0mm. X 118.0mm. X 230.0mm. X 16.0mm.	...
2	CYLINDRICAL SKIRT SEGMENT 14.0mm. THK X 3038.1mm. X 2000.0mm.	SA-36
1	CYLINDRICAL SKIRT SEGMENT 14.0mm. THK X 1636.4mm. X 2000.0mm.	SA-36
2	CYLINDRICAL SKIRT SEGMENT 14.0mm. THK X 3054.2mm. X 1000.0mm.	SA-516 70
1	CYLINDRICAL SKIRT SEGMENT 14.0mm. THK X 1604.1mm. X 1000.0mm.	SA-516 70
2	CYLINDRICAL SEGMENT 16.0mm. THK X 3048.0mm. X 2725.0mm.	SA-516 70
1	CYLINDRICAL SEGMENT 16.0mm. THK X 1600.9mm. X 2725.0mm.	SA-516 70
1	NAMEPLATE	...

**Minimum Design Metal Temperature Results Summary (°C):**

Description	Notes	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm.	E*	PWHT reqd
BOTTOM HEAD	[10]	D	-46	-48		0.738	16.000	0.85	No
BOTTOM HEAD	[7]	D	-40	-48		0.568	20.000	0.85	No
SHELL	[8]	D	-46	-48		0.743	16.000	0.85	No
TOP HEAD	[10]	D	-46	-48		0.738	16.000	0.85	No
TOP HEAD	[7]	D	-40	-48		0.568	20.000	0.85	No
D1-2"	[1]	B	-46	-48		0.671	16.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
N1-16"	[1]	B	-14	-28		0.751	16.000	1.00	No
Nozzle Flg	[5]	A	-18	-44					
K1-2"	[1]	B	-14	-28		0.751	16.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
K2-2"	[1]	B	-14	-28		0.751	16.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
K3-2"	[1]	B	-14	-28		0.751	16.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
K4-2"	[1]	B	-14	-28		0.751	16.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
M1-24"	[1]	B	-18	-32		0.751	14.000	1.00	No
Nozzle Flg	[5]	A	-18	-44					
N2-14"	[1]	B	-14	-32		0.671	16.000	1.00	No
Nozzle Flg	[5]	A	-18	-44					
K5-2"	[1]	B	-14	-32		0.671	16.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
K6-2"	[1]	B	-14	-32		0.671	16.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
Noz1. Bolting[22]			-48			1.000	64.000	1.00	No

Warmest MDMT: -14 -28

Required Minimum Design Metal Temperature -12.0 °C

Warmest Computed Minimum Design Metal Temperature -28.0 °C

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.

UCS-66(g) was not considered.

UCS-66(i) was not considered.

Notes:

Impact test temps were not entered in and not considered in the analysis.

UCS-66(i) applies to impact tested materials not by specification and



UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).
The Basic MDMT includes the (30F) PWHT credit if applicable.

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Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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Units used in this Analysis (barsUniCode):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm.
Force	Pounds	0.4536	Kgf
Mass	Pounds	0.4536	kg.
Area	sq. inches	6.4516	cm2
Moment	ft. lbs.	0.1383	Kg-m.
Stress	lbs./sq.in.	0.0069	N./mm2
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0689	bars
Elast. Modulus	lbs./sq.in.	6.8946	KPa.
Pipe Density	lbs./cu.in.	0.0277	kg./cm3
Ins. Density	lbs./cu.ft.	0.1600E-04	kg./cm3
Fluid Density	lbs./cu.ft.	0.1600E-04	kg./cm3
Wind Speed	miles/hr	1.6093	Km/hr
Tray Weight	lbs./sq.ft.	0.0005	Kgs/cm2
Inertia	in.4	41.6231	cm4
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8825	Kgs/m2
Elevation	Feet	304.8000	mm.
Volume	in.3	16.3900	cm3
Diameter	inches	25.4000	mm.
Thickness	inches	25.4000	mm.

PV Elite Vessel Analysis Program: Input Data

Design Internal Pressure (for Hydrotest)	10.799	bars
Design Internal Temperature	82.0	°C
Projection of Nozzle from Vessel Top	0	mm.
Projection of Nozzle from Vessel Bottom	0	mm.
Minimum Design Metal Temperature	-12.0	°C
Type of Construction	Welded	
Special Service	None	
Degree of Radiography	RT-1	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	N	

Shop Pressure Test:

Type of Pressure Test	UG-99(c)
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	No

Field Pressure Test:

Type of Pressure Test	UG-99(b)
Pressure Test Position	Vertical
Test Performed in Corroded Condition	Yes

Load Case 1	NP+EW+WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW



Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed	[V] 97.2 Km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	300 mm.
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment	H or Hh 0 mm.
Distance Upwind of Crest	Lh 0 mm.
Distance from Crest to the Vessel	x 0 mm.
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm.

Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	1.250
Table Value Fa	1.300
Table Value Fv	1.500
Max. Mapped Res. Acceleration	[Ss] 0.147
Max. Eff. Ground Acceleration	[S] 0.065
Force Modification Factor R	1.250
Site Class	C
Component Elevation Ratio	z/h 0.000
Amplification Factor	Ap 2.500
Force Factor	0.000
Consider Vertical Acceleration	No
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000

Overall M.A.W.P. + Static Head	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y
Use ASME VIII-1 Appendix 1-9	N

Perform Blast Load Analysis	No
Material Database Year	Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	Yes
Do not use Nozzle MDMT Interpretation VIII-1 01-37	No
Use Table G instead of exact equation for "A"	Yes
Shell Head Joints are Tapered	Yes
Compute "K" in corroded condition	Yes
Use Code Case 2286	No
Use the MAWP to compute the MDMT	Yes
For thickness ratios <= 0.35, MDMT will be -155F (-104C)	Yes
For PWHT & P1 Materials the MDMT can be < -55F (-48C)	No
Using Metric Material Databases, ASME II D	Yes
Calculate B31.3 type stress for Nozzles with Loads	Yes
Reduce the MDMT due to lower membrane stress	Yes
Consider Longitudinal Stress in MDMT Calculations	Yes

Complete Listing of Vessel Elements and Details:

Element#	1/5
Element From Node	10



Element To Node	20	
Element Type	Skirt Support	
Description	SKIRT	
Distance "FROM" to "TO"	2000	mm.
Skirt Inside Diameter	2455	mm.
Diameter of Skirt at Base	2455	mm.
Skirt Thickness	14	mm.
Internal Corrosion Allowance	3	mm.
Nominal Thickness	14	mm.
External Corrosion Allowance	0	mm.
Design Temperature Internal Pressure	82	°C
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.28	
Material Name	SA-36	
Allowable Stress, Ambient	114	N./mm2
Allowable Stress, Operating	114	N./mm2
Allowable Stress, Hydrotest	225	N./mm2
Material Density	0.00775	kg./cm3
P Number Thickness	32.004	mm.
Yield Stress, Operating	230.08	N./mm2
UCS-66 Chart Curve Designation	A	
External Pressure Chart Name	CS-2	
UNS Number	K02600	
Product Form	Plate; sheet	
Efficiency, Longitudinal Seam	0.7	
Efficiency, Head-to-Skirt or Circ. Seam	0.7	
Weld is pre-Heated	No	

Element#	2/5	
Element From Node	20	
Element To Node	30	
Element Type	Skirt Support	
Description	SKIRT	
Distance "FROM" to "TO"	1000	mm.
Skirt Inside Diameter	2455	mm.
Diameter of Skirt at Base	2455	mm.
Skirt Thickness	14	mm.
Internal Corrosion Allowance	3	mm.
Nominal Thickness	14	mm.
External Corrosion Allowance	0	mm.
Design Temperature Internal Pressure	82	°C
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.28	
Material Name	SA-516 70	
Allowable Stress, Ambient	138	N./mm2
Allowable Stress, Operating	138	N./mm2
Allowable Stress, Hydrotest	234	N./mm2
Material Density	0.00775	kg./cm3
P Number Thickness	32.004	mm.
Yield Stress, Operating	242.6	N./mm2
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K02700	
Product Form	Plate	
Efficiency, Longitudinal Seam	0.7	
Efficiency, Head-to-Skirt or Circ. Seam	0.5	
Weld is pre-Heated	No	

Element#	3/5	
Element From Node	30	
Element To Node	40	
Element Type	Elliptical	
Description	BOTTOM HEAD	
Distance "FROM" to "TO"	50	mm.
Inside Diameter	2450	mm.
Element Thickness	16	mm.



Internal Corrosion Allowance	3 mm.
Nominal Thickness	20 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	10.799 bars
Design Temperature Internal Pressure	82 °C
Design External Pressure	0 bars
Design Temperature External Pressure	0 °C
Effective Diameter Multiplier	1.28
Material Name	SA-516 70
Efficiency, Longitudinal Seam	0.85
Efficiency, Circumferential Seam	0.85
Elliptical Head Factor	2.0
Weld is pre-Heated	No

Element From Node	30
Detail Type	Platform
Detail ID	PLAT:[1 OF 1]
Dist. from "FROM" Node / Offset dist	0 mm.
Platform Start Angle (degrees)	360.0
Platform End Angle (degrees)	0.0
Platform Wind Area	30000 cm2
Platform Weight	3339.2 Kg
Platform Railing Weight	0 Kg/mm.
Platform Grating Weight	0.017 Kgs/cm2
Platform Width	1500 mm.
Platform Height	1000 mm.
Platform Clearance or End Offset	85 mm.
Platform Force Coefficient	1.0
Ladder Layout Angle	180.0
Ladder Start Elevation	0 mm.
Ladder End Elevation	3000 mm.
Unit Weight of Ladder	0.003773 Kg/mm.
Platform Length (top head platform)	0 mm.

Element From Node	30
Detail Type	Liquid
Detail ID	Liquid: 20
Dist. from "FROM" Node / Offset dist	-612.5 mm.
Height/Length of Liquid	662.5 mm.
Liquid Density	0.0009874 kg./cm3

Element From Node	30
Detail Type	Nozzle
Detail ID	D1-2"
Dist. from "FROM" Node / Offset dist	0 mm.
Nozzle Diameter	2 in.
Nozzle Schedule	XXS
Nozzle Class	300
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	29.048 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-106 B

Element From Node	30
Detail Type	Weight
Detail ID	Anti Swirl Baff
Dist. from "FROM" Node / Offset dist	-612 mm.
Miscellaneous Weight	0 Kg
Offset from Element Centerline	0 mm.

Element#	4/5
Element From Node	40
Element To Node	50
Element Type	Cylinder
Description	SHELL
Distance "FROM" to "TO"	2725 mm.
Inside Diameter	2450 mm.



Element Thickness	16	mm.
Internal Corrosion Allowance	3	mm.
Nominal Thickness	16	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	10.799	bars
Design Temperature Internal Pressure	82	°C
Design External Pressure	0	bars
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.28	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	0.85	
Efficiency, Circumferential Seam	0.85	
Weld is pre-Heated	No	
Element From Node	40	
Detail Type	Liquid	
Detail ID	Liquid: 30	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Liquid	1125	mm.
Liquid Density	0.0009874	kg./cm ³
Element From Node	40	
Detail Type	Nozzle	
Detail ID	N1-16"	
Dist. from "FROM" Node / Offset dist	1625	mm.
Nozzle Diameter	16	in.
Nozzle Schedule	None	
Nozzle Class	150	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	496.87	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	
Element From Node	40	
Detail Type	Nozzle	
Detail ID	K1-2"	
Dist. from "FROM" Node / Offset dist	1075	mm.
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	135.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	9.3929	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	
Element From Node	40	
Detail Type	Nozzle	
Detail ID	K2-2"	
Dist. from "FROM" Node / Offset dist	550	mm.
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	9.3383	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	
Element From Node	40	
Detail Type	Nozzle	
Detail ID	K3-2"	
Dist. from "FROM" Node / Offset dist	1275	mm.
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	270.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	9.3383	Kgf



Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	
Element From Node	40	
Detail Type	Nozzle	
Detail ID	K4-2"	
Dist. from "FROM" Node / Offset dist	275	mm.
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	270.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	9.3383	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	
Element From Node	40	
Detail Type	Nozzle	
Detail ID	M1-24"	
Dist. from "FROM" Node / Offset dist	750	mm.
Nozzle Diameter	24	in.
Nozzle Schedule	None	
Nozzle Class	150	
Layout Angle	0.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	457.35	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	
Element From Node	40	
Detail Type	Weight	
Detail ID	Demisting Pad	
Dist. from "FROM" Node / Offset dist	2475	mm.
Miscellaneous Weight	110	Kgf
Offset from Element Centerline	0	mm.
Element From Node	40	
Detail Type	Weight	
Detail ID	Davit Weight-M1	
Dist. from "FROM" Node / Offset dist	750	mm.
Miscellaneous Weight	50	Kgf
Offset from Element Centerline	1650	mm.
Element From Node	40	
Detail Type	Weight	
Detail ID	Clip Weight	
Dist. from "FROM" Node / Offset dist	1360	mm.
Miscellaneous Weight	1025	Kgf
Offset from Element Centerline	0	mm.
Element From Node	40	
Detail Type	Weight	
Detail ID	Demisting Suppo	
Dist. from "FROM" Node / Offset dist	2475	mm.
Miscellaneous Weight	64	Kgf
Offset from Element Centerline	0	mm.

Element#	5/5	
Element From Node	50	
Element To Node	60	
Element Type	Elliptical	
Description	TOP HEAD	
Distance "FROM" to "TO"	50	mm.
Inside Diameter	2450	mm.
Element Thickness	16	mm.
Internal Corrosion Allowance	3	mm.
Nominal Thickness	20	mm.
External Corrosion Allowance	0	mm.



Design Internal Pressure	10.799	bars
Design Temperature Internal Pressure	82	°C
Design External Pressure	0	bars
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.28	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	0.85	
Efficiency, Circumferential Seam	0.85	
Elliptical Head Factor	2.0	
Weld is pre-Heated	No	
Element From Node	50	
Detail Type	Platform	
Detail ID	SQR PLAT	
Dist. from "FROM" Node / Offset dist	930	mm.
Platform Start Angle (degrees)	0.0	
Platform End Angle (degrees)	0.0	
Platform Wind Area	50000	cm2
Platform Weight	4265.2	Kgf
Platform Railing Weight	0	Kgf/mm.
Platform Grating Weight	0.017	Kgs/cm2
Platform Width	5000	mm.
Platform Height	1000	mm.
Platform Clearance or End Offset	2500	mm.
Platform Force Coefficient	1.0	
Ladder Layout Angle	180.0	
Ladder Start Elevation	0	mm.
Ladder End Elevation	3440	mm.
Unit Weight of Ladder	0.003773	Kgf/mm.
Platform Length (top head platform)	5000	mm.
Element From Node	50	
Detail Type	Nozzle	
Detail ID	N2-14"	
Dist. from "FROM" Node / Offset dist	0	mm.
Nozzle Diameter	14	in.
Nozzle Schedule	80	
Nozzle Class	150	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	113.49	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	
Element From Node	50	
Detail Type	Nozzle	
Detail ID	K5-2"	
Dist. from "FROM" Node / Offset dist	650	mm.
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	270.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	18.163	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	
Element From Node	50	
Detail Type	Nozzle	
Detail ID	K6-2"	
Dist. from "FROM" Node / Offset dist	650	mm.
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	18.163	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	



Element From Node	50
Detail Type	For./Mom.
Detail ID	N2 Force
Dist. from "FROM" Node / Offset dist	630 mm.
Force in X Direction	1784.6 Kgf
Force in Y Direction	-2676.9 Kgf
Force in Z Direction	1784.6 Kgf
Moment about X Axis	1748.3 Kg-m.
Moment about Y Axis	0 Kg-m.
Moment about Z Axis	1748.3 Kg-m.
Force/Moment Combination Method	SRSS

XY Coordinate Calculations:

From	To	X (Horiz.) mm.	Y (Vert.) mm.	DX (Horiz.) mm.	DY (Vert.) mm.
	SKIRT	...	2000	...	2000
	SKIRT	...	3000	...	1000
	BOTTOM HEAD	...	3050	...	50
	SHELL	...	5775	...	2725
	TOP HEAD	...	5825	...	50

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd bars	Nominal Thickness mm.	Total Corr Allowance mm.	Element Diameter mm.	Allowable Stress(SE) N./mm ²	Element#
	SKIRT	...	14	3	2455	...	1
	SKIRT	...	14	3	2455	...	2
	BOTTOM HEAD	10.98	20	3	2450	117.3	3
	SHELL	10.915	16	3	2450	117.3	4
	TOP HEAD	10.806	20	3	2450	117.3	5

Element Required Thickness and MAWP :

From	To	Design Pressure bars	M.A.W.P. Corroded bars	M.A.P. New & Cold bars	Minimum Thickness mm.	Required Thickness mm.	Element#
	SKIRT	...	No Calc	No Calc	14	No Calc	1
	SKIRT	...	No Calc	No Calc	14	No Calc	2
	BOTTOM HEAD	10.7994	12.3	15.3	16	14.4686	3
	SHELL	10.7994	12.2	15.2	16	14.4921	4
	TOP HEAD	10.7994	12.4	15.3	16	14.2874	5

Minimum 10.8 15.2

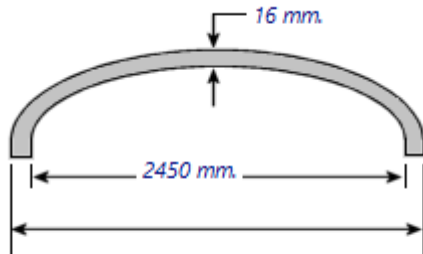
MAWP: 10.8 bars, limited by: DESIGN (user specified)

Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:

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Internal Pressure Results for: BOTTOM HEAD

Elliptical Head Design Information:							
Design Pressure		P	10.980 bars	Int. Design Temperature		82 °C	
Material			SA-516 70	Ext. Design Temperature		0 °C	
Length			50.000 mm.	External Pressure Chart		CS-2	
				UNS Number		K02700	
Int. Corr. All.		c	3.0000 mm.	Allowable Stress (ope)		S	138.0 N./mm^2
Ext. Corr. All.		ce	0.0000 mm.	Allowable Stress (amb)		Sa	138.0 N./mm^2
Inside Diameter		Di	2450.000 mm.	Actual Stress		121.8 N./mm^2	
Spec. Min. Thk t _e , or s _a			16.0000 mm.	Spec. Nominal Thk.		20.000 mm.	
Surface Area			71119.5 cm^2	SG of Contents		0.989	
Element Volume			2161132.0 cm3	Weight of Contents		2136.1 Kg	
Empty Weight			1191.9 Kg	Operating Weight		3328.0 Kg	
				Comp. All. Ope. Stress		111.8 N./mm^2	
				Aspect Ratio		2.00	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 30.625 mm.

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 15.313 mm.

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (11 \cdot 2456 \cdot 1) / (2 \cdot 138 \cdot 0.85 - 0.2 \cdot 11)$$

$$= 11.4686 + 3.0000 = 14.4686 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.180 bars

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 0.85 \cdot 13) / (1 \cdot 2456 + 0.2 \cdot 13)$$

$$= 12.4 - 0.18 = 12.3 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 0.85 \cdot 16) / (1 \cdot 2450 + 0.2 \cdot 16)$$

$$= 15.3 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (11 \cdot (1 \cdot 2456 + 0.2 \cdot 13)) / (2 \cdot 0.85 \cdot 13)$$

$$= 121.758 \text{ N./mm}^2$$

Straight Flange Required Thickness:
 $= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c$ per UG-27 (c)(1)
 $= (11 \cdot 1228) / (138 \cdot 0.85 - 0.6 \cdot 11) + 3$
 $= 14.560 \text{ mm.}$

Straight Flange Maximum Allowable Working Pressure:
 Less Operating Hydrostatic Head Pressure of 0.121 bars
 $= (S \cdot E \cdot t) / (R + 0.6 \cdot t)$ per UG-27 (c)(1)
 $= (138 \cdot 0.85 \cdot 17) / (1228 + 0.6 \cdot 17)$
 $= 16.1 - 0.12 = 16 \text{ bars}$

Factor K, corroded condition [Kcor]:
 $= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6$
 $= (2 + (2456 / (2 \cdot 616))^2) / 6$
 $= 0.996755$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 3.517 %

MDMT Calculations in the Knuckle Portion:

Govrn. thk, $t_g = 16$, $t_r = 11.3$, $c = 3 \text{ mm.}$, $E^* = 0.85$
 Thickness Ratio $= t_r \cdot E^* / (t_g - c) = 0.74$, Temp. Reduction = 15 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -46 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, $t_g = 20$, $t_r = 11.4$, $c = 3 \text{ mm.}$, $E^* = 0.85$
 Thickness Ratio $= t_r \cdot E^* / (t_g - c) = 0.57$, Temp. Reduction = 25 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -40 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	10.799 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	16.000 mm.	Basic MDMT	-46 °C
Stress/Thk. Ratio	0.738	Temperature Difference	3 °C
UCS-66(c) red.	No	Computed Min. Temp.	-48 °C
		Joint Efficiency E^*	85 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

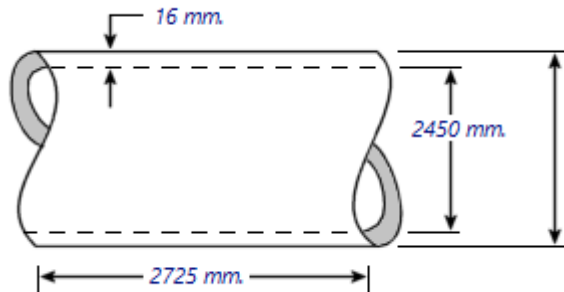
Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 30.625 mm.
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 15.313 mm.

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Internal Pressure Calculation Results:

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Internal Pressure Results for: SHELL

Cylindrical Shell Design Information:					
Design Pressure	P	10.915 bars	Int. Design Temperature		82 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		2725.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm.	Allowable Stress (ope)	S	138.0 N./mm^2
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm^2
Inside Diameter	Di	2450.000 mm.	Actual Stress		122.1 N./mm^2
Spec. Min. Thk t _e , or sa		16.0000 mm.	Spec. Nominal Thk.		16.000 mm.
Surface Area		212480.1 cm^2	SG of Contents		0.989
Element Volume		12848914.0 cm3	Weight of Contents		5243.0 Kgf
Empty Weight		2617.8 Kgf	Operating Weight		7860.8 Kgf
			Comp. All. Ope. Stress		111.8 N./mm^2

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (10.9 \cdot 1228) / (138 \cdot 0.85 - 0.6 \cdot 10.9)$$

$$= 11.4921 + 3.0000 = 14.4921 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.116 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 0.85 \cdot 13) / (1228 + 0.6 \cdot 13)$$

$$= 12.3 - 0.12 = 12.2 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 0.85 \cdot 16) / (1225 + 0.6 \cdot 16)$$

$$= 15.2 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (10.9 \cdot (1228 + 0.6 \cdot 13)) / (0.85 \cdot 13)$$

$$= 122.082 \text{ N./mm}^2$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 0.649 %

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 16, tr = 11.4, c = 3 mm., E* = 0.85



Thickness Ratio = $tr * E^*/(tg - c) = 0.74$, Temp. Reduction = 14 °C

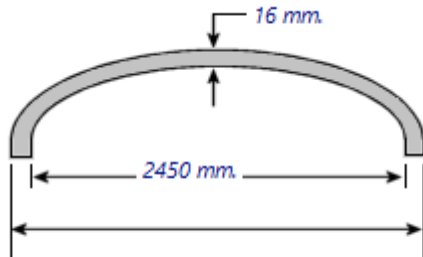
Min Metal Temp. w/o impact per UCS-66, Curve D -46 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	10.799 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	16.000 mm.	Basic MDMT	-46 °C
Stress/Thk. Ratio	0.743	Temperature Difference	3 °C
UCS-66(c) red.	No	Computed Min. Temp.	-48 °C
		Joint Efficiency E*	85 %

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: TOP HEAD

Elliptical Head Design Information:							
Design Pressure	P	10.806	bars	Int. Design Temperature		82	°C
Material		SA-516	70	Ext. Design Temperature		0	°C
Length		50.000	mm.	External Pressure Chart		CS-2	
				UNS Number		K02700	
Int. Corr. All.	c	3.0000	mm.	Allowable Stress (ope)	S	138.0	N./mm^2
Ext. Corr. All.	ce	0.0000	mm.	Allowable Stress (amb)	Sa	138.0	N./mm^2
Inside Diameter	Di	2450.000	mm.	Actual Stress		119.8	N./mm^2
Spec. Min. Thk t _e , or sa		16.0000	mm.	Spec. Nominal Thk.		20.000	mm.
Surface Area		71119.5	cm^2	SG of Contents		0.000	
Element Volume		2161132.0	cm3	Weight of Contents		0.0	Kgf
Empty Weight		1191.9	Kgf	Operating Weight		1191.9	Kgf
				Comp. All. Ope. Stress		111.8	N./mm^2
				Aspect Ratio		2.00	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 30.625 mm.

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 15.313 mm.

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (10.8 \cdot 2456 \cdot 1) / (2 \cdot 138 \cdot 0.85 - 0.2 \cdot 10.8)$$

$$= 11.2874 + 3.0000 = 14.2874 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.007 bars

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 0.85 \cdot 13) / (1 \cdot 2456 + 0.2 \cdot 13)$$

$$= 12.4 - 0.007 = 12.4 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 0.85 \cdot 16) / (1 \cdot 2450 + 0.2 \cdot 16)$$

$$= 15.3 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (10.8 \cdot (1 \cdot 2456 + 0.2 \cdot 13)) / (2 \cdot 0.85 \cdot 13)$$

$$= 119.837 \text{ N./mm}^2$$



Straight Flange Required Thickness:
 $= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c$ per UG-27 (c)(1)
 $= (10.8 \cdot 1228) / (138 \cdot 0.85 - 0.6 \cdot 10.8) + 3$
 $= 14.377 \text{ mm.}$

Straight Flange Maximum Allowable Working Pressure:
 Less Operating Hydrostatic Head Pressure of 0.007 bars
 $= (S \cdot E \cdot t) / (R + 0.6 \cdot t)$ per UG-27 (c)(1)
 $= (138 \cdot 0.85 \cdot 17) / (1228 + 0.6 \cdot 17)$
 $= 16.1 - 0.007 = 16.1 \text{ bars}$

Factor K, corroded condition [Kcor]:
 $= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6$
 $= (2 + (2456 / (2 \cdot 616))^2) / 6$
 $= 0.996755$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 3.517 %

MDMT Calculations in the Knuckle Portion:

Govrn. thk, tg = 16, tr = 11.3, c = 3 mm., $E^* = 0.85$
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.74$, Temp. Reduction = 15 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -46 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, tg = 20, tr = 11.4, c = 3 mm., $E^* = 0.85$
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.57$, Temp. Reduction = 25 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -40 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	10.799 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	16.000 mm.	Basic MDMT	-46 °C
Stress/Thk. Ratio	0.738	Temperature Difference	3 °C
UCS-66(c) red.	No	Computed Min. Temp.	-48 °C
		Joint Efficiency E^*	85 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 30.625 mm.
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 15.313 mm.

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Hydrostatic Test Pressure Results:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	14.039 bars
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	14.039 bars
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	19.521 bars
UG100	= 1.10 * M.A.W.P. * Sa/S	11.879 bars
PED	= max(1.43*DP, 1.25*DP*ratio)	15.443 bars
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	15.443 bars
App 27-4	= M.A.W.P.	10.799 bars

Horizontal Test performed per: UG-99c

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & bars):

From To	Stress	Allowable	Ratio	Pressure
BOTTOM HEAD	178.2	234.0	0.762	19.76
SHELL	179.4	234.0	0.767	19.76
TOP HEAD	178.2	234.0	0.762	19.76

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
D1-2"	Nozzle	118.00	118.00	1.000
D1-2"	Pad	138.00	138.00	1.000
N1-16"	Nozzle	118.00	118.00	1.000
N1-16"	Pad	138.00	138.00	1.000
K1-2"	Nozzle	138.00	138.00	1.000
K2-2"	Nozzle	138.00	138.00	1.000
K3-2"	Nozzle	138.00	138.00	1.000
K4-2"	Nozzle	138.00	138.00	1.000
M1-24"	Nozzle	118.00	118.00	1.000
M1-24"	Pad	138.00	138.00	1.000
N2-14"	Nozzle	118.00	118.00	1.000
N2-14"	Pad	138.00	138.00	1.000
K5-2"	Nozzle	138.00	138.00	1.000
K6-2"	Nozzle	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
BOTTOM HEAD	138.00	138.00	1.000
SHELL	138.00	138.00	1.000
TOP HEAD	138.00	138.00	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
D1-2"	4.59	216.00	0.021
N1-16"	24.31	216.00	0.113
K1-2"	4.18	225.00	0.019
K2-2"	4.18	225.00	0.019
K3-2"	4.18	225.00	0.019
K4-2"	4.18	225.00	0.019
M1-24"	42.23	216.00	0.196
N2-14"	17.65	216.00	0.082
K5-2"	4.18	225.00	0.019



K6-2"	4.18	225.00	0.019
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Elements Suitable for Test Pressure.

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Hydrostatic Test Pressure Results:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	14.039 bars
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	14.039 bars
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	19.429 bars
UG100	= 1.10 * M.A.W.P. * Sa/S	11.879 bars
PED	= max(1.43*DP, 1.25*DP*ratio)	15.443 bars
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	15.443 bars
App 27-4	= M.A.W.P.	10.799 bars

UG-99(b), Test Pressure Calculation:
 = Test Factor • MAWP • Stress Ratio
 = 1.3 • 10.8 • 1
 = 14.039 bars

Vertical Test performed per: UG-99b

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & bars):

From To	Stress	Allowable	Ratio	Pressure
BOTTOM HEAD	160.1	234.0	0.684	14.44
SHELL	160.7	234.0	0.687	14.37
TOP HEAD	156.4	234.0	0.668	14.10

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
D1-2"	Nozzle	118.00	118.00	1.000
D1-2"	Pad	138.00	138.00	1.000
N1-16"	Nozzle	118.00	118.00	1.000
N1-16"	Pad	138.00	138.00	1.000
K1-2"	Nozzle	138.00	138.00	1.000
K2-2"	Nozzle	138.00	138.00	1.000
K3-2"	Nozzle	138.00	138.00	1.000
K4-2"	Nozzle	138.00	138.00	1.000
M1-24"	Nozzle	118.00	118.00	1.000
M1-24"	Pad	138.00	138.00	1.000
N2-14"	Nozzle	118.00	118.00	1.000
N2-14"	Pad	138.00	138.00	1.000
K5-2"	Nozzle	138.00	138.00	1.000
K6-2"	Nozzle	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
BOTTOM HEAD	138.00	138.00	1.000
SHELL	138.00	138.00	1.000
TOP HEAD	138.00	138.00	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
D1-2"	3.35	216.00	0.016
N1-16"	17.68	216.00	0.082
K1-2"	3.04	225.00	0.014
K2-2"	3.04	225.00	0.014



K3-2"	3.04	225.00	0.014
K4-2"	3.04	225.00	0.014
M1-24"	30.72	216.00	0.142
N2-14"	12.60	216.00	0.058
K5-2"	2.98	225.00	0.013
K6-2"	2.98	225.00	0.013

Elements Suitable for Test Pressure.

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External Pressure Calculation Results Summary:

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External Pressure Calculations:

From	To	Section Length mm.	Outside Diameter mm.	Corroded Thickness mm.	Factor A	Factor B N./mm ²
10	20	3000	...	...	No Calc	No Calc
20	30	3000	...	...	No Calc	No Calc
30	40	No Calc	2482	13	0.00072746	72.7296
40	50	3233.33	2482	13	0.00038605	38.5965
50	60	No Calc	2482	13	0.00072746	72.7296

External Pressure Calculations:

From	To	External Actual T. mm.	External Required T. mm.	External Design Pressure bars	External M.A.W.P. bars
10	20	...	No Calc	...	No Calc
20	30	...	No Calc	...	No Calc
30	40	16	No Calc	...	4.23
40	50	16	No Calc	...	2.7
50	60	16	No Calc	...	4.23

Minimum

2.7

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm.	Allowable Length Bet. Stiffeners mm.	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	3000	No Calc	No Calc	No Calc
20	30	3000	No Calc	No Calc	No Calc
30	40	No Calc	No Calc	No Calc	No Calc
40	50	3233.33	No Calc	No Calc	No Calc
50	60	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Element and Detail Weights:

From	To	Element Metal Wgt. kg.	Element ID Volume cm3	Corroded Metal Wgt. kg.	Corroded ID Volume cm3	Extra due Misc % kg.
10	20	2549.85	...	2190.75	...	178.49
20	30	841.646	...	662.097	...	58.9152
30	40	1191.94	2161132	1013.15	2176468	83.4361
40	50	2617.8	12848914	2129.54	12911926	183.246
50	60	1191.94	2161132	1013.15	2176468	83.4361
Total		8393	17171178.00	7008	17264862.00	587

Weight of Details:

From	Type	Weight of Detail kg.	X Offset, Dtl. Cent. mm.	Y Offset, Dtl. Cent. mm.	Z Offset Dtl. Cent. mm.	Description
30	Plat	3572.97	...	...	0.19047E-04	PLAT:[1 OF 1]
30	Liqd	2136.06	...	-306.25	...	Liquid: 20
30	Nozl	31.0808	...	-734.497	...	D1-2"
30	Wght	...	...	-612	...	Anti Swirl Baff
40	Liqd	5242.95	...	562.5	...	Liquid: 30
40	Nozl	531.65	1350.01	1625	...	N1-16"
40	Nozl	10.0504	922.772	1075	922.772	K1-2"
40	Nozl	9.99194	1305	550	...	K2-2"
40	Nozl	9.99194	-1305	1275	...	K3-2"
40	Nozl	9.99194	-1305	275	...	K4-2"
40	Nozl	489.361	...	750	-1400.01	M1-24"
40	Wght	110	...	2475	...	Demisting Pad
40	Wght	50	...	750	-1650	Davit Weight-M1
40	Wght	1025	...	1360	...	Clip Weight
40	Wght	64	...	2475	...	Demisting Suppo
50	Plat	4563.73	...	930	...	SQR PLAT
50	Nozl	121.436	...	779.506	...	N2-14"
50	Nozl	19.4348	-650	801.169	...	K5-2"
50	Nozl	19.4348	650	801.169	...	K6-2"
50	Forc	...	...	630	...	N2 Force

Total Weight of Each Detail Type:

Platforms	8136.7
Liquid	7379.0
Nozzles	1252.4
Weights	1249.0
Sum of the Detail Weights	18017.1 kg.

Weight Summation Results: (kg.)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	8980.7	8980.7	8980.7	8980.7	8980.7	8980.7
Nozzles	1252.4	1252.4	1252.4	1252.4	1252.4	1252.4
Wld Weights	1089.0	1089.0	1089.0	1089.0	1089.0	1089.0
Platforms	...	...	...	8136.7	8136.7	8136.7
Empty Weights	...	...	...	160.0	160.0	...
Test Weights	...	50.0	...	...	...	...
Ope Weights	...	...	...	...	...	160.0
Ope. Liquid	...	...	...	...	...	7379.0
Test Liquid	...	17160.7	...	...	...	...
Totals	11322.1	28532.8	11322.1	19618.8	19618.8	26997.8

Field Installation Options:

Platforms installed before lifting.



Miscellaneous Weight Percent: 7.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	11322.1 kg.
Shop Test Wt.	- Fabricated weight + water (full)	28532.8 kg.
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	11322.1 kg.
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	19618.8 kg.
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	19618.8 kg.
Operating Wt.	- Empty weight + operating liq. uncorroded	26997.8 kg.
Field Test Wt.	- Empty weight + test medium (full)	36669.5 kg.

Weights used in Vortex Shedding Calculations:

Mass of the Upper 1/3 of the Vertical Vessel (operating)	10499.9 kg.
Mass of the Upper 1/3 of the Vertical Vessel (filled w/water)	18757.4 kg.
Mass of the Upper 1/3 of the Vertical Vessel (empty)	9694.6 kg.

Surface Areas of Elements:

From	To	Outside Surface Area cm^2	Inside Surface Area cm^2
10	20	156012	154252
20	30	78005.8	77126.1
30	40	71119.5	68914.7
40	50	212480	209741
50	60	71119.5	68914.7

Total	588736.2	578948.2 cm2
Total	633.7	623.2 ft²

Element and Detail Weights:

From	To	Total Ele. Empty Wgt. kg.	Total. Ele. Oper. Wgt. kg.	Total. Ele. Hydro. Wgt. kg.	Total Dtl. Offset Mom. Kg-m.	Oper. Wgt. No Liquid kg.
10	20	2728.34	2728.34	2728.34	...	2728.34
20	30	900.562	900.562	900.562	...	900.562
30	40	1306.46	7015.49	7039.24	0.68055E-04	4879.43
40	50	5111.09	10354	17842.2	758.35	5111.09
50	60	1435.69	5999.42	8159.23	...	5999.42

Empty Support Force + the Sum of the Y forces	14159.07	Kgf
Operating Support Force + the Sum of the Y forces	29674.78	Kgf
Hydro Support Force + the Sum of the Y forces	39346.47	Kgf

Cumulative Vessel Weight

From	To	Cumulative Ope Wgt. No Liquid kg.	Cumulative Oper. Wgt. kg.	Cumulative Hydro. Wgt. kg.
10	20	19618.8	26997.8	36669.5
20	30	16890.5	24269.5	33941.2
30	40	15989.9	23368.9	33040.6
40	50	11110.5	16353.5	26001.4
50	60	5999.42	5999.42	8159.23

The cumulative operating weights no liquid in the column above are the cumulative operating weights minus the operating liquid weight minus any weights absent in the empty condition.

Cumulative Vessel Moment

From	To	Cumulative Empty Mom. Kg-m.	Cumulative Oper. Mom. Kg-m.	Cumulative Hydro. Mom. Kg-m.
10	20	758.35	758.35	758.35
20	30	758.35	758.35	758.35
30	40	758.35	758.35	758.35
40	50	758.35	758.35	758.35
50	60	...	...	...

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Nozzle Flange MAWP Results: (bars & °C)

Nozzle Description	Flange Rating		Des. Temp	Class	Grade/Group	Equiv. Press	Max Pressure		
	Ope.	Ambient					UG-44	50%	DNV
D1-2"	47.82	51.00	82	300	GR 1.1	53.407	18.318	47.817	18.318
N1-16"	18.20	19.60	82	150	GR 1.1	18.554	18.204	18.204	8.753
K1-2"	47.82	51.00	82	300	GR 1.1	...	...	...	...
K2-2"	47.82	51.00	82	300	GR 1.1	...	...	...	...
K3-2"	47.82	51.00	82	300	GR 1.1	...	...	...	...
K4-2"	47.82	51.00	82	300	GR 1.1	...	...	...	...
M1-24"	18.20	19.60	82	150	GR 1.1	...	...	...	...
N2-14"	18.20	19.60	82	150	GR 1.1	20.976	18.204	18.204	6.330
K5-2"	47.82	51.00	82	300	GR 1.1	...	...	...	...
K6-2"	47.82	51.00	82	300	GR 1.1	...	...	...	...
Min Rating	18.204	19.600, [for Core Elements]					...	...	...

Selected Method for Derating ASME B16.5 Flange MAWP: ASME UG-44(b)

The UG-44(b) Method is based on the paper PVP 2013-97814. PV Elite uses the maximum loads from each load category to determine ME and FE. In many cases, the computed maximum allowable pressure will be greater than the flange rating. In these cases, the minimum of the rating from the table and the UG-44(b) method will be used. SA-193 B8 Cl. 2 bolts or ones with higher allowable stresses at the specified bolt size shall be used.

The axial force FE must put the nozzle in tension (-P for 107/297; +Fr for PD 5500 AnnexG) to be considered.

Nozzle Description	Gasket Dimension G mm.	Fm Value	Resolved Moment ME Kg-m.	Axial Force FE Kgf	MAWP PD bars	Maximum Pressure Pmax bars	Pass Fail
D1-2"	77.851	0.50	50	0	10.799	18.318	Pass
N1-16"	442.976	1.20	3229	0	10.799	18.204	Pass
N2-14"	389.001	1.20	2472	0	10.799	18.204	Pass

Evaluating Flange on Nozzle [D1-2"]:
 $= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot [(PR - PD) + FM \cdot PR]$
 $= 16(50) + 4 \cdot \text{abs}(0)77.851 \leq \pi \cdot 77.851^3 \cdot [(47.817 - 10.799) + .500 \cdot 47.817]$
 $= 0.80732E+03 \leq 0.92097E+03$ Check Passes

Maximum Pressure for Flange on Nozzle D1-2" [Pmax]:
 $= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$
 $= \min(47.817, 0.500 \cdot 47.817 + 47.817 - 807.3 / (\pi \cdot 77.851^3))$
 $= \min(47.817, 18.318)$
 $= 18.318 \text{ bars}$

Evaluating Flange on Nozzle [N1-16"]:
 $= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot [(PR - PD) + FM \cdot PR]$
 $= 16(3229) + 4 \cdot \text{abs}(0)442.976 \leq \pi \cdot 442.976^3 \cdot [(18.204 - 10.799) + 1.200 \cdot 18.204]$
 $= 0.51668E+05 \leq 0.81456E+05$ Check Passes

Maximum Pressure for Flange on Nozzle N1-16" [Pmax]:
 $= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$
 $= \min(18.204, 1.200 \cdot 18.204 + 18.204 - 51668.2 / (\pi \cdot 442.976^3))$
 $= \min(18.204, 21.496)$
 $= 18.204 \text{ bars}$

Evaluating Flange on Nozzle [N2-14"]:
 $= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot [(PR - PD) + FM \cdot PR]$
 $= 16(2472) + 4 \cdot \text{abs}(0)389.001 \leq \pi \cdot 389.001^3 \cdot [(18.204 - 10.799) + 1.200 \cdot 18.204]$
 $= 0.39559E+05 \leq 0.55161E+05$ Check Passes

Maximum Pressure for Flange on Nozzle N2-14" [Pmax]:
 $= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$



$$= \min(18.204, 1.200 \cdot 18.204 + 18.204 - 39558.5/(\pi \cdot 389.001^3))$$

$$= \min(18.204, 19.073)$$

$$= 18.204 \text{ bars}$$

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

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The Natural Frequencies for the vessel have been computed iteratively by solving a system of matrices. These matrices describe the mass and the stiffness of the vessel. This is the generalized eigenvalue/eigenvector problem and is referenced in some mathematical texts.

The Natural Frequency for the Vessel (Empty.) is 19.7032 Hz.

The Natural Frequency for the Vessel (Ope...) is 16.9785 Hz.

The Natural Frequency for the Vessel (Filled) is 14.0231 Hz.

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Forces/Moments Applied to Vessel (Combined w/Wind Loads)

From	To	X and Z Dir Force Res. Kgf	X,Z Moment and For Res Kg-m.
10	20	...	...
20	30	...	...
30	40	...	...
40	50	...	...
50	60	2523.84	4062.46

Forces/Moments Applied to Vessel (Combined w/Seismic Loads)

From	To	X and Z Dir Force Res. Kgf	X,Z Moment and For Res Kg-m.
10	20	...	...
20	30	...	...
30	40	...	...
40	50	...	...
50	60	2523.84	4062.46

User Input Forces and Moments:

From Node	Distance From	- - - Fx	Forces - - - Fy	- - - Fz	- - - Mx	Moments - - - My	- - - Mz
50	630.00	1785.	-2677.	1785.	1748.		1748.

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Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed [V]	97.2 Km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	300 mm.
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm.
Distance Upwind of Crest Lh	0 mm.
Distance from Crest to the Vessel x	0 mm.
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm.

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Static Gust-Effect Factor, Operating Case [G]:
 $= \min(0.85, 0.925((1 + 1.7 \cdot gQ \cdot I_{zbar} \cdot Q) / (1 + 1.7 \cdot gV \cdot I_{zbar})))$
 $= \min(0.85, 0.925((1 + 1.7 \cdot 3.4 \cdot 0.23 \cdot 0.95) / (1 + 1.7 \cdot 3.4 \cdot 0.23)))$
 $= \min(0.85, 0.9)$
 $= 0.850$

Natural Frequency of Vessel (Operating)	16.979 Hz
Natural Frequency of Vessel (Empty)	19.703 Hz
Natural Frequency of Vessel (Test)	14.023 Hz

User Entered Importance Factor is	1.000
Force Coefficient [Cf]	0.528
Structure Height to Diameter ratio	2.676
Height to top of Structure	6453.500 mm.

This is classified as a rigid structure. Static analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of $[\alpha]$ and $[Zg]$:
 Exposure Category: C from Table 26.11-1
 $\alpha = 9.5 : Zg = 274320 \text{ mm.}$

Effective Height $[z]$:
 $= \text{Centroid Height} + \text{Vessel Base Elevation}$
 $= 1000 + 300 = 1300 \text{ mm.}$
 $= 4.27 \text{ ft. Imperial Units}$

Velocity Pressure coefficient evaluated at height z $[Kz]$:
 Because $z (4.27 \text{ ft.}) < 15 \text{ ft.}$
 $= 2.01(15 / Zg)^{2/\alpha}$
 $= 2.01(15/900)^{2/9.5}$
 $= 0.85$

Type of Hill: No Hill

Wind Directionality Factor $[Kd]$:
 $= 1, \text{ per Table 26.6-1}$

As there is No Hill Present: $[Kzt]$:

K1 = 0, K2 = 0, K3 = 0

Topographical Factor [Kzt]:

$$= (1 + K1 \cdot K2 \cdot K3)^2$$

$$= (1 + 0 \cdot 0 \cdot 0)^2$$

$$= 1$$

Velocity Pressure evaluated at height z, Imperial Units [qz]:

$$= \max(16, 0.00256 \cdot Kz \cdot Kzt \cdot Kd \cdot Ke \cdot V(\text{mph})^2)$$

$$= \max(16, 0.00256 \cdot 0.85 \cdot 1 \cdot 1 \cdot 1 \cdot 60.4^2)$$

$$= 16 \text{ psf } [78.1] \text{ Kgs/m}^2$$

Force on the first element [F]:

$$= qz \cdot G \cdot Cf \cdot \text{WindArea}$$

$$= 16 \cdot 0.85 \cdot 0.53 \cdot 68.4$$

$$= 491 \text{ lbs. } [223] \text{ Kg}$$

Element	Hgt (z) mm.	K1	K2	K3	Kz	Kzt	qz Kgs/m ²
SKIRT	1300.0	0.000	0.000	0.000	0.849	1.000	78.120
SKIRT	2800.0	0.000	0.000	0.000	0.849	1.000	78.120
BOTTOM HEAD	3325.0	0.000	0.000	0.000	0.849	1.000	78.120
SHELL	4712.5	0.000	0.000	0.000	0.854	1.000	78.120
TOP HEAD	6364.9	0.000	0.000	0.000	0.910	1.000	78.120

Wind Vibration Calculations

This evaluation is based on work by Kanti Mahajan and Ed Zorilla

Nomenclature

- Cf - Correction factor for natural frequency
- D - Average outer insulated diameter of vessel, mm.
- Df - Damping Factor < 0.75 Unstable, > 0.95 Stable
- Dr - Average outer insulated of top half of vessel, mm.
- f - Natural frequency of vibration (Hertz)
- f1 - Natural frequency of bare vessel based on a unit value of $(D/L^2)(10^4)$
- L - Total height of structure mm.
- Lc - Total length of conical section(s) of vessel, mm.
- tb - Uncorroded plate thickness at bottom of vessel, mm.
- V30 - Design Wind Speed provided by user, Km/hr
- Vc - Critical wind velocity, Km/hr
- Vw - Maximum wind speed at top of structure, Km/hr
- W - Total corroded weight of structure, Kg
- Ws - Cor. vessel weight excl. weight of parts which do not effect stiff. Kg
- Z - Maximum amplitude of vibration at top of vessel, mm.
- Dl - Logarithmic decrement (taken as 0.03 for Welded Structures)
- Vp - Vib. Chance, <= 0.32037E-06 (High); 0.32037E-06 < 0.40047E-06 (Probable)
- P30 - wind pressure 30 feet above the base

Check other Conditions and Basic Assumptions:

:: Total Cone Length / Total Length < 0.5
 :: 0/5825 = 0

:: $[D / L^2] \cdot 10^4 < 8$ (English Units, ft.)
 :: $[8.14/19.1^2] \cdot 10^4 = 223$ [Geometry Violation]

Compute the vibration possibility. If $Vp > 0.40047E-06$ no chance. [Vp]:

$$= W / (L \cdot Dr^2)$$

$$= 25516 / (5825 \cdot 2482^2)$$

$$= 0.71107E-06$$

Since Vp is > 0.40047E-06 no further vibration analysis is required !

Platform Load Calculations

ID	Wind Area	Elevation	Pressure	Force	Cf
----	-----------	-----------	----------	-------	----



	cm2	mm.	Kgs/m2	Kgf	
PLAT:[1 OF 1]	30000.00	3300.00	78.12	234.36	1.00
SQR PLAT	50000.00	7005.00	78.12	390.60	1.00

Wind Loads on Masses/Equipment/Piping

ID	Wind Area cm2	Elevation mm.	Pressure Kgs/m2	Force Kgf
Anti Swirl Ba	0.00	2688.00	78.12	0.00
Demisting Pad	0.00	5825.00	78.12	0.00
Davit Weight-	0.00	4100.00	78.12	0.00
Clip Weight	0.00	4710.00	78.12	0.00
Demisting Sup	0.00	5825.00	78.12	0.00

The Natural Frequency for the Vessel (Ope...) is 16.9785 Hz.

Wind Load Calculation:

From	To	Wind Height mm.	Wind Diameter mm.	Wind Area cm^2	Wind Pressure Kgs/m^2	Element Wind Load Kgf
10	20	1300	3178.24	63564.8	78.12	245.118
20	30	2800	3178.24	31782.4	78.12	122.559
30	40	3325	3176.96	1588.48	78.12	261.984
40	50	4712.5	3176.96	86572.2	78.12	333.838
50	60	6364.91	3176.96	17270.7	78.12	539.469

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	1.250
Table Value Fa	1.300
Table Value Fv	1.500
Max. Mapped Res. Acceleration [Ss]	0.147
Max. Eff. Ground Acceleration [S]	0.065
Force Modification Factor R	1.250
Site Class	C
Component Elevation Ratio z/h	0.000
Amplification Factor Ap	2.500
Force Factor	0.000
Consider Vertical Acceleration	No
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000

Seismic Analysis Results:

$$\begin{aligned}
 SMS &= Fa * Ss = 1.3 * 0.15 = 0.19 \\
 SM1 &= Fv * S1 = 1.5 * 0.065 = 0.097 \\
 SDS &= 2/3 * Sms = 2/3 * 0.19 = 0.13 \\
 SD1 &= 2/3 * Sm1 = 2/3 * 0.097 = 0.065
 \end{aligned}$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
 $= Ct \cdot h_{nx}$ where $Ct = 0.020$, $x = 0.75$ and h_n = Structural Height (ft.)
 $= 0.020(21.10.75)$
 $= 0.197$ seconds

The Coefficient C_u from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
 $= (1/\text{Natural Frequency}) = (1/17)$
 $= 0.059$

Check the Value of T which is the smaller of $C_u \cdot T_a$ and T:
 $= \text{Minimum Value of } (1.7 \cdot 0.2, 0.059) \text{ per 12.8.2}$
 $= 0.059$

As the time period is < 0.06 second, use section 15.4.2.

Base Shear per equation 15.4-5, [V]:
 $= 0.3 \cdot Sds \cdot W \cdot I$
 $= 0.3 \cdot 0.13 \cdot 26998 \cdot 1.25$
 $= 1289.822$ Kgf

Distribute the Base shear force to each element according to the equations
 $F_x = C_{vx} \cdot V$ (eqn. 12.8-11) and the vertical distribution factor
 $C_{vx} = W_x \cdot h_x^k / (\text{Sum of } W_i \cdot h_i^k)$ and k is an exponent which is related to the period of Vibration.

In this case, the value of k was 1

The Natural Frequency for the Vessel (Ope...) is 16.9785 Hz.

Earthquake Load Calculation:

From	To	Earthquake Height mm.	Earthquake Weight Kgf	Element Ope Load Kgf	Element Emp Load Kgf	Element Vertical Load Kgf
10	20	1000	2728.34	32.9856	37.5937	-2676.93
20	30	2500	900.562	27.2194	31.0219	-2676.93



30	40	3025	7015.49	256.571	54.455	-2676.93
40	50	4412.5	10354	552.356	310.752	-2676.93
50	60	5800	5999.42	420.69	114.737	-2676.93

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Bending Moments due to user defined forces and moments.

User Force/Moment Shear and Bending

From	To	Distance to Support mm.	Cumulative Shr Wind Cas Kgf	Cumulative Shr Eqk Cas Kgf	Wind Bending Kg-m.	Earthquake Bending Kg-m.
10	20	1000	2523.84	2523.84	18637.9	18637.9
20	30	2500	2523.84	2523.84	13590.1	13590.1
30	40	3025	2523.84	2523.84	11066.2	11066.2
40	50	4412.5	2523.84	2523.84	10940	10940
50	60	5800	2523.84	2523.84	4062.46	4062.46

Note:

The Wind Shears/Moments and the Earthquake Shears/Moments calculated and printed in the Wind/Earthquake Shear and Bending report have been factored by the input Scalar/Load reductions factors of;
 Wind: 1.100; Earthquake: 1.000.

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The following table is for the Operating Case.

Wind/Earthquake Shear, Bending:

From	To	Distance to Support mm.	Cumulative Wind Shear Kgf	Earthquake Shear Kgf	Wind Bending Kg-m.	Earthquake Bending Kg-m.
10	20	1000	1502.97	1289.82	6089.03	5866
20	30	2500	1257.85	1256.84	3328.16	3319.29
30	40	3025	1135.29	1229.62	2131.56	2076.03
40	50	4412.5	873.307	973.046	2081.35	2020.97
50	60	5800	539.469	420.69	156.401	121.965

Note:

The Wind Shears/Moments and the Earthquake Shears/Moments calculated and printed in the Wind/Earthquake Shear and Bending report have been factored by the input Scalar/Load reductions factors of;
 Wind: 1.100; Earthquake: 1.000.

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Wind Deflection Calculations:

The following table is for Wind Deflection; Applied Forces Case.

Wind Load Deflection:

From	To	Cumulative Shear Kgf	Centroid Deflection mm.	Elem. End Deflection mm.	Elem. Ang. Rotation
10	20	2523.84	0.0058433	0.022268	0.21163E-04
20	30	2523.84	0.033931	0.047618	0.29259E-04
30	40	2523.84	0.048351	0.049089	0.29607E-04
40	50	2523.84	0.09518	0.15008	0.42538E-04
50	60	2523.84	0.15114	0.15221	0.42538E-04

Allowable deflection at the Tower Top (For)(6.000"/100ft. Criteria)
 Allowable deflection : 29.125 Actual deflection : 0.152 mm.

The following table is for Wind Deflection; Operating Case.

Wind Load Deflection:

From	To	Cumulative Shear Kgf	Centroid Deflection mm.	Elem. End Deflection mm.	Elem. Ang. Rotation
10	20	1502.97	0.0018615	0.0068204	0.62083E-05
20	30	1257.85	0.010184	0.013999	0.80170E-05
30	40	1135.29	0.0142	0.014402	0.80844E-05
40	50	873.307	0.02643	0.039676	0.99053E-05
50	60	539.469	0.039924	0.040172	0.99054E-05

Critical Wind Velocity for Tower Vibration:

From	To	1st Crit. Wind Speed Km/hr	2nd Crit. Wind Speed Km/hr
10	20	968.695	6054.34
20	30	968.695	6054.34
30	40	968.305	6051.91
40	50	968.305	6051.91
50	60	968.305	6051.91

Allowable deflection at the Tower Top (Ope)(6.000"/100ft. Criteria)
 Allowable deflection : 29.125 Actual deflection : 0.040 mm.

Total Deflection in the Operating Condition + Applied Forces:
 Allowable deflection : 29.125 Actual deflection : 0.192 mm.

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Longitudinal Stress Constants:

From	To	Metal Area New cm ²	Metal Area Corroded cm ²	Section Modulus New mm. ³	Section Modulus Corroded mm. ³
10	20	1085.92	854.262	66652764	52560468
20	30	1085.92	854.262	66652764	52560468
30	40	1239.54	1008.35	75928520	61916416
40	50	1239.54	1008.35	75928520	61916416
50	60	1239.54	1008.35	75928520	61916416

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Longitudinal Allowable Stresses:

From	To	Tensile N./mm ²	Hydrotest Tensile N./mm ²	Compressive N./mm ²	Hydrotest Compressive N./mm ²
10	20	95.7583	188.997	-106.054	-106.054
20	30	82.7985	140.397	-106.054	-106.054
30	40	140.76	238.68	-111.788	-111.788
40	50	140.76	238.68	-111.788	-111.788
50	60	140.76	238.68	-111.788	-111.788

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Longitudinal Stress Report



Longitudinal Operating and Empty Stresses are computed in the corroded condition. Stresses due to loads in the hydrostatic test cases have also been computed in the corroded condition.

Longitudinal Pressure Stresses due to:

From	To	Longitudinal Stress Internal Pressure N./mm ²	Longitudinal Stress External Pressure N./mm ²	Longitudinal Stress Hydrotest Pressure N./mm ²
10	20	...	...	...
20	30	...	...	...
30	40	50.8262	...	66.0312
40	50	50.8262	...	66.0312
50	60	50.8262	...	66.0312

Longitudinal Stresses due to Weight Loads for these Conditions:

From	To	Wght. Str. Empty N./mm ²	Wght. Str. Operating N./mm ²	Wght. Str. Hydrotest N./mm ²	Wght. Str. Emp. Mom. N./mm ²	Wght. Str. Opr. Mom. N./mm ²
10	20	-2.25222	-3.09932	-4.20962	0.14149	0.14149
20	30	-1.93901	-2.78611	-3.89641	0.14149	0.14149
30	40	-1.55512	-1.55512	-1.55512	0.12011	0.12011
40	50	-1.08056	-1.08056	-1.08056	0.12011	0.12011
50	60	-0.58348	-0.58348	-0.58348	...	...

Longitudinal Stresses due to Weight Loads and Bending for these Conditions:

From	To	Wght. Str. Hyd. Mom. N./mm ²	Bend. Str. Oper. Wind N./mm ²	Bend. Str. Oper. Equ. N./mm ²	Bend. Str. Hyd. Wind N./mm ²	Bend. Str. Hyd. Equ. N./mm ²
10	20	0.14149	1.13608	1.09447	0.56804	...
20	30	0.14149	0.62096	0.61931	0.31048	...
30	40	0.12011	0.33761	0.32881	0.1688	...
40	50	0.12011	0.32965	0.32009	0.16483	...
50	60	...	0.024772	0.019317	0.012386	...

Longitudinal Stresses due to these Conditions:

From	To	Vortex Shedding Operating Case N./mm ²	Vortex Shedding Empty Case N./mm ²	Vortex Shedding Test Case N./mm ²	Earthquake Empty Case N./mm ²
10	20	...	...	...	0.4379
20	30	...	...	...	0.24021
30	40	...	...	...	0.12544
40	50	...	...	...	0.12185
50	60	...	...	...	0.0052685

Longitudinal Stresses due to Applied Axial Forces:

From	To	Longitudinal Stress Y Forces Wind N./mm ²	Longitudinal Stress Y Forces Seismic N./mm ²
10	20	-0.30731	-0.30731
20	30	-0.30731	-0.30731
30	40	-0.26035	-0.26035

40	50	-0.26035	-0.26035
50	60	-0.26035	-0.26035

Longitudinal Stresses due to User Forces and Moments:

From	To	Wind For/Mom Corroded N./mm ²	Earthquake For/Mom Corroded N./mm ²	Wind For/Mom No Corrosion N./mm ²	Earthquake For/Mom No Corrosion N./mm ²
10	20	3.47743	3.47743	2.7422	2.7422
20	30	2.53562	2.53562	1.99952	1.99952
30	40	1.75273	1.75273	1.42927	1.42927
40	50	1.73274	1.73274	1.41297	1.41297
50	60	0.64343	0.64343	0.52469	0.52469

Bending Stresses due to Blast Load:

From	To	Bending Stress N./mm ²
10	20	...
20	30	...
30	40	...
40	50	...
50	60	...

Stress Combination Load Cases for Vertical Vessels:

Load Case Definition Key

IP = Longitudinal Stress due to Internal Pressure
 EP = Longitudinal Stress due to External Pressure
 HP = Longitudinal Stress due to Hydrotest Pressure
 NP = No Pressure
 EW = Longitudinal Stress due to Weight (No Liquid)
 OW = Longitudinal Stress due to Weight (Operating)
 HW = Longitudinal Stress due to Weight (Hydrotest)
 WI = Bending Stress due to Wind Moment (Operating)
 EQ = Bending Stress due to Earthquake Moment (Operating)
 EE = Bending Stress due to Earthquake Moment (Empty)
 HI = Bending Stress due to Wind Moment (Hydrotest)
 HE = Bending Stress due to Earthquake Moment (Hydrotest)
 WE = Bending Stress due to Wind Moment (Empty) (no CA)
 WF = Bending Stress due to Wind Moment (Filled) (no CA)
 CW = Longitudinal Stress due to Weight (Empty) (no CA)
 VO = Bending Stress due to Vortex Shedding Loads (Ope)
 VE = Bending Stress due to Vortex Shedding Loads (Emp)
 VF = Bending Stress due to Vortex Shedding Loads (Test No CA.)
 FW = Axial Stress due to Vertical Forces for the Wind Case
 FS = Axial Stress due to Vertical Forces for the Seismic Case
 BW = Bending Stress due to Lat. Forces for the Wind Case, Corroded
 BS = Bending Stress due to Lat. Forces for the Seismic Case, Corroded
 BN = Bending Stress due to Lat. Forces for the Wind Case, UnCorroded
 BU = Bending Stress due to Lat. Forces for the Seismic Case, UnCorroded
 BL = Bending Stress due to Blast Loads, Corroded

General Notes:

Case types HI and HE are in the Corroded condition.

Case types WE, WF, and CW are in the Un-Corroded condition.

A blank stress and stress ratio indicates that the corresponding stress comprising those components that did not contribute to that type of stress.

An asterisk (*) in the final column denotes overstress.

Analysis of Load Case 1 : NP+EW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	2.20	95.76	-7.31	106.05	0.0229	0.0690
20	1.05	82.80	-5.54	106.05	0.0127	0.0523
30	0.39	140.76	-4.03	111.79	0.0028	0.0360
40	0.84	140.76	-3.52	111.79	0.0060	0.0315
50		140.76	-1.51	111.79		0.0135

Analysis of Load Case 2 : NP+EW+EE+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	1.50	95.76	-6.62	106.05	0.0156	0.0624
20	0.67	82.80	-5.16	106.05	0.0081	0.0487
30	0.18	140.76	-3.81	111.79	0.0013	0.0341
40	0.63	140.76	-3.32	111.79	0.0045	0.0297
50		140.76	-1.49	111.79		0.0134

Analysis of Load Case 3 : NP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	1.35	95.76	-8.16	106.05	0.0141	0.0770
20	0.20	82.80	-6.39	106.05	0.0025	0.0603
30	0.39	140.76	-4.03	111.79	0.0028	0.0360
40	0.84	140.76	-3.52	111.79	0.0060	0.0315



50 140.76 -1.51 111.79 0.0135

Analysis of Load Case 4 : NP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	1.31	95.76	-8.12	106.05	0.0136	0.0766
20	0.20	82.80	-6.39	106.05	0.0025	0.0603
30	0.39	140.76	-4.02	111.79	0.0027	0.0359
40	0.83	140.76	-3.51	111.79	0.0059	0.0314
50		140.76	-1.51	111.79		0.0135

Analysis of Load Case 5 : NP+HW+HI

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		189.00	-4.92	106.05		0.0464
20		140.40	-4.35	106.05		0.0410
30		238.68	-1.84	111.79		0.0165
40		238.68	-1.37	111.79		0.0122
50		238.68	-0.60	111.79		0.0053

Analysis of Load Case 6 : NP+HW+HE

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		189.00	-4.35	106.05		0.0410
20		140.40	-4.04	106.05		0.0381
30		238.68	-1.68	111.79		0.0150
40		238.68	-1.20	111.79		0.0107
50		238.68	-0.58	111.79		0.0052

Analysis of Load Case 7 : IP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	1.35	95.76	-8.16	106.05	0.0141	0.0770
20	0.20	82.80	-6.39	106.05	0.0025	0.0603
30	51.22	140.76		111.79	0.3639	
40	51.67	140.76		111.79	0.3671	
50	50.65	140.76		111.79	0.3598	

Analysis of Load Case 8 : IP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	1.31	95.76	-8.12	106.05	0.0136	0.0766
20	0.20	82.80	-6.39	106.05	0.0025	0.0603
30	51.21	140.76		111.79	0.3638	
40	51.66	140.76		111.79	0.3670	
50	50.65	140.76		111.79	0.3598	

Analysis of Load Case 9 : EP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	1.35	95.76	-8.16	106.05	0.0141	0.0770
20	0.20	82.80	-6.39	106.05	0.0025	0.0603
30	0.39	140.76	-4.03	111.79	0.0028	0.0360
40	0.84	140.76	-3.52	111.79	0.0060	0.0315
50		140.76	-1.51	111.79		0.0135

Analysis of Load Case 10 : EP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	1.31	95.76	-8.12	106.05	0.0136	0.0766
20	0.20	82.80	-6.39	106.05	0.0025	0.0603
30	0.39	140.76	-4.02	111.79	0.0027	0.0359
40	0.83	140.76	-3.51	111.79	0.0059	0.0314



50 140.76 -1.51 111.79 0.0135

Analysis of Load Case 11 : HP+HW+HI

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		189.00	-4.92	106.05		0.0464
20		140.40	-4.35	106.05		0.0410
30	64.77	238.68		111.79	0.2713	
40	65.24	238.68		111.79	0.2733	
50	65.46	238.68		111.79	0.2743	

Analysis of Load Case 12 : HP+HW+HE

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		189.00	-4.35	106.05		0.0410
20		140.40	-4.04	106.05		0.0381
30	64.60	238.68		111.79	0.2706	
40	65.07	238.68		111.79	0.2726	
50	65.45	238.68		111.79	0.2742	

Analysis of Load Case 13 : IP+WE+EW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		95.76	-2.39	106.05		0.0226
20		82.80	-2.08	106.05		0.0196
30	49.39	140.76		111.79	0.3509	
40	49.87	140.76		111.79	0.3543	
50	50.24	140.76		111.79	0.3569	

Analysis of Load Case 14 : IP+WF+CW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		95.76	-2.44	106.05		0.0230
20		82.80	-2.19	106.05		0.0207
30	49.56	140.76		111.79	0.3521	
40	49.95	140.76		111.79	0.3548	
50	50.35	140.76		111.79	0.3577	

Analysis of Load Case 15 : IP+VO+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		95.76	-3.24	106.05		0.0306
20		82.80	-2.93	106.05		0.0276
30	49.39	140.76		111.79	0.3509	
40	49.87	140.76		111.79	0.3543	
50	50.24	140.76		111.79	0.3569	

Analysis of Load Case 16 : IP+VE+EW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		95.76	-2.39	106.05		0.0226
20		82.80	-2.08	106.05		0.0196
30	49.39	140.76		111.79	0.3509	
40	49.87	140.76		111.79	0.3543	
50	50.24	140.76		111.79	0.3569	

Analysis of Load Case 17 : NP+VO+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		95.76	-3.24	106.05		0.0306
20		82.80	-2.93	106.05		0.0276
30		140.76	-1.68	111.79		0.0150
40		140.76	-1.20	111.79		0.0107



50 140.76 -0.58 111.79 0.0052

Analysis of Load Case 18 : FS+BS+IP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.21	79.80	-7.03	88.38	0.0027	0.0795
20		69.00	-5.77	88.38		0.0653
30	50.88	117.30		93.16	0.4338	
40	51.34	117.30		93.16	0.4377	
50	50.63	117.30		93.16	0.4316	

Analysis of Load Case 19 : FS+BS+EP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.21	79.80	-7.03	88.38	0.0027	0.0795
20		69.00	-5.77	88.38		0.0653
30	0.06	117.30	-3.69	93.16	0.0005	0.0396
40	0.51	117.30	-3.19	93.16	0.0044	0.0343
50		117.30	-1.49	93.16		0.0160

Analysis of Load Case 20 : BL+IP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		95.76	-3.24	106.05		0.0306
20		82.80	-2.93	106.05		0.0276
30	49.39	140.76		111.79	0.3509	
40	49.87	140.76		111.79	0.3543	
50	50.24	140.76		111.79	0.3569	

Absolute Maximum of the all of the Stress Ratio's 0.4377

Governing Element: SHELL

Governing Load Case 18 : FS+BS+IP+OW

Element Description	Long Pressure Stress	Weight + Axial Load Stress	Eccentric Moment Stress	Wind or Earthquake Stress	Combined Stress	Which Side
SKIRT	...	-3.4	0.1	3.5	0.2	+ Tens
SKIRT	...	-3.4	-0.1	-3.5	-7.0	- Comp
SKIRT	...	-3.1	0.1	2.5	...	+ Tens
SKIRT	...	-3.1	-0.1	-2.5	-5.8	- Comp
BOTTOM HEAD	50.8	-1.8	0.1	1.8	50.9	+ Tens
BOTTOM HEAD	50.8	-1.8	-0.1	-1.8	...	- Comp
SHELL	50.8	-1.3	0.1	1.7	51.3	+ Tens
SHELL	50.8	-1.3	-0.1	-1.7	...	- Comp
TOP HEAD	50.8	-0.8	...	0.6	50.6	+ Tens
TOP HEAD	50.8	-0.8	...	-0.6	...	- Comp

Skirt Support Analysis Per EN 13445-3 Section 16.12:

The analysis of this skirt has been conducted referencing the EN 13445-3 design code. It is important to note that the selected design code does not explicitly cover the analysis of this component. Therefore, the results provided herein should be regarded as informational only.

Item: SKIRT

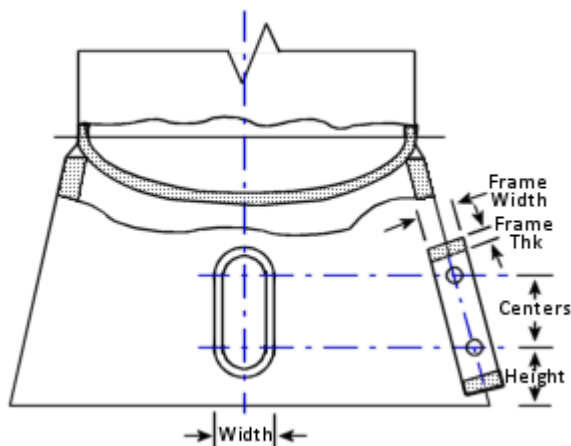
Total vessel (empty) weight	Fg	16227.341 Kgf
Head Weight below section 2-2	DeltaFg	4879.431 Kgf
Weight of vessel contents (liquid)	Ff	7379.010 Kgf
Vessel Gross Weight	Fg + Ff	23606.350 Kgf
Skirt Mean Diameter	Dz	2472.000 mm.
Skirt Analysis Thickness (Corroded)	ez	11.000 mm.
Skirt Height	H	1000.000 mm.
Additional Axial Force at section 1-1	F1	-2676.933 Kgf
Moment at section 1-1	M1	13197.803 Kg-m.
Applied Moment - Bottom of Skirt	Mb	16918.283 Kg-m.
Weight at Bottom of Skirt	W	24269.504 Kgf

Structural Type B Skirt per EN 13445 Figure 16.12-2:

Type of Head:		Elliptical
Head Mean Diameter at Skirt Top	Db	2469.000 mm.
Head Corroded Thickness	eb	13.000 mm.

Per EN 13445 Section 7.5.4 [R, r]:

$$\begin{aligned}
 h_i &= 616 \text{ mm.} \\
 K &= D_i / (2 * h_i) = 2456 / (2 * 616) = 2 \\
 R &= D_i (0.44 * K + 0.02) = 2456 * (0.44 * 2 + 0.02) = 2205 \text{ mm.} \\
 r &= D_i ((0.5 / K) - 0.08) = 2456 * ((0.5 / 2) - 0.08) = 419 \text{ mm.}
 \end{aligned}$$



There are no Access Openings in the Skirt - Plain Skirt Considered

Moment of Inertia	Ixx =	6525255.500 cm ⁴
Moment of Inertia	Iyy =	6525255.500 cm ⁴
Section Modulus	W =	52793380.000 mm. ³
Section Metal Area	A =	854.262 cm ²

Analysis of the Section at weakest section Modulus [Z min]:

$$\begin{aligned}
 &\text{Radius of Gyration of Skirt at 4-4 [rg]:} \\
 &= \sqrt{(\min(I_{xx}, I_{yy}) / \text{Area})} \\
 &= \sqrt{(\min(6525255, 6525255) / 854)} \\
 &= 874 \text{ mm.}
 \end{aligned}$$

Skirt Slenderness Ratio [rg/L]:

$$\begin{aligned}
 &= r_g / L \\
 &= 874/1000 \\
 &= 0.874
 \end{aligned}$$

Correct the Moment [M4]:
 top of skirt is the critical end
 Controlling Moment [M4] = 13198 Kg-m.

Axial Stress at Residual (Weakest) Surface Area [Sigma]:
 Downwind Side:
 $= -M / Z - W / A$
 $= -13198/211173520 - 24270/854$
 $= -3.4 \text{ N./mm}^2$

Upwind Side:
 $= +(M) / Z - W / A$
 $= +13198/211173520 - 24270/854$
 $= -2.17 \text{ N./mm}^2$

16.12.3.4 Checking at connection areas:

Force at the joint, 16.12-1 [Fzp]:
 $= F1 - Fg - Ff + 4 \cdot M1 / Dz$
 $= -2677 - 16227 - 7379 + 4 \cdot 13198/2472$
 $= -4928 \text{ Kgf}$

Force at the joint, 16.12-2 [Fzq]:
 $= F1 - Fg - Ff - 4 \cdot M1 / Dz$
 $= -2677 - 16227 - 7379 - 4 \cdot 13198/2472$
 $= -47639 \text{ Kgf}$

16.12.3.4.1 Membrane stresses:

Membrane stress at 1-1, 16.12-3 & 4:

Stress [σ_{1pm}]:
 $= (Fzp + \Delta Fg) / (\pi \cdot Db \cdot eb) + P \cdot Db / (4 \cdot eb)$
 $= (-4928 + 4879 + 7379) / (\pi \cdot 2469 \cdot 13) + 10.8 \cdot 2469 / (4 \cdot 13)$
 $= 51.99 \text{ N./mm}^2$

Stress [σ_{1qm}]:
 $= (Fzq + \Delta Fg + Ff) / (\pi \cdot Db \cdot eb) + P \cdot Db / (4 \cdot eb)$
 $= (-47639 + 4879 + 7379) / (\pi \cdot 2469 \cdot 13) + 10.8 \cdot 2469 / (4 \cdot 13)$
 $= 47.84 \text{ N./mm}^2$

Membrane stress at 2-2, 16.12-9:

Stress [σ_{2pm} & σ_{2qm}]:
 $= (Ff + \Delta Fg) / (\pi \cdot Db \cdot eb) + P \cdot Db / (4 \cdot eb)$
 $= (7379 + 4879) / (\pi \cdot 2469 \cdot 13) + 10.8 \cdot 2469 / (4 \cdot 13)$
 $= 52.47 \text{ N./mm}^2$

Membrane stress at 3-3, 16.12 12 & 13:

Stress [σ_{3pm}]:
 $= Fzp / (\pi \cdot Dz \cdot ez)$
 $= -4928 / (\pi \cdot 2472 \cdot 11)$
 $= -0.57 \text{ N./mm}^2$

Stress [σ_{3qm}]:
 $= Fzq / (\pi \cdot Dz \cdot ez)$
 $= -47639 / (\pi \cdot 2472 \cdot 11)$
 $= -5.47 \text{ N./mm}^2$

16.12.3.4.2 Bending stresses:

Cosine of angle, 16.12-25 [$\cos(\gamma)$]:
 $= 1 - 0.5(Db + eb - Dz + ez) / (r + eb)$
 $= 1 - 0.5(2469 + 13 - 2472 + 11) / (419 + 13)$
 $= 0.98$

$$\gamma = \arccos(\cos(\gamma)) = \arccos(0.98) = 12.7 \text{ deg}$$

Dimension [a]:

$$= 0.5 \cdot \sqrt{(eb^2 + ez^2 + 2 \cdot eb \cdot ez \cdot \cos(\gamma))}$$

$$= 0.5 \cdot \sqrt{(13^2 + 11^2 + 2 \cdot 13 \cdot 11 \cdot 0.98)}$$

$$= 11.9 \text{ mm.}$$

Bending moment at verification points p and q [Mp & Mq]:

$$M_p = a \cdot F_{zp} = 11.92 \cdot -4928.0 = -58.78 \text{ Kg-m.}$$

$$M_q = a \cdot F_{zq} = 11.92 \cdot -47638.5 = -568.21 \text{ Kg-m.}$$

Correction factor [C]:

$$= 0.63 - 0.057 \left(\frac{eb}{ez} \right)^2$$

$$= 0.63 - 0.057 \left(\frac{13}{11} \right)^2$$

$$= 0.550$$

Bending stresses in sections 1-1 to 3-3 at the outer surface (o), 16.12-26 to 29:

Stress [$\sigma_{1pb(o)}$ & $\sigma_{2pb(o)}$]:

$$= C \cdot 6 \cdot M_p / (\pi \cdot Db \cdot eb^2)$$

$$= 0.55 \cdot 6 \cdot -58.8 / (\pi \cdot 2469 \cdot 13^2)$$

$$= -1.45 \text{ N./mm}^2$$

Stress [$\sigma_{1qb(o)}$ & $\sigma_{2qb(o)}$]:

$$= C \cdot 6 \cdot M_q / (\pi \cdot Db \cdot eb^2)$$

$$= 0.55 \cdot 6 \cdot -568 / (\pi \cdot 2469 \cdot 13^2)$$

$$= -14.04 \text{ N./mm}^2$$

Stress [$\sigma_{3pb(o)}$]:

$$= C \cdot 6 \cdot M_p / (\pi \cdot Dz \cdot ez^2)$$

$$= 0.55 \cdot 6 \cdot -58.8 / (\pi \cdot 2472 \cdot 11^2)$$

$$= -2.03 \text{ N./mm}^2$$

Stress [$\sigma_{3qb(o)}$]:

$$= C \cdot 6 \cdot M_q / (\pi \cdot Dz \cdot ez^2)$$

$$= 0.55 \cdot 6 \cdot -568 / (\pi \cdot 2472 \cdot 11^2)$$

$$= -19.58 \text{ N./mm}^2$$

Geometry variables, 16.12-32 to 36:

$$eb/Db = 0.51/97.2 = 0.0053$$

$$y = 125 \cdot eb/Db = 125 \cdot 13/2469 = 0.66$$

As this an elliptical head, it will be treated as a Korbbogen type head

Geometric value [α]:

$$= 1.51861 \cdot 2.71828 - 4.2335 \cdot RYY + 3.994$$

$$= 1.51861 \cdot 2.71828 - 4.2335 \cdot 0.66 + 3.994 = 4.09$$

Bending stress in sections 1-1 and 2-2 Equation 16.12-31 [$\sigma_{1b(p)}$ & $\sigma_{2b(p)}$]:

$$= (P + Ph) \cdot Db \left(\left(\frac{y}{yA} \right) \cdot \alpha - 1 \right) / (4 \cdot eb)$$

$$= (10.8 + 0) \cdot 2469 \left(\left(\frac{12.7}{40} \right) \cdot 4.09 - 1 \right) / (4 \cdot 13)$$

$$= 15.05 \text{ N./mm}^2$$

16.12.3.4.3 Total stresses and strength conditions:

Structural shape B and C:

Stress [σ_{1pitot}]:

$$= \sigma_{1pm} - \sigma_{1pb(o)} + \sigma_{1b(p)}$$

$$= 52 - 1.45 + 15.1$$

$$= 68.50 \text{ N./mm}^2$$

Stress [σ_{1potot}]:

$$= \sigma_{1pm} + \sigma_{1pb(o)} - \sigma_{1b(p)}$$

$$= 52 + 1.45 - 15.1$$

$$= 35.49 \text{ N./mm}^2$$

Stress [σ_{1qitot}]:

$$= \sigma_{1qm} - \sigma_{1qb(o)} + \sigma_{1b(p)}$$



$$= 47.8 - -14 + 15.1$$

$$= 76.93 \text{ N./mm}^2$$

Stress [σ_{1qotot}]:

$$= \sigma_{1qm} + \sigma_{1qb(o)} - \sigma_{1b(p)}$$

$$= 47.8 + -14 - 15.1$$

$$= 18.75 \text{ N./mm}^2$$

Stress [σ_{2pitot}]:

$$= \sigma_{2pm} + \sigma_{2pb(o)} + \sigma_{2b(p)}$$

$$= 52.5 + -1.45 + 15.1$$

$$= 66.07 \text{ N./mm}^2$$

Stress [σ_{2potot}]:

$$= \sigma_{2pm} - \sigma_{2pb(o)} - \sigma_{2b(p)}$$

$$= 52.5 - -1.45 - 15.1$$

$$= 38.87 \text{ N./mm}^2$$

Stress [σ_{2qitot}]:

$$= \sigma_{2qm} + \sigma_{2qb(o)} + \sigma_{2b(p)}$$

$$= 52.5 + -14 + 15.1$$

$$= 53.49 \text{ N./mm}^2$$

Stress [σ_{2qotot}]:

$$= \sigma_{2pm} - \sigma_{2pb(o)} - \sigma_{2b(p)}$$

$$= 52.5 - -14 - 15.1$$

$$= 51.46 \text{ N./mm}^2$$

Stress [σ_{3pitot}]:

$$= \sigma_{3pm} - \sigma_{3pb}$$

$$= -0.57 - -2.03$$

$$= 1.46 \text{ N./mm}^2$$

Stress [σ_{3potot}]:

$$= \sigma_{3pm} + \sigma_{3pb}$$

$$= -0.57 + -2.03$$

$$= -2.59 \text{ N./mm}^2$$

Stress [σ_{3qitot}]:

$$= \sigma_{3qm} - \sigma_{3qb}$$

$$= -5.47 - -19.6$$

$$= 14.11 \text{ N./mm}^2$$

Stress [σ_{3qotot}]:

$$= \sigma_{3qm} + \sigma_{3qb}$$

$$= -5.47 + -19.6$$

$$= -25.05 \text{ N./mm}^2$$

Allowable stresses:

For σ_{1pitot} and σ_{1potot} , the stress must not exceed:

$$= fs(3 - (1 / 1.5)) \cdot (\sigma_{1pm} / f)^2$$

$$= 138 (3 - (1 / 1.5)) \cdot (52 / 138)^2$$

$$= 400.94 \text{ N./mm}^2$$

For σ_{1qitot} and σ_{1qotot} , the stress must not exceed:

$$= fs(3 - (1 / 1.5)) \cdot (\sigma_{1qm} / f)^2$$

$$= 138 (3 - (1 / 1.5)) \cdot (47.8 / 138)^2$$

$$= 402.94 \text{ N./mm}^2$$

For σ_{2pitot} and σ_{2potot} , the stress must not exceed:

$$= fs(3 - (1 / 1.5)) \cdot (\sigma_{2pm} / f)^2$$

$$= 138 (3 - (1 / 1.5)) \cdot (52.5 / 138)^2$$

$$= 400.70 \text{ N./mm}^2$$

For σ_{2qitot} and σ_{2qotot} , the stress must not exceed:

$$= fs(3 - (1 / 1.5)) \cdot (\sigma_{2qm} / f)^2$$

$$= 138 (3 - (1 / 1.5)) \cdot (52.5 / 138)^2$$

$$= 400.70 \text{ N./mm}^2$$

For σ_{3pitot} & σ_{3potot} , the stress must not exceed:

$$= fs(3 - (1 / 1.5) \cdot (\sigma_{3pm} / f)^2)$$

$$= 138 (3 - (1/1.5) \cdot (-0.57/138)^2)$$

$$= 414.00 \text{ N./mm}^2$$

For σ_{3qitot} & σ_{3qotot} , the stress must not exceed:

$$= fs(3 - (1 / 1.5) \cdot (\sigma_{3qm} / f)^2)$$

$$= 138 (3 - (1/1.5) \cdot (-5.47/138)^2)$$

$$= 413.86 \text{ N./mm}^2$$

Please note: None of the following stresses in the table below are in the region where there is an access opening.

Summary of Stresses: (N./mm2)

Stress	Actual	Allowed	Result	Equation
Stot1pi	68.50	400.94	Pass	46
Stot1po	35.49	400.94	Pass	47
Stot1qi	76.93	402.94	Pass	48
Stot1qo	18.75	402.94	Pass	49
Stot2pi	66.07	400.70	Pass	50
Stot2po	38.87	400.70	Pass	51
Stot2qi	53.49	400.70	Pass	52
Stot2qo	51.46	400.70	Pass	53
Stot3pi	1.46	414.00	Pass	54
Stot3po	-2.59	414.00	Pass	55
Stot3qi	14.11	413.86	Pass	56
Stot3qo	-25.05	413.86	Pass	57

The stresses are satisfactory in the region without openings.

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Shop/Field Installation Options :

Platform(s) installed in the Field before being lifted.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Platforms	5078.069 mm.
Center of Gravity of the Liquid	3346.542 mm.
Center of Gravity of the Nozzles	4488.004 mm.
Center of Gravity of the Added Weights (Operating)	4540.913 mm.
Center of Gravity of the Added Weights (Empty)	4540.913 mm.
Center of Gravity of Bare Shell New and Cold	3184.001 mm.
Center of Gravity of Bare Shell Corroded	3165.163 mm.
Vessel CG in the Operating Condition	3959.875 mm.
Vessel CG in the Fabricated (Shop/Empty) Condition	3473.838 mm.
Vessel CG in the Test Condition	4036.216 mm.

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Skirt Data :

Skirt Outside Diameter at Base	SOD, D6	2483.0000	mm.
Skirt Thickness	STHK	14.0000	mm.
Skirt Internal Corrosion Allowance	SCA	3.0000	mm.
Skirt External Corrosion Allowance		0.0000	mm.
Skirt Material		SA-36	

Basing Input: Type of Geometry: Continuous Top Ring W/Gussets

Thickness of Basing	TBA, ea4	32.0000	mm.
Design Temperature of the Basing		38.00	°C
Basing Material		SA-516 70	
Basing Ope. All. Stress	BASOPE, f4	138.00	N./mm2
Basing Yield Stress		262.00	N./mm2
Inside Diameter of Basing	DI, D4	2279.0000	mm.
Outside Diameter of Basing	DOU, D5	2739.0000	mm.
Nominal Diameter of Bolts	BND, d(b0)	42.0000	mm.
Bolt Corrosion Allowance	BCA	1.5000	mm.
Root Area of a Single Bolt	Area	8.6100	cm2
Bolt Material		SA-193 B7	
Bolt Material Class/Thickness Range		<= 64	
Bolt Operating Allowable Stress	SA, fB	172.00	N./mm2
Number of Bolts	RN, nb	16	
Diameter of Bolt Circle	DC, DBC	2611.0000	mm.
Ultimate Comp. Strength of Concrete	FPC	41.0	N./mm2
Allowable Comp. Strength of Concrete	FC	22.5	N./mm2
Modular ratio Steel/Concrete		9.833	
Thickness of Gusset Plates	TGA, e7	16.0000	mm.
Width of Gussets at Top Plate	TWDT	118.0000	mm.
Width of Gussets at Base Plate	BWDT	118.0000	mm.
Gusset Plate Elastic Modulus	E	201734288.0	KPa.
Gusset Plate Yield Stress	SY	262.0	N./mm2
Height of Chair	h1	300.0000	mm.
Distance between Gussets	RG, b6	90.0000	mm.
Dist. from Bolt Center to Gusset (Rg/2)	CG	45.0000	mm.
Number of Gussets per bolt	NG	2	
Thickness of Top Plate or Ring	TTA, e5	38.0000	mm.
Radial Width of the Top Plate	TOPWTH, b5	128.0000	mm.
Anchor Bolt Hole Dia. in Top Plate	dia,H	58.0000	mm.
External Corrosion Allowance	CA	0.0000	mm.
Dead Weight of Vessel	DW	19618.8	Kgf
Operating Weight of Vessel	ROW	29674.8	Kgf
Earthquake Moment on Basing	EQMOM	5948.5	Kg-m.
Wind Moment on Basing	WIMOM	6171.5	Kg-m.
User Moment at the Base [Earthquake Case]		18637.9	Kg-m. (UEQMOM)
User Moment at the Base [WindLoad Case]		18637.9	Kg-m. (UWIMOM)
Percent Bolt Preload	ppl	100.0	
Use AISC A5.2 Increase in Fc and Bolt Stress		No	
Use Allowable Weld Stress per AISC J2.5		No	
Factor for Increase of Allowables	Fact	1.0000	

Results for Brownell and Young Basing Analysis : Analyze Option

Note: This analysis is based on Neutral Axis shift method for Steel on Concrete (or a material with significantly different Young's modulus).

Calculation of Bolt Loads for Multiple Load Conditions:

Bolt Load, Earthquake Operating Case:

$$= ((4 \cdot M/DC) - W) / nb \quad \text{per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 24586/2611) - 29675) / 16$$

$$= 499.4 \text{ Kgf}$$

Bolt Load, Windload Empty Case:

$$= ((4 \cdot M/DC) - W) / nb \text{ per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 24809/2611) - 19619) / 16$$

$$= 1149.2 \text{ Kgf}$$

Bolt Load, Windload Vortex Shedding Case:

$$= ((4 \cdot M/DC) - W) / nb \text{ per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 0/2611) - 29675) / 16$$

$$= -1854.7 \text{ Kgf}$$

Governing Bolt Load Condition, Wind + Dead Weight Condition:

Area Available in 1 Bolt (corroded) Abss : 8.6100 cm²
 Area Available in all the Bolts Abss * RN : 137.7600 cm²

Trial#	k	knew	Cc	Ct	z	j	Ft	Fc	Fs
2	0.274	0.410	1.437	2.500	0.443	0.780	1030.2	20649.1	121.3
4	0.479	0.444	1.951	2.049	0.398	0.785	2164.5	21783.3	310.9
6	0.462	0.453	1.911	2.087	0.402	0.785	2066.3	21685.1	291.3
8	0.449	0.451	1.881	2.116	0.405	0.785	1993.1	21612.0	277.2
10	0.450	0.450	1.884	2.113	0.404	0.785	1999.2	21618.0	278.4
12	0.450	0.450	1.884	2.113	0.404	0.785	2000.7	21619.6	278.7
13	0.450	0.450	1.884	2.113	0.404	0.785	2000.0	21618.8	278.5

The Actual Stress in a Single Bolt [Sbolt]:

$$= 2 \cdot Ft / (T1 \cdot Dc \cdot Ct)$$

$$= 2 \cdot 2000 / (1.68 \cdot 2611 \cdot 2.11)$$

$$= 4.23 \text{ N./mm}^2, \text{ Should be } \leq 172 \text{ N./mm}^2$$

Thickness of the Band of Bolting Steel [T1]

$$= RN \cdot \text{Bolt Area} / (\pi \cdot Dc)$$

$$= 16 \cdot 8.61 / (\pi \cdot 2611)$$

$$= 1.68 \text{ mm.}$$

Check the Bearing Stress in the Concrete [fc(max)]

$$= fc[(2kd + t3) / (2kd)]$$

$$= 0.35[(2 \cdot 0.45 \cdot 2611 + 230) / (2 \cdot 0.45 \cdot 2611)]$$

$$= 0.39 \text{ N./mm}^2, \text{ Should be } \leq 22.5 \text{ N./mm}^2$$

Values for table 10.3, l = 128, b = 256, l/b = 0.49934795

Maximum Moment per unit width [Mmax]:

$$= \max(Mx, My) = \max(75.7, 206) = 206 \text{ Kgf}$$

Reqd Thickness of Basing, Brownell & Young Method [T]:

$$= \sqrt{(6 \cdot Mmax / fallow)} + Ca$$

$$= \sqrt{(6 \cdot 206 / 173)} + 0$$

$$= 8.38 \text{ mm.}$$

Nomenclature:

a = (Dc - Ds) / 2	Skirt Distance to Bolt Circle
P = Sa * Abss	Maximum Load on one Bolt
l	Radial Top Plate Width
lavg	Average Gusset Plate Width
g1 = Gamma 1	Constant Term f(b/l)
g2 = Gamma 2	Constant Term f(b/l)
g = Flat distance / 2	Nut 1/2 Dimension (from Tema)
Fb	Allowable Bending Stress

Values for table 10.6, l = 128, b = 90, b/l = 0.703125
 As b/l (0.7) is less than 1, inverting b/l = 1.42 0.

Moment Term, based on geometry [Mo]:

$$= P / (4 \cdot \pi) [1.3(\ln((2l \sin(\pi \cdot a/l)) / (\pi \cdot g))) + 1] - [(0.7 - g2)P / (4\pi)]$$

$$= 15101 / (4 \cdot \pi) [1.3(\ln((2 \cdot 128 \cdot \sin(\pi \cdot 64/128)) / (\pi \cdot 34.9))) + 1] - [(0.7 - 0.08) \cdot 15101 / (4 \cdot \pi)]$$

$$= 1779.8973 \text{ Kgf}$$

Required Thickness of Continuous Top Ring [Tc]:

$$= \sqrt{\frac{6 \cdot \text{Abs}(\text{Mo})}{F_b}} + C_a$$

$$= \sqrt{\frac{6 \cdot \text{Abs}(1780)}{175}} + 0$$

$$= 24.4871 \text{ mm.}$$

Required Gusset Plate Thickness [tg]:

$$= P / (\text{Stress Term} \cdot l_{\text{avg}}) + C_a$$

$$= 15101 / (124 \cdot 118) + 0$$

$$= 10.1 \text{ (not less than } 9.52 + 0 \text{) mm.}$$

Bolt spacing [m]:

$$= \pi \cdot \text{Bolt Circle Diameter} / \text{number of Bolts}$$

$$= \pi \cdot 2611 / 16 = 513 \text{ mm.}$$

Req. Skirt Thk. to withstand Local Bending, (Brownell and Young) [t]:

$$= 1.76 \left(\frac{P \cdot a}{m \cdot (h + t_{ba})} \cdot 1.5 \cdot \text{Sktope} \right)^{2/3} \cdot r^{1/3} + C_a$$

$$= 1.76 (15101 \cdot 64 / (513 \cdot 262 \cdot 171))^{2/3} \cdot 12421^{1/3} + C_a$$

$$= 10.5 + 3 = 13.5 \text{ mm.}$$

Summary of Basing Thickness Calculations

Required Basing Thickness (tension)	8.3779	mm.
Actual Basing Thickness as entered by user	32.0000	mm.
Required Thickness of Chair Cap	24.4871	mm.
Actual Top Ring Thickness as entered by user	38.0000	mm.
Required Gusset thickness, + CA	10.1120	mm.
Actual Gusset Thickness as entered by user	16.0000	mm.
Required Thickness of Skirt for Local Stress	13.4845	mm.
Given Thickness of Skirt	14.0000	mm.
Required Gusset Height to meet local stress	137.7968	mm.

Weld Size Calculations per Steel Plate Engineering Data - Vol. 2

Compute the Weld load at the Skirt/Base Junction [W]:

$$= \text{SkirtStress} \cdot (\text{SkirtThickness} - C_A)$$

$$= 8.16 \cdot (14 - 3)$$

$$= 9.15 \text{ Kg/mm.}$$

Results for Computed Minimum Basing Weld Size [BWeld]:

$$= W / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$$

$$= 9.15 / [(0.4 \cdot 230) \cdot 2 \cdot 0.707]$$

$$= 0.69 \text{ mm.}$$

Results for Computed Minimum Gusset and Top Plate to Skirt Weld Size

Vertical Plate Load [Wv]:

$$= \text{Bolt Load} / (\text{Cmwth} + 2 \cdot (\text{Hg} + \text{Tta}))$$

$$= 15101 / (184 + 2 \cdot (230 + 38))$$

$$= 21 \text{ Kg/mm.}$$

Horizontal Plate Load [Wh]:

$$= \text{Bolt Load} \cdot e / (\text{Cmwth} \cdot (\text{Hg} + \text{Tta}) + 0.6667 \cdot (\text{Hg} + \text{Tta})^2)$$

$$= 15101 \cdot 64 / (184 \cdot (268) + 0.6667 \cdot (268)^2)$$

$$= 9.94 \text{ Kg/mm.}$$

Resultant Weld Load [Wr]:

$$= \sqrt{W_v^2 + W_h^2}$$

$$= \sqrt{21^2 + 9.94^2}$$

$$= 23.2 \text{ Kg/mm.}$$

Results for Computed Min Gusset and Top Plate to Skirt Weld Size [GsWeld]:

$$= W_r / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$$

$$= 23.2 / [(0.4 \cdot 230) \cdot 2 \cdot 0.707]$$

$$= 1.75 \text{ mm.}$$

Results for Computed Minimum Gusset to Top Plate Weld Size



Weld Load [Wv]:
 = Bolt Load / (2 • TopWth)
 = 15101/(2 • 118)
 = 64 Kgf/mm.

Weld Load [Wh]:
 = Bolt Load • e / (2 • Hgt • TopWth)
 = 15101 • 64/(2 • 268 • 118)
 = 15.3 Kgf/mm.

Resultant Weld Load [Wr]:
 = $\sqrt{Wv^2 + Wh^2}$
 = $\sqrt{64^2 + 15.3^2}$
 = 65.8 Kgf/mm.

Results for Computed Min Gusset to Top Plate Weld Size [GtpWeld]:
 = Wr / [(0.4 • Yield) • 2 • 0.707]
 = 65.8/[(0.4 • 230) • 2 • 0.707]
 = 4.96 mm.

Note: The calculated weld sizes need not exceed the component thickness framing into the weld. At the same time, the weld must meet a minimum size specification which is 3/16 in. (4.76 mm) or 1/4 in. (6.35 mm), depending on the component thickness.

Summary of Required Weld Sizes:

Required Basing to Skirt Double Fillet Weld Size	6.3500 mm.
Required Gusset to Skirt Double Fillet Weld Size	6.3500 mm.
Required Top Plate to Skirt Weld Size	6.3500 mm.
Required Gusset to Top Plate Double Fillet Weld Size	4.9575 mm.

Review notes about nozzle loadings and supports in the Warnings report.

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Input, Nozzle Desc: D1-2"

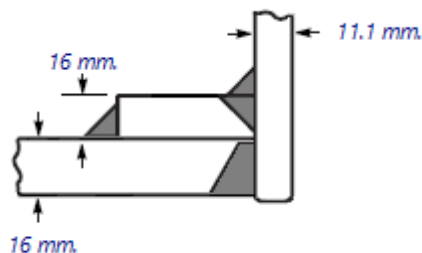
From: 30

Pressure for Reinforcement Calculations	P	10.915	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Elliptical Head	D	2450.00	mm.
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	16.0000	mm.
Head Internal Corrosion Allowance	cas	3.0000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	0.0000	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm2
Allowable Stress At Ambient	Sna	118.00	N./mm2
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		XXS	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	160.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	14.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm2
Pad Allowable Stress At Ambient	Spa	138.00	N./mm2
Diameter of Pad along vessel surface	Dp	170.0000	mm.
Thickness of Pad	te	16.0000	mm.
Weld leg size between Pad and Shell	Wp	12.0000	mm.
Groove weld depth between Pad and Nozzle	Wgpn	16.0000	mm.
Reinforcing Pad Width		54.8375	mm.
Flange Class		300	
Flange Grade		GR 1.1	
Piping Exit Angle [North Clockwise reference]		0.0	deg
Bend Radius Multiplier		1.50	
Horizontal Run Length		1400.0	mm.
Centerline Distance to Tangent		850.0	mm.

The Pressure Design option was Overall MAWP.

**Reinforcement CALCULATION, Description: D1-2"**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	2.375 in.
Actual Thickness Used in Calculation	0.436 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 10.9 \cdot 2456 \cdot 0.9 / (2 \cdot 138 \cdot 1 - 0.2 \cdot 10.9)$
 $= 8.7289$ mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot Ro / (Sn \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 10.9 \cdot 30.2 / (118 \cdot 1 + 0.4 \cdot 10.9)$
 $= 0.2780$ mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	88.3524 mm.
Parallel to Vessel Wall, opening length	d	44.1762 mm.
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		32.5000 mm.

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 118/138)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 118/138)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, Sp/Sv)$
 $= \min(1, 138/138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.86, 1)$
 $= 0.855$

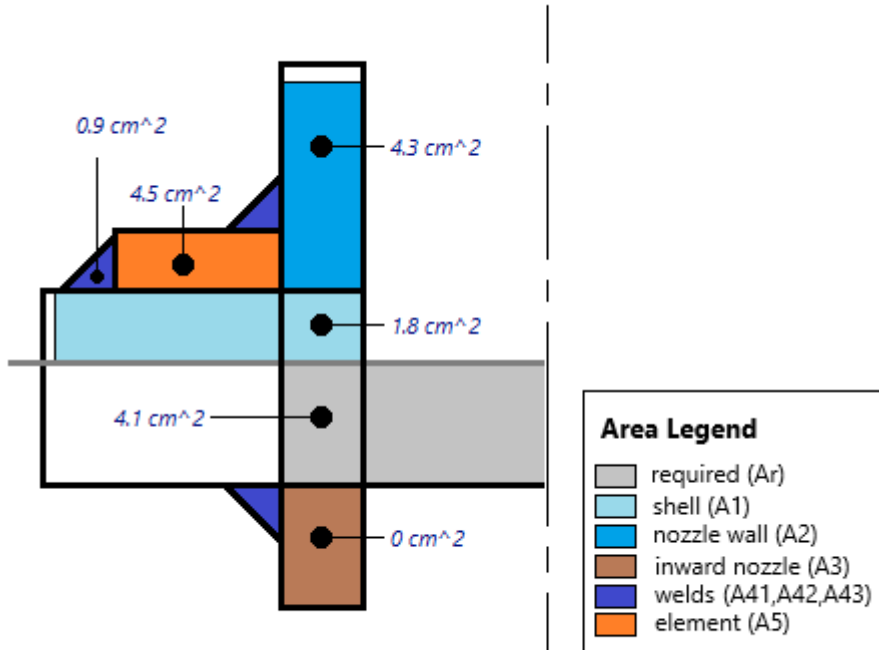


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	4.060	NA	NA
Area in Shell	A1	1.787	NA	NA
Area in Nozzle Wall	A2	4.333	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	0.855	NA	NA
Area in Element	A5	4.484	NA	NA
TOTAL AREA AVAILABLE	Atot	11.459	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.
 The area available with the given pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (44.2 \cdot 8.73 \cdot 1 + 2 \cdot 8.07 \cdot 8.73 \cdot 1 \cdot (1 - 0.86))$
 $= 4.060 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 44.2(1 \cdot 13 - 1 \cdot 8.73) - 2 \cdot 8.07$
 $(1 \cdot 13 - 1 \cdot 8.73) \cdot (1 - 0.86)$
 $= 1.787 \text{ cm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 32.5) \cdot (8.07 - 0.28) \cdot 0.86$
 $= 4.333 \text{ cm}^2$

Area Available in Welds [A41 + A42 + A43]:
 $= Wo^2 \cdot fr3 + (Wi \cdot can / 0.707)^2 \cdot fr2 + Wp^2 \cdot fr4$
 $= 10^2 \cdot 0.86 + (0)^2 \cdot 0.86 + 0^2 \cdot 1$

= 0.855 cm²

Area Available in Element [A5]:

= (min(Dp,DL)-(Nozzle OD))*(min(tp,Tlwp,te)) * fr4
 = (88.4 - 60.3) * 16 * 1
 = 4.484 cm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures ta = 3.2780 mm.
 Wall Thickness per UG16(b), tr16b = 4.5000 mm.
 Wall Thickness, shell/head, internal pressure trb1 = 12.6899 mm.
 Wall Thickness tb1 = max(trb1, tr16b) = 12.6899 mm.
 Wall Thickness tb2 = max(trb2, tr16b) = 4.5000 mm.
 Wall Thickness per table UG-45 tb3 = 6.4200 mm.

Determine Nozzle Thickness candidate [tb]:

= min[tb3, max(tb1,tb2)]
 = min[6.42, max(12.7, 4.5)]
 = 6.4200 mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= max(ta, tb)
 = max(3.28, 6.42)
 = 6.4200 mm.

Available Nozzle Neck Thickness = 0.875 * 11.074 = 9.690 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	36.3,	Allowable	:	118.0 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	258.7 N./mm ²	Passed
Occasional	:	1.3,	Allowable	:	156.9 N./mm ²	Passed
Shear	:	26.5,	Allowable	:	82.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 9.69, tr = 0.28, c = 3 mm., E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.042, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 9.69, tr = 0.28, c = 3 mm., E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.042, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: D

Govrn. thk, tg = 9.69, tr = 0.28, c = 3 mm., E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.042, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve D	-48 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Shell to Pad Weld Junction at Pad OD, Curve: D

Govrn. thk, tg = 16, tr = 8.73, c = 3 mm., E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.67, Temp. Reduction = 18 °C

Min Metal Temp. w/o impact per UCS-66, Curve D	-46 °C
--	--------



Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), min(Curve:B, Curve:D)

Govrn. thk, tg = 9.69, tr = 0.28, c = 3 mm., E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.042, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the Nozzle : -104 °C
 Gov. MDMT of the Reinforcement Pad : -48 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -48 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 10.92/51.00 = 0.214

Weld Size Calculations, Description: D1-2"

Intermediate Calc. for nozzle/shell Welds Tmin 8.0744 mm.
 Intermediate Calc. for pad/shell Welds TminPad 16.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	5.6521 = 0.7 * tmin.	7.0700 = 0.7 * Wo mm.
Pad Weld	8.0000 = 0.5*TminPad	8.4840 = 0.7 * Wp mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
 $= \max(0, (4.06 - 1.79 + 2 \cdot 8.07 \cdot 0.86 \cdot (1 \cdot 13 - 8.73)) \cdot 138)$
 $= 4029.23 \text{ Kg}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$
 $= (4.33 + 4.48 + 0.86 - 0 \cdot 0.86) \cdot 138$
 $= 13611.08 \text{ Kg}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$
 $= (4.33 + 0 + 0.86 + (1.8)) \cdot 138$
 $= 9826.71 \text{ Kg}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S$
 $= (4.33 + 0 + 0.86 + 4.48 + (1.8)) \cdot 138$
 $= 16137.05 \text{ Kg}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$
 $= (3.14/2.0) \cdot 60.3 \cdot 10 \cdot 0.49 \cdot 118$
 $= 5587. \text{ Kg}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$
 $= (3.14/2.0) \cdot 170 \cdot 12 \cdot 0.49 \cdot 138$
 $= 22095. \text{ Kg}$

Shear, Nozzle Wall [Snw]:



$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.14 \cdot 26.1) \cdot (11.1 - 3) \cdot 0.7 \cdot 118$$

$$= 5582. \text{ Kg}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$$

$$= (3.14/2) \cdot 60.3 \cdot 16 \cdot 0.74 \cdot 138$$

$$= 15788. \text{ Kg}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$

$$= (3.14/2.0) \cdot 60.3 \cdot (16 - 3) \cdot 0.74 \cdot 138$$

$$= 12828. \text{ Kg}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (22095 + 5582) = 27677 \text{ Kg}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (5587 + 15788 + 12828 + 0) = 34202 \text{ Kg}$$

Failure Path [Path 3-3]:

$$= (Spew + Tngw + Sinw)$$

$$= (22095 + 12828 + 0) = 34923 \text{ Kg}$$

Summary of Failure Path Calculations:

Path 1-1 = 27676	Kgf, must exceed W = 4029 Kg	or W1 = 13611 Kg
Path 2-2 = 34201	Kgf, must exceed W = 4029 Kg	or W2 = 9826 Kg
Path 3-3 = 34922	Kgf, must exceed W = 4029 Kg	or W3 = 16137 Kg

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 10.8 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 0.2050 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 176.2050 mm.

Input Echo, WRC107/537 Item 1, Description: D1-2" :

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	3.0000	mm.
Vessel Diameter	Dv	4410.000	mm.
Vessel Thickness	Tv	16.000	mm.
Design Temperature	T1	82.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh	138.00	N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Attachment Type	Type	Round	
WRC107 Attachment Classification	Holsol	Hollow	
Diameter Basis for Nozzle	Nbasis	OD	
Corrosion Allowance for Nozzle	Can	3.0000	mm.
Nozzle Diameter	Dn	60.325	mm.
Nozzle Thickness	Tn	11.074	mm.
Nozzle Material		SA-106 B	
Nozzle UNS Number		K03006	
Nozzle Cold S.I. Allowable	SNmc	118.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh	118.00	N./mm ²



Thickness of Reinforcing Pad	Tpad	16.000	mm.
Diameter of Reinforcing Pad	Dpad	170.000	mm.
Design Internal Pressure	Dp	10.915	bars
Include Pressure Thrust	No		

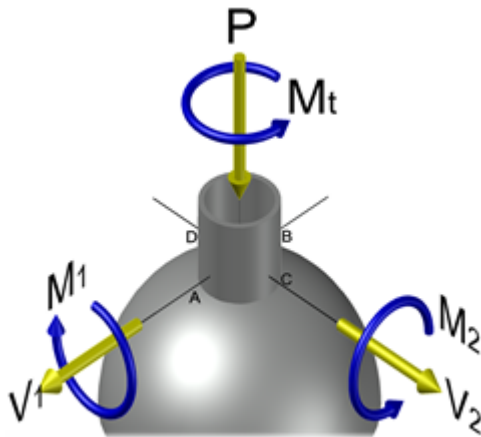
External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	382.4	Kgf
Longitudinal Shear (SUS)	(V1) V1	254.9	Kgf
Circumferential Shear (SUS)	(Vc) V2	254.9	Kgf
Circumferential Moment (SUS)	(Mc) M1	35.7	Kg-m.
Longitudinal Moment (SUS)	(Ml) M2	35.7	Kg-m.
Torsional Moment (SUS)	Mt	51.0	Kg-m.

Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$\begin{aligned}
 &= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))} \\
 &= 60.325 + 2 \cdot 1.65 \cdot \sqrt{(2214.5 (16.0 - 3.0))} \\
 &= 620.242 \text{ mm.}
 \end{aligned}$$

WRC 107 Stress Calculation for SUSTained loads:

Radial Load	P	382.4	Kgf
Circumferential Shear	(VC) V2	254.9	Kgf
Longitudinal Shear	(VL) V1	254.9	Kgf
Circumferential Moment	(MC) M1	35.7	Kg-m.
Longitudinal Moment	(ML) M2	35.7	Kg-m.
Torsional Moment	MT	51.0	Kg-m.

$$\text{Dimensionless Param: } U = 0.12 \text{ TAU} = 5.00 (3.24) \text{ RHO} = 3.59$$

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
$N(x) * T / P$	SP 3	0.12162	(A,B,C,D)
$M(x) / P$	SP 3	0.06171	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / MC$	SM 3	0.49052	(A,B,C,D)



M(x) * SQRT(Rm * T) / MC	SM 3	0.26576	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / ML	SM 3	0.49052	(A,B,C,D)
M(x) * SQRT(Rm * T) / ML	SM 3	0.26576	(A,B,C,D)
N(y) * T / P	SP 3	0.30417	(A,B,C,D)
M(y) / P	SP 3	0.26416	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / MC	SM 3	0.10032	(A,B,C,D)
M(y) * SQRT(Rm * T) / MC	SM 3	1.44989	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / ML	SM 3	0.10032	(A,B,C,D)
M(y) * SQRT(Rm * T) / ML	SM 3	1.44989	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (N./mm2)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb. P		-0.5	-0.5	-0.5	-0.5	-0.5	-0.5	-0.5	-0.5
Rad. Bend. P		-1.7	1.7	-1.7	1.7	-1.7	1.7	-1.7	1.7
Rad. Memb. MC		0.0	0.0	0.0	0.0	-0.8	-0.8	0.8	0.8
Rad. Bend. MC		0.0	0.0	0.0	0.0	-2.6	2.6	2.6	-2.6
Rad. Memb. ML		-0.8	-0.8	0.8	0.8	0.0	0.0	0.0	0.0
Rad. Bend. ML		-2.6	2.6	2.6	-2.6	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-5.6	2.9	1.2	-0.7	-5.6	2.9	1.2	-0.7
Tang. Memb. P		-1.4	-1.4	-1.4	-1.4	-1.4	-1.4	-1.4	-1.4
Tang. Bend. P		-7.1	7.1	-7.1	7.1	-7.1	7.1	-7.1	7.1
Tang. Memb. MC		0.0	0.0	0.0	0.0	-0.2	-0.2	0.2	0.2
Tang. Bend. MC		0.0	0.0	0.0	0.0	-14.3	14.3	14.3	-14.3
Tang. Memb. ML		-0.2	-0.2	0.2	0.2	0.0	0.0	0.0	0.0
Tang. Bend. ML		-14.3	14.3	14.3	-14.3	0.0	0.0	0.0	0.0
Tot. Tang. Str.		-22.8	19.8	6.0	-8.4	-22.8	19.8	6.0	-8.4
Shear VC		0.9	0.9	-0.9	-0.9	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-0.9	-0.9	0.9	0.9
Shear MT		3.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0
Tot. Shear		3.9	3.9	2.1	2.1	2.1	2.1	3.9	3.9
Str. Int.		23.7	20.7	6.8	8.9	23.1	20.1	9.2	11.0

Unitless Prm: U = 0.50 TAU = 0.00 (10.03) RHO = 0.00 (1.61)

Dimensionless Loads for Spherical Shells at Pad edge:

Curves read for 1979	Figure	Value	Location
N(x) * T / P	SR 2	0.13487	(A,B,C,D)
M(x) / P	SR 2	0.08337	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / MC	SR 3	0.14270	(A,B,C,D)
M(x) * SQRT(Rm * T) / MC	SR 3	0.24387	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / ML	SR 3	0.14270	(A,B,C,D)
M(x) * SQRT(Rm * T) / ML	SR 3	0.24387	(A,B,C,D)
N(y) * T / P	SR 2	0.04020	(A,B,C,D)
M(y) / P	SR 2	0.02520	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / MC	SR 3	0.04325	(A,B,C,D)
M(y) * SQRT(Rm * T) / MC	SR 3	0.07306	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / ML	SR 3	0.04325	(A,B,C,D)
M(y) * SQRT(Rm * T) / ML	SR 3	0.07306	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Edge of Reinforcing Pad (N./mm2)

Stress Intensity Values at



Type of Stress	Load	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb. P		-3.0	-3.0	-3.0	-3.0	-3.0	-3.0	-3.0	-3.0
Rad. Bend. P		-11.1	11.1	-11.1	11.1	-11.1	11.1	-11.1	11.1
Rad. Memb. MC		0.0	0.0	0.0	0.0	-1.7	-1.7	1.7	1.7
Rad. Bend. MC		0.0	0.0	0.0	0.0	-17.9	17.9	17.9	-17.9
Rad. Memb. ML		-1.7	-1.7	1.7	1.7	0.0	0.0	0.0	0.0
Rad. Bend. ML		-17.9	17.9	17.9	-17.9	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-33.7	24.2	5.5	-8.0	-33.7	24.2	5.5	-8.0
Tang. Memb. P		-0.9	-0.9	-0.9	-0.9	-0.9	-0.9	-0.9	-0.9
Tang. Bend. P		-3.4	3.4	-3.4	3.4	-3.4	3.4	-3.4	3.4
Tang. Memb. MC		0.0	0.0	0.0	0.0	-0.5	-0.5	0.5	0.5
Tang. Bend. MC		0.0	0.0	0.0	0.0	-5.3	5.3	5.3	-5.3
Tang. Memb. ML		-0.5	-0.5	0.5	0.5	0.0	0.0	0.0	0.0
Tang. Bend. ML		-5.3	5.3	5.3	-5.3	0.0	0.0	0.0	0.0
Tot. Tang. Str.		-10.1	7.3	1.6	-2.4	-10.1	7.3	1.6	-2.4
Shear VC		0.7	0.7	-0.7	-0.7	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-0.7	-0.7	0.7	0.7
Shear MT		0.8	0.8	0.8	0.8	0.8	0.8	0.8	0.8
Tot. Shear		1.6	1.6	0.1	0.1	0.1	0.1	1.6	1.6
Str. Int.		33.8	24.4	5.5	8.0	33.7	24.2	6.1	8.4

WRC 107/537 Stress Summations:**Vessel Stress Summation at Attachment Junction (N./mm2)**

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		41.6	41.6	41.6	41.6	41.6	41.6	41.6	41.6
Rad. Pl (SUS)		-1.3	-1.3	0.3	0.3	-1.3	-1.3	0.3	0.3
Rad. Q (SUS)		-4.3	4.3	1.0	-1.0	-4.3	4.3	1.0	-1.0
Long. Pm (SUS)		41.6	41.6	41.6	41.6	41.6	41.6	41.6	41.6
Long. Pl (SUS)		-1.5	-1.5	-1.2	-1.2	-1.5	-1.5	-1.2	-1.2
Long. Q (SUS)		-21.3	21.3	7.2	-7.2	-21.3	21.3	7.2	-7.2
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		0.9	0.9	-0.9	-0.9	-0.9	-0.9	0.9	0.9
Shear Q (SUS)		3.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0
Pm (SUS)		41.6	41.6	41.6	41.6	41.6	41.6	41.6	41.6
Pm+Pl (SUS)		41.0	41.0	42.3	42.3	41.0	41.0	42.3	42.3
Pm+Pl+Q (Total)		36.8	62.2	48.4	41.4	36.2	61.6	49.8	42.5

Vessel Stress Summation Comparison (N./mm2):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	41.56	138.00	Passed
Pm+Pl (SUS)	42.26	207.00	Passed
Pm+Pl+Q (TOTAL)	62.23	414.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



WRC 107/537 Stress Summations:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm2)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Pm (SUS)	92.7	92.7	92.7	92.7	92.7	92.7	92.7	92.7
Rad.	Pl (SUS)	-4.7	-4.7	-1.3	-1.3	-4.7	-4.7	-1.3	-1.3
Rad.	Q (SUS)	-29.0	29.0	6.8	-6.8	-29.0	29.0	6.8	-6.8
Long.	Pm (SUS)	92.7	92.7	92.7	92.7	92.7	92.7	92.7	92.7
Long.	Pl (SUS)	-1.4	-1.4	-0.4	-0.4	-1.4	-1.4	-0.4	-0.4
Long.	Q (SUS)	-8.7	8.7	2.0	-2.0	-8.7	8.7	2.0	-2.0
Shear	Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear	Pl (SUS)	0.7	0.7	-0.7	-0.7	-0.7	-0.7	0.7	0.7
Shear	Q (SUS)	0.8	0.8	0.8	0.8	0.8	0.8	0.8	0.8
Pm (SUS)		92.7	92.7	92.7	92.7	92.7	92.7	92.7	92.7
Pm+Pl (SUS)		91.4	91.4	92.7	92.7	91.4	91.4	92.7	92.7
Pm+Pl+Q (Total)		82.7	117.1	98.2	90.3	82.6	116.9	98.8	90.8

Vessel Stress Summation Comparison (N./mm2):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	92.70	138.00	Passed
Pm+Pl (SUS)	92.74	207.00	Passed
Pm+Pl+Q (TOTAL)	117.07	414.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: N1-16"

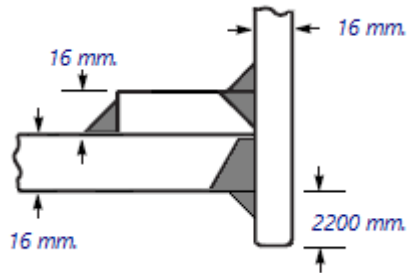
From: 40

Pressure for Reinforcement Calculations	P	10.915	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	2450.00	mm.
Shell Finished (Minimum) Thickness	t	16.0000	mm.
Shell Internal Corrosion Allowance	cas	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		1675.00	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm2
Allowable Stress At Ambient	Sna	118.00	N./mm2
Diameter Basis (for tr calc only)		Outside	
Layout Angle		90.00	deg
Diameter		16.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.0000	mm.
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	250.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	14.0000	mm.
Inside Projection	h	2200.0000	mm.
Weld leg size, Inside Element to Shell	Wi	14.0000	mm.
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm2
Pad Allowable Stress At Ambient	Spa	138.00	N./mm2
Diameter of Pad along vessel surface	Dp	850.0001	mm.
Thickness of Pad	te	16.0000	mm.
Weld leg size between Pad and Shell	Wp	12.0000	mm.
Groove weld depth between Pad and Nozzle	Wgpn	16.0000	mm.
Reinforcing Pad Width		221.8000	mm.
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was Overall MAWP.

**Reinforcement CALCULATION, Description: N1-16"**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	16.000 in.
Actual Thickness Used in Calculation	0.630 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 10.9 \cdot 1228 / (138 \cdot 1 - 0.6 \cdot 10.9)$
 $= 9.7600 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 10.9 \cdot 203 / (118 \cdot 1 + 0.4 \cdot 10.9)$
 $= 1.8729 \text{ mm.}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	760.8000 mm.
Parallel to Vessel Wall, opening length	d	380.4000 mm.
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		32.5000 mm.
Normal to Vessel Wall, Inward		25.0000 mm.

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118 / 138)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118 / 138)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.86, 1)$
 $= 0.855$

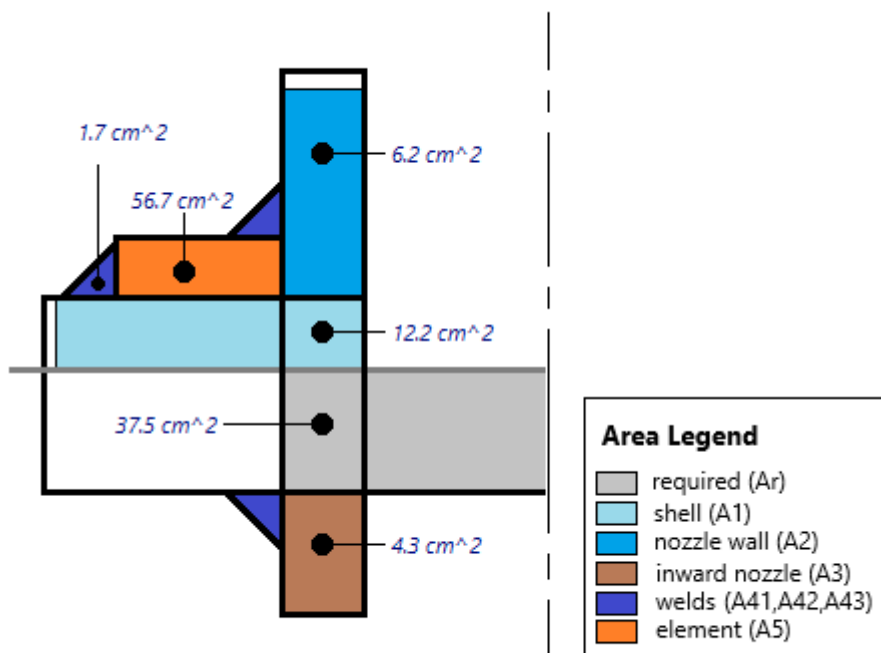


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	37.495	NA	NA
Area in Shell	A1	12.203	NA	NA
Area in Nozzle Wall	A2	6.184	NA	NA
Area in Inward Nozzle	A3	4.275	NA	NA
Area in Welds	A41+A42+A43	1.669	NA	NA
Area in Element	A5	56.703	NA	NA
TOTAL AREA AVAILABLE	Atot	81.034	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 488.6738 16.0000 mm.
 Based on given Pad Diameter: 850.0001 3.7144 mm.
 Based on Shell or Nozzle Thickness: 488.6738 16.0000 mm.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (380 \cdot 9.76 \cdot 1 + 2 \cdot 13 \cdot 9.76 \cdot 1 \cdot (1 - 0.86))$
 $= 37.495 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 380(1 \cdot 13 - 1 \cdot 9.76) - 2 \cdot 13$
 $(1 \cdot 13 - 1 \cdot 9.76) \cdot (1 - 0.86)$
 $= 12.203 \text{ cm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 32.5) \cdot (13 - 1.87) \cdot 0.86$

$$= 6.184 \text{ cm}^2$$

Area Available in Inward Nozzle [A3]:

$$= 2 \cdot t_i \cdot \min(h, t_i, 2.5 \cdot t_i) \cdot fr_2$$

$$= 2 \cdot 10 \cdot (25) \cdot 0.86$$

$$= 4.275 \text{ cm}^2$$

Area Available in Welds [A41 + A42 + A43]:

$$= W_o^2 \cdot fr_3 + (W_i - can/0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$$

$$= 10^2 \cdot 0.86 + (9.76)^2 \cdot 0.86 + 0^2 \cdot 1$$

$$= 1.669 \text{ cm}^2$$

Area Available in Element [A5]:

$$= (\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, t_{lp}, t_e)) \cdot fr_4$$

$$= (761 - 406) \cdot 16 \cdot 1$$

$$= 56.703 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	$t_a = 4.8729 \text{ mm.}$
Wall Thickness per UG16(b),	$tr_{16b} = 4.5000 \text{ mm.}$
Wall Thickness, shell/head, internal pressure	$trb_1 = 12.7600 \text{ mm.}$
Wall Thickness	$tb_1 = \max(trb_1, tr_{16b}) = 12.7600 \text{ mm.}$
Wall Thickness	$tb_2 = \max(trb_2, tr_{16b}) = 4.5000 \text{ mm.}$
Wall Thickness per table UG-45	$tb_3 = 11.3312 \text{ mm.}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb_3, \max(tb_1, tb_2)]$$

$$= \min[11.3, \max(12.8, 4.5)]$$

$$= 11.3312 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(t_a, t_b)$$

$$= \max(4.87, 11.3)$$

$$= 11.3312 \text{ mm.}$$

Available Nozzle Neck Thickness = 16.0000 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	30.3,	Allowable	:	118.0 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	264.7 N./mm ²	Passed
Occasional	:	7.7,	Allowable	:	156.9 N./mm ²	Passed
Shear	:	14.5,	Allowable	:	82.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

$$\text{Govrn. thk, } t_g = 16, t_r = 1.87, c = 3 \text{ mm., } E^* = 1$$

$$\text{Thickness Ratio} = t_r \cdot E^* / (t_g - c) = 0.14, \text{ Temp. Reduction} = 78 \text{ }^\circ\text{C}$$

Min Metal Temp. w/o impact per UCS-66, Curve A	6 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

$$\text{Govrn. thk, } t_g = 16, t_r = 1.87, c = 3 \text{ mm., } E^* = 1$$

$$\text{Thickness Ratio} = t_r \cdot E^* / (t_g - c) = 0.14, \text{ Temp. Reduction} = 78 \text{ }^\circ\text{C}$$

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

$$\text{Govrn. thk, } t_g = 16, t_r = 1.87, c = 3 \text{ mm., } E^* = 1$$

$$\text{Thickness Ratio} = t_r \cdot E^* / (t_g - c) = 0.14, \text{ Temp. Reduction} = 78 \text{ }^\circ\text{C}$$



Min Metal Temp. w/o impact per UCS-66, Curve B -14 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Shell to Pad Weld Junction at Pad OD, min(Curve:B, Curve:D)

Govrn. thk, tg = 16, tr = 9.76, c = 3 mm., E* = 1
 Thickness Ratio = $tr * E* / (tg - c) = 0.75$, Temp. Reduction = 14 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -14 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -28 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:B, Curve:D)

Govrn. thk, tg = 16, tr = 9.76, c = 3 mm., E* = 1
 Thickness Ratio = $tr * E* / (tg - c) = 0.75$, Temp. Reduction = 14 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -14 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -28 °C

Gov. MDMT of the Nozzle : -28 °C
 Gov. MDMT of the Reinforcement Pad : -28 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -28 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -44 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $10.92/19.60 = 0.557$

Weld Size Calculations, Description: N1-16"

Intermediate Calc. for nozzle/shell Welds	Tmin	13.0000	mm.
Intermediate Calc. for pad/shell Welds	TminPad	13.0000	mm.
Intermediate Calc. for Inward Weld	TminIns	13.0000	mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = $0.7 * W_o$ mm.
Pad Weld	6.5000 = $0.5 * T_{minPad}$	8.4840 = $0.7 * W_p$ mm.
Inward Weld	6.0000 = Min per Code	6.8980 = $0.7 * W_i$ -Can mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - tr)) \cdot S_v)$
 $= \max(0, (37.5 - 12.2 + 2 \cdot 13 \cdot 0.86 \cdot (1 \cdot 13 - 9.76)) \cdot 138)$
 $= 36604.43 \text{ Kg}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i-Can/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (6.18 + 56.7 + 1.67 - 95.2 \cdot 0.86) \cdot 138$
 $= 89697.10 \text{ Kg}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (6.18 + 4.28 + 1.67 + (2.89)) \cdot 138$
 $= 21134.19 \text{ Kg}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (6.18 + 4.28 + 1.67 + 56.7 + (2.89)) \cdot 138$
 $= 100925.63 \text{ Kg}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$
 $= (3.14/2.0) \cdot 406 \cdot 10 \cdot 0.49 \cdot 118$
 $= 37637. \text{ Kg}$

Shear, Inward Nozzle Weld [Sinw]:
 $= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$
 $= (3.14/2.0) \cdot 406 \cdot 9.76 \cdot 0.49 \cdot 118$
 $= 36724. \text{ Kg}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$
 $= (3.14/2.0) \cdot 850 \cdot 12 \cdot 0.49 \cdot 138$
 $= 110474. \text{ Kg}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$
 $= (3.14 \cdot 197) \cdot (16 - 3) \cdot 0.7 \cdot 118$
 $= 67661. \text{ Kg}$

Tension, Pad Groove Weld [Tpgw]:
 $= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$
 $= (3.14/2) \cdot 406 \cdot 16 \cdot 0.74 \cdot 138$
 $= 106357. \text{ Kg}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$
 $= (3.14/2.0) \cdot 406 \cdot (16 - 3) \cdot 0.74 \cdot 138$
 $= 86417. \text{ Kg}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (SPEW + SNW) = (110474 + 67661) = 178135 \text{ Kg}$

Failure Path [Path 2-2]:
 $= (Sonw + Tpgw + Tngw + Sinw)$
 $= (37637 + 106357 + 86417 + 36724) = 267135 \text{ Kg}$

Failure Path [Path 3-3]:
 $= (Spew + Tngw + Sinw)$
 $= (110474 + 86417 + 36724) = 233614 \text{ Kg}$

Summary of Failure Path Calculations:

Path 1-1 = 178135 Kg	must exceed W = 36604 Kg	or W1 = 89697 Kg
Path 2-2 = 267135 Kg	must exceed W = 36604 Kg	or W2 = 21134 Kg
Path 3-3 = 233614 Kg	must exceed W = 36604 Kg	or W3 = 100925 Kg

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 10.8 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 16.9707 mm.
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 2466.0000 mm.

Input Echo, WRC297 Item 1, Description: N1-16" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	2450.000	mm.
Shell Thickness	16.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 N./mm ²
Vessel Hot S.I. Allowable	Smh	138.00 N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	3.0000	mm.
Nozzle Diameter	406.400	mm.
Nozzle Thickness	16.0000	mm.
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	SNmc	118.00 N./mm ²
Nozzle Hot S.I. Allowable	SNmh	118.00 N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter of Reinforcing Pad	850.000	mm.
Thickness of Reinforcing Pad	16.0000	mm.

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSstained loads.*

Design Internal Pressure	Dp	10.92	bars
Radial Load	P	3059.35	Kgf
Circumferential Shear	Vc	2039.57	Kgf
Longitudinal Shear	Vl	2039.57	Kgf
Circumferential Moment	Mc	2283.43	Kg-m.
Longitudinal Moment	Ml	2283.43	Kg-m.
Torsional Moment	Mt	3262.05	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$
 $= 406.4 + 2 \cdot 1.65 \cdot \sqrt{1242.5(16.0 - 3.0)}$
 $= 825.805 \text{ mm.}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2485.000	mm.
Nozzle Outside Diameter	(d) =	406.400	mm.
Vessel Thickness used	(T) =	29.000	mm.
Nozzle Thickness used	(t) =	13.000	mm.
T / t	=	2.231	
d / t	=	31.262	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.514$$

Nr/P	=	0.086
Mr/P	=	0.097
M0/P	=	0.064
N0/P	=	0.173
MrD/Mc	=	0.189
NrDL/Mc	=	0.115
M0d/Mc	=	0.143
N0DL/Mc	=	0.165
MrD/Ml	=	0.123
NrDL/Ml	=	0.083
M0D/Ml	=	0.085
N0DL/Ml	=	0.297

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to	Top	Top	Bottom	Bottom



longitudinal plane)	Outside	Inside	Outside	Inside
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Outplane Membrane (P)	-6	-6	-6	-6
Outplane Bending (P)	-13	13	-13	13
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-19	-19	19	19
Outplane Bending (ML)	-33	33	33	-33
Normal Pressure Stress	45	46	45	46

Outplane Stress Summary	-26	68	78	40
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses parallel to longitudinal plane)	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside

Inplane Membrane (P)	-3	-3	-3	-3
Inplane Bending (P)	-20	20	-20	20
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-5	-5	5	5
Inplane Bending (ML)	-48	48	48	-48
Inplane Pressure Stress	22	22	22	22

Inplane Stress Summary	-54	83	52	-2
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside

Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	4	4	4	4

Shear Stress Summary	5	5	3	3
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	55	85	78	43
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left	Left	Right	Right
	Outside	Inside	Outside	Inside

Outplane Membrane (P)	-6	-6	-6	-6
Outplane Bending (P)	-13	13	-13	13
Outplane Membrane (Mc)	-10	-10	10	10
Outplane Bending (Mc)	-56	56	56	-56
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	22	22	22	22

Outplane Stress Summary	-63	75	70	-15
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left	Left	Right	Right
	Outside	Inside	Outside	Inside

Inplane Membrane (P)	-3	-3	-3	-3
Inplane Bending (P)	-20	20	-20	20
Inplane Membrane (Mc)	-7	-7	7	7
Inplane Bending (Mc)	-74	74	74	-74
Inplane Membrane (ML)	0	0	0	0



Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	45	46	45	46

Inplane Stress Summary	-59	131	103	-2
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	4	4	4	4

Shear Stress Summary	3	3	5	5
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Left	Left	Right	Right
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	65	131	104	17
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses in the	Top	Top	Bottom	Bottom
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	-6	-6	-6	-6
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-19	-19	19	19
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	18	19	18	19

Hoop Stress Summary	-7	-5	31	33
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to	Top	Top	Bottom	Bottom
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	-1	-1	-1	-1
Axial Bending (P)	-57	57	-57	57
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-14	-14	14	14
Axial Bending (ML)	-160	160	160	-160
Axial Pressure Stress	9	9	9	9

Axial Stress Summary	-224	210	125	-81
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	2	2	-2	-2
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	9	9	9	9

Shear Stress Summary	11	11	7	7
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	225	217	125	115
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Nozzle Stresses (N./mm²):



CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	-6	-6	-6	-6
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-10	-10	10	10
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	18	19	18	19
Hoop Stress Summary	1	2	23	24

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	-1	-1	-1	-1
Axial Bending (P)	-57	57	-57	57
Axial Membrane (Mc)	-14	-14	14	14
Axial Bending (Mc)	-257	257	257	-257
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	9	9	9	9
Axial Stress Summary	-322	307	222	-178

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-2	-2	2	2
Shear due to Torsion	9	9	9	9
Shear Stress Summary	7	7	11	11

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	324	307	223	204

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$$

$$= 406.4 + 2 \cdot 1.65 \cdot \sqrt{(1234.5(16.0 - 3.0))}$$

$$= 824.453 \text{ mm.}$$

WRC 297 Curve Access Parameters:

$$\text{Vessel Mean Diameter (D)} = 2469.000 \text{ mm.}$$

$$\text{Nozzle Outside Diameter (d)} = 850.000 \text{ mm.}$$

$$\text{Vessel Thickness used (T)} = 13.000 \text{ mm.}$$

$$\text{Nozzle Thickness used (t)} = 234.800 \text{ mm.}$$

$$T / t = 0.055$$

$$d / t = 10.000$$

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 4.744$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

$$\text{Nr/P} = 0.048$$

$$\text{Mr/P} = 0.072$$

$$\text{M0/P} = 0.020$$

$$\text{N0/P} = 0.024$$



MrD/Mc = 0.242
 NrDL/Mc = 0.104
 MØd/Mc = 0.072
 NØDL/Mc = 0.047
 MrD/Ml = 0.050
 NrDL/Ml = 0.053
 MØD/Ml = 0.015
 NØDL/Ml = 0.025

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE
 (Stresses Normal to longitudinal plane)

	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Outplane Membrane (P)	-4	-4	-4	-4
Outplane Bending (P)	-21	21	-21	21
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-3	-3	3	3
Outplane Bending (ML)	-14	14	14	-14
Normal Pressure Stress	102	103	102	103

Outplane Stress Summary	59	130	95	109
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE
 (Stresses parallel to longitudinal plane)

	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Inplane Membrane (P)	-8	-8	-8	-8
Inplane Bending (P)	-76	76	-76	76
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-8	-8	8	8
Inplane Bending (ML)	-46	46	46	-46
Inplane Pressure Stress	51	51	51	51

Inplane Stress Summary	-88	157	21	80
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE
 (Shear stress normal to longitudinal plane)

	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2

Shear Stress Summary	3	3	1	1
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE
 (Stress Intensities)

	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Two * Max Shear Stress	148	158	95	110
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE
 (Stresses Normal to circumferential plane)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Outplane Membrane (P)	-4	-4	-4	-4
Outplane Bending (P)	-21	21	-21	21
Outplane Membrane (Mc)	-7	-7	7	7
Outplane Bending (Mc)	-67	67	67	-67
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	51	51	51	51



Outplane Stress Summary -48 128 100 7

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE
(Stresses parallel to circumferential plane)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Inplane Membrane (P)	-8	-8	-8	-8
Inplane Bending (P)	-76	76	-76	76
Inplane Membrane (Mc)	-16	-16	16	16
Inplane Bending (Mc)	-226	226	226	-226
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	102	103	102	103

Inplane Stress Summary -224 381 260 -38

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE
(Shear stress normal to circumferential plane)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	2	2	2	2

Shear Stress Summary 1 1 3 3

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE
(Stress Intensities)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Two * Max Shear Stress 224 381 260 46

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	45	46	45	46	45	46	45	46
Circ. Pl (SUS)	-25	-25	13	13	-10	-10	4	4
Circ. Q (SUS)	-46	46	20	-20	-94	94	54	-54
Long. Pm (SUS)	22	22	22	22	22	22	22	22
Long. Pl (SUS)	-8	-8	2	2	-16	-16	4	4
Long. Q (SUS)	-68	68	28	-28	-69	69	43	-43
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	4	4	4	4	4	4	4	4
Pm (SUS)	45.0	46.0	45.0	46.0	45.0	46.0	45.0	46.0
Pm+Pl (SUS)	20.2	21.1	58.0	59.0	35.0	36.0	49.0	50.0
Pm+Pl+Q (Total)	54.9	83.5	78.3	43.4	64.6	130.2	103.7	18.7

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	46.00	138.00	Passed
Pm+Pl (SUS)	59.03	207.00	Passed
Pm+Pl+Q (TOTAL)	130.16	414.00	Passed

Because only sustained loads were specified, the $Pm+Pl+Q$ allowable was $3 * Smh$.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	102	103	102	103	102	103	102	103
Circ. Pl (SUS)	-7	-7	-1	-1	-24	-24	8	8
Circ. Q (SUS)	-35	35	-7	7	-302	302	150	-150
Long. Pm (SUS)	51	51	51	51	51	51	51	51
Long. Pl (SUS)	-16	-16	0	0	-11	-11	3	3
Long. Q (SUS)	-122	122	-30	30	-88	88	46	-46
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	102.0	103.0	102.0	103.0	102.0	103.0	102.0	103.0
Pm+Pl (SUS)	95.0	96.0	101.0	102.0	78.0	79.0	110.0	111.0
Pm+Pl+Q (Total)	147.1	157.3	94.0	109.0	224.0	381.0	260.1	47.4

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	103.00	138.00	Passed
Pm+Pl (SUS)	111.02	207.00	Passed
Pm+Pl+Q (TOTAL)	381.00	414.00	Passed

Because only sustained loads were specified, the $Pm+Pl+Q$ allowable was $3 * Smh$.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	18	19	18	19	18	19	18	19
Circ. Pl (SUS)	-25	-25	13	13	-16	-16	4	4
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	9	9	9	9	9	9	9	9
Long. Pl (SUS)	-15	-15	13	13	-15	-15	13	13
Long. Q (SUS)	-217	217	103	-103	-314	314	200	-200
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	9	9	9	9	9	9	9	9
Pm (SUS)	18.0	19.0	18.0	19.0	18.0	19.0	18.0	19.0
Pm+Pl (SUS)	8.6	8.0	31.4	32.4	8.9	9.8	24.0	24.6
Pm+Pl+Q (Total)	223.6	218.1	125.5	113.9	322.3	308.2	222.6	202.2

Nozzle Stress Summation Comparison (N./mm²):

Type of	Max. S.I.	S.I. Allowable	Result
---------	-----------	----------------	--------



Stress Int.			
Pm (SUS)	19.00	118.00	Passed
Pm+Pl (SUS)	32.39	177.00	Passed
Pm+Pl+Q (TOTAL)	322.30	353.99	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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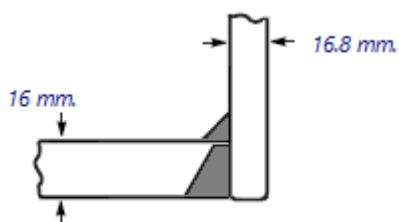
Input, Nozzle Desc: K1-2"
From: 40

Pressure for Reinforcement Calculations	P	10.915	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2450.00	mm.
Shell Finished (Minimum) Thickness	t	16.0000	mm.
Shell Internal Corrosion Allowance	cas	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		1125.00	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		135.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm.
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	160.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	14.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Overall MAWP.


Reinforcement CALCULATION, Description: K1-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.660	in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 10.9 \cdot 1228 / (138 \cdot 1 - 0.6 \cdot 10.9)$
 $= 9.7600$ mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 10.9 \cdot 28.4 / (138 \cdot 1 - 0.6 \cdot 10.9)$
 $= 0.2257$ mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 113.6000 mm.
 Parallel to Vessel Wall, opening length d 56.8000 mm.
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 32.5000 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$

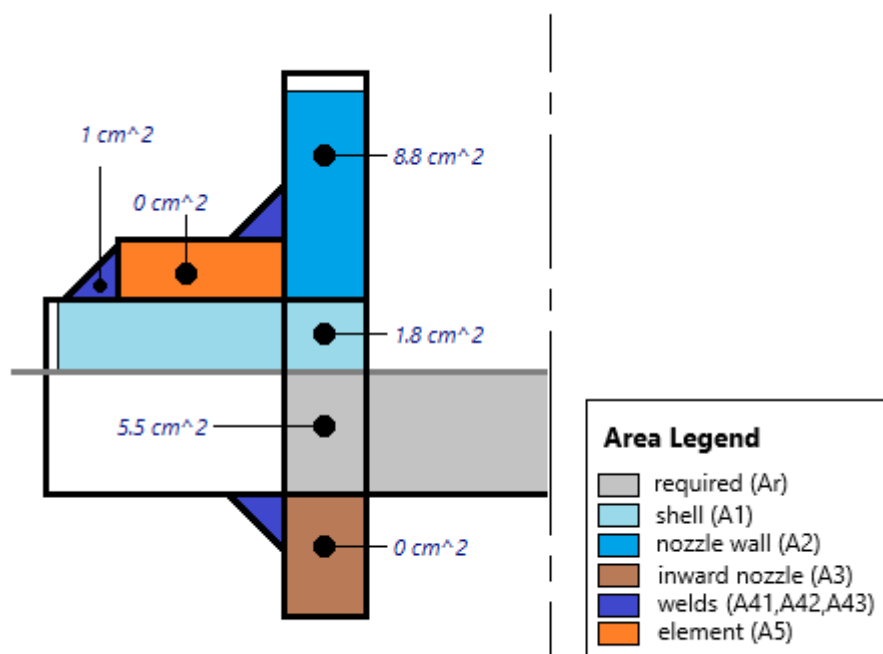


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	5.544	NA	NA
Area in Shell	A1	1.840	NA	NA
Area in Nozzle Wall	A2	8.800	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA



Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
TOTAL AREA AVAILABLE	Atot	11.640	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (56.8 \cdot 9.76 \cdot 1 + 2 \cdot 13.8 \cdot 9.76 \cdot 1 \cdot (1 - 1))$
 $= 5.544 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 56.8(1 \cdot 13 - 1 \cdot 9.76) - 2 \cdot 13.8$
 $(1 \cdot 13 - 1 \cdot 9.76) \cdot (1 - 1)$
 $= 1.840 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2$
 $= (2 \cdot 32.5)(13.8 - 0.23)1$
 $= 8.800 \text{ cm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10^2 \cdot 1 + (0)^2 \cdot 1$
 $= 1.000 \text{ cm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 3.2257 \text{ mm.}$
 Wall Thickness per UG16(b), $tr16b = 4.5000 \text{ mm.}$
 Wall Thickness, shell/head, internal pressure $trb1 = 12.7600 \text{ mm.}$
 Wall Thickness $tb1 = \max(trb1, tr16b) = 12.7600 \text{ mm.}$
 Wall Thickness $tb2 = \max(trb2, tr16b) = 4.5000 \text{ mm.}$
 Wall Thickness per table UG-45 $tb3 = 7.8000 \text{ mm.}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[7.8, \max(12.8, 4.5)]$
 $= 7.8000 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.23, 7.8)$
 $= 7.8000 \text{ mm.}$

Available Nozzle Neck Thickness = 16.7640 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:B, Curve:D)

Govrn. thk, $tg = 16$, $tr = 9.76$, $c = 3 \text{ mm.}$, $E* = 1$
 Thickness Ratio $= tr \cdot E* / (tg - c) = 0.75$, Temp. Reduction = 14 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -14 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -28 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -28 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $10.92/51.00 = 0.214$

Weld Size Calculations, Description: K1-2"

Intermediate Calc. for nozzle/shell Welds Tmin 13.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (5.54 - 1.84 + 2 \cdot 13.8 \cdot 1 \cdot (1 \cdot 13 - 9.76)) \cdot 138)$
 $= 6466.43 \text{ Kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (8.8 + 0 + 1 - 0 \cdot 1) \cdot 138$
 $= 13790.22 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (8.8 + 0 + 1 + (3.58)) \cdot 138$
 $= 18826.02 \text{ Kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (8.8 + 0 + 1 + 0 + (3.58)) \cdot 138$
 $= 18826.02 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.14/2.0) \cdot 84.3 \cdot 10 \cdot 0.49 \cdot 138$
 $= 9134. \text{ Kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.14 \cdot 35.3) \cdot (16.8 - 3) \cdot 0.7 \cdot 138$
 $= 15028. \text{ Kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng}$
 $= (3.14/2.0) \cdot 84.3 \cdot (16 - 3) \cdot 0.74 \cdot 138$
 $= 17932. \text{ Kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (\text{SONW} + \text{SNW}) = (9134 + 15028) = 24161 \text{ Kgf}$

Failure Path [Path 2-2]:
 $= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$
 $= (9134 + 0 + 17932 + 0) = 27065 \text{ Kgf}$

Failure Path [Path 3-3]:
 $= (\text{Sonw} + \text{Tngw} + \text{Sinw})$
 $= (9134 + 17932 + 0) = 27065 \text{ Kgf}$

Summary of Failure Path Calculations:



Path 1-1 = 24161 Kgf, must exceed W = 6466 Kgf or W1 = 13790 Kgf
Path 2-2 = 27065 Kgf, must exceed W = 6466 Kgf or W2 = 18826 Kgf
Path 3-3 = 27065 Kgf, must exceed W = 6466 Kgf or W3 = 18826 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 10.8 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 0.7258 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 176.7258 mm.

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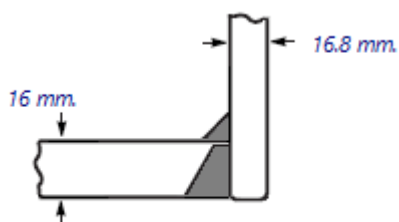
Input, Nozzle Desc: K2-2"
From: 40

Pressure for Reinforcement Calculations	P	10.915	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2450.00	mm.
Shell Finished (Minimum) Thickness	t	16.0000	mm.
Shell Internal Corrosion Allowance	cas	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		600.00	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm.
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	160.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	14.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Overall MAWP.


Reinforcement CALCULATION, Description: K2-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.660	in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 10.9 \cdot 1228 / (138 \cdot 1 - 0.6 \cdot 10.9)$
 $= 9.7600$ mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 10.9 \cdot 28.4 / (138 \cdot 1 - 0.6 \cdot 10.9)$
 $= 0.2257$ mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 113.6000 mm.
 Parallel to Vessel Wall, opening length d 56.8000 mm.
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 32.5000 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$

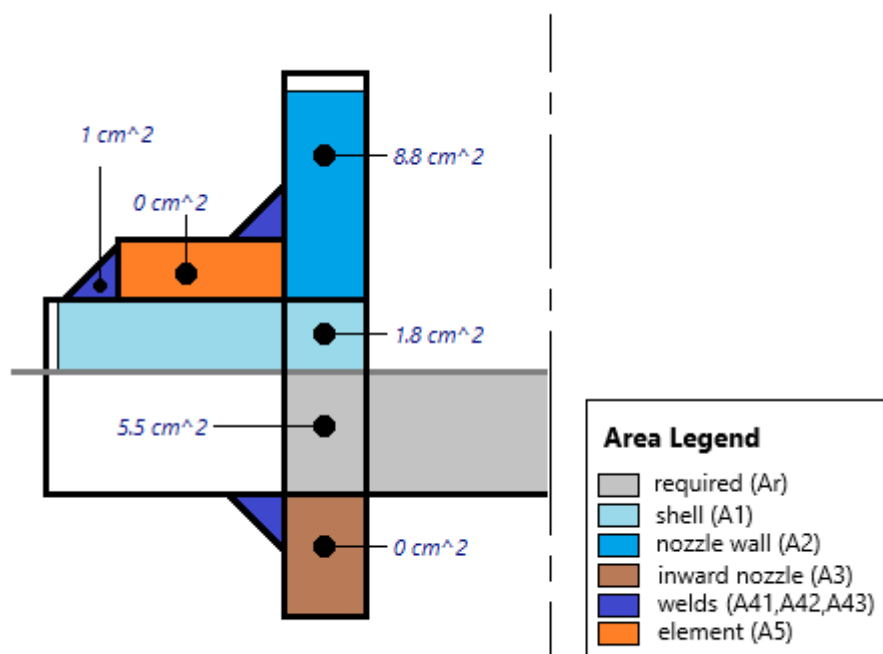


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	5.544	NA	NA
Area in Shell	A1	1.840	NA	NA
Area in Nozzle Wall	A2	8.800	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA



Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
TOTAL AREA AVAILABLE	Atot	11.640	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (56.8 \cdot 9.76 \cdot 1 + 2 \cdot 13.8 \cdot 9.76 \cdot 1 \cdot (1 - 1))$
 $= 5.544 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 56.8(1 \cdot 13 - 1 \cdot 9.76) - 2 \cdot 13.8$
 $(1 \cdot 13 - 1 \cdot 9.76) \cdot (1 - 1)$
 $= 1.840 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tnp)(tn - trn)fr2$
 $= (2 \cdot 32.5)(13.8 - 0.23)1$
 $= 8.800 \text{ cm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10^2 \cdot 1 + (0)^2 \cdot 1$
 $= 1.000 \text{ cm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 3.2257 \text{ mm.}$
 Wall Thickness per UG16(b), $tr16b = 4.5000 \text{ mm.}$
 Wall Thickness, shell/head, internal pressure $trb1 = 12.7600 \text{ mm.}$
 Wall Thickness $tb1 = \max(trb1, tr16b) = 12.7600 \text{ mm.}$
 Wall Thickness $tb2 = \max(trb2, tr16b) = 4.5000 \text{ mm.}$
 Wall Thickness per table UG-45 $tb3 = 7.8000 \text{ mm.}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[7.8, \max(12.8, 4.5)]$
 $= 7.8000 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.23, 7.8)$
 $= 7.8000 \text{ mm.}$

Available Nozzle Neck Thickness = 16.7640 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:B, Curve:D)

Govrn. thk, $tg = 16$, $tr = 9.76$, $c = 3 \text{ mm.}$, $E* = 1$
 Thickness Ratio $= tr \cdot E* / (tg - c) = 0.75$, Temp. Reduction = 14 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -14 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -28 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -28 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $10.92/51.00 = 0.214$

Weld Size Calculations, Description: K2-2"

Intermediate Calc. for nozzle/shell Welds Tmin 13.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_r1 \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (5.54 - 1.84 + 2 \cdot 13.8 \cdot 1 \cdot (1 \cdot 13 - 9.76)) \cdot 138)$
 $= 6466.43 \text{ Kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i - \text{Can}/707)^2 \cdot f_r2) \cdot S_v$
 $= (8.8 + 0 + 1 - 0 \cdot 1) \cdot 138$
 $= 13790.22 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_r1)) \cdot S_v$
 $= (8.8 + 0 + 1 + (3.58)) \cdot 138$
 $= 18826.02 \text{ Kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_r1)) \cdot S$
 $= (8.8 + 0 + 1 + 0 + (3.58)) \cdot 138$
 $= 18826.02 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.14/2.0) \cdot 84.3 \cdot 10 \cdot 0.49 \cdot 138$
 $= 9134. \text{ Kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.14 \cdot 35.3) \cdot (16.8 - 3) \cdot 0.7 \cdot 138$
 $= 15028. \text{ Kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng}$
 $= (3.14/2.0) \cdot 84.3 \cdot (16 - 3) \cdot 0.74 \cdot 138$
 $= 17932. \text{ Kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (\text{SONW} + \text{SNW}) = (9134 + 15028) = 24161 \text{ Kgf}$

Failure Path [Path 2-2]:
 $= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$
 $= (9134 + 0 + 17932 + 0) = 27065 \text{ Kgf}$

Failure Path [Path 3-3]:
 $= (\text{Sonw} + \text{Tngw} + \text{Sinw})$
 $= (9134 + 17932 + 0) = 27065 \text{ Kgf}$

Summary of Failure Path Calculations:



Path 1-1 = 24161 Kgf, must exceed W = 6466 Kgf or W1 = 13790 Kgf
Path 2-2 = 27065 Kgf, must exceed W = 6466 Kgf or W2 = 18826 Kgf
Path 3-3 = 27065 Kgf, must exceed W = 6466 Kgf or W3 = 18826 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 10.8 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 0.7258 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 176.7258 mm.

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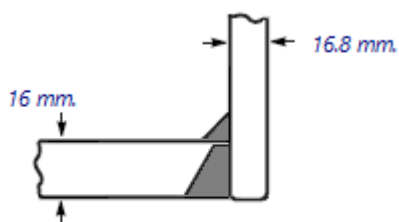
Input, Nozzle Desc: K3-2"
From: 40

Pressure for Reinforcement Calculations	P	10.915	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2450.00	mm.
Shell Finished (Minimum) Thickness	t	16.0000	mm.
Shell Internal Corrosion Allowance	cas	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		1325.00	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm.
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	160.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	14.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Overall MAWP.


Reinforcement CALCULATION, Description: K3-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.660	in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 10.9 \cdot 1228 / (138 \cdot 1 - 0.6 \cdot 10.9)$
 $= 9.7600$ mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 10.9 \cdot 28.4 / (138 \cdot 1 - 0.6 \cdot 10.9)$
 $= 0.2257$ mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 113.6000 mm.
 Parallel to Vessel Wall, opening length d 56.8000 mm.
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 32.5000 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$

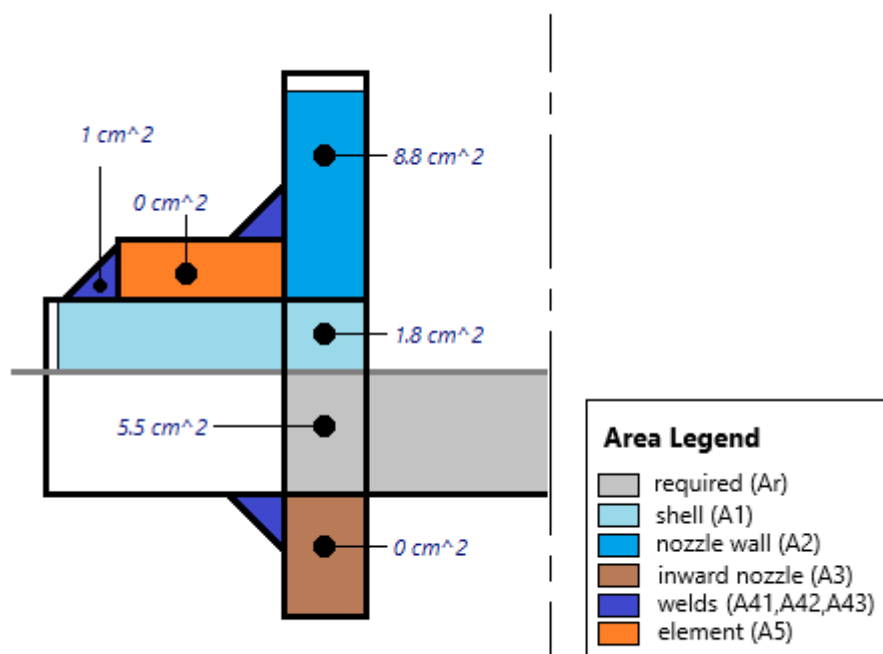


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	5.544	NA	NA
Area in Shell	A1	1.840	NA	NA
Area in Nozzle Wall	A2	8.800	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA



Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
TOTAL AREA AVAILABLE	Atot	11.640	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (56.8 \cdot 9.76 \cdot 1 + 2 \cdot 13.8 \cdot 9.76 \cdot 1 \cdot (1 - 1))$
 $= 5.544 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 56.8(1 \cdot 13 - 1 \cdot 9.76) - 2 \cdot 13.8$
 $(1 \cdot 13 - 1 \cdot 9.76) \cdot (1 - 1)$
 $= 1.840 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2$
 $= (2 \cdot 32.5)(13.8 - 0.23)1$
 $= 8.800 \text{ cm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10^2 \cdot 1 + (0)^2 \cdot 1$
 $= 1.000 \text{ cm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 3.2257 \text{ mm.}$
 Wall Thickness per UG16(b), $tr16b = 4.5000 \text{ mm.}$
 Wall Thickness, shell/head, internal pressure $trb1 = 12.7600 \text{ mm.}$
 Wall Thickness $tb1 = \max(trb1, tr16b) = 12.7600 \text{ mm.}$
 Wall Thickness $tb2 = \max(trb2, tr16b) = 4.5000 \text{ mm.}$
 Wall Thickness per table UG-45 $tb3 = 7.8000 \text{ mm.}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[7.8, \max(12.8, 4.5)]$
 $= 7.8000 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.23, 7.8)$
 $= 7.8000 \text{ mm.}$

Available Nozzle Neck Thickness = 16.7640 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:B, Curve:D)

Govrn. thk, $t_g = 16$, $t_r = 9.76$, $c = 3 \text{ mm.}$, $E* = 1$
 Thickness Ratio $= tr \cdot E* / (t_g - c) = 0.75$, Temp. Reduction = 14 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -14 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -28 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -28 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $10.92/51.00 = 0.214$

Weld Size Calculations, Description: K3-2"

Intermediate Calc. for nozzle/shell Welds Tmin 13.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (5.54 - 1.84 + 2 \cdot 13.8 \cdot 1 \cdot (1 \cdot 13 - 9.76)) \cdot 138)$
 $= 6466.43 \text{ Kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (8.8 + 0 + 1 - 0 \cdot 1) \cdot 138$
 $= 13790.22 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (8.8 + 0 + 1 + (3.58)) \cdot 138$
 $= 18826.02 \text{ Kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (8.8 + 0 + 1 + 0 + (3.58)) \cdot 138$
 $= 18826.02 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.14/2.0) \cdot 84.3 \cdot 10 \cdot 0.49 \cdot 138$
 $= 9134. \text{ Kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.14 \cdot 35.3) \cdot (16.8 - 3) \cdot 0.7 \cdot 138$
 $= 15028. \text{ Kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng}$
 $= (3.14/2.0) \cdot 84.3 \cdot (16 - 3) \cdot 0.74 \cdot 138$
 $= 17932. \text{ Kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (\text{SONW} + \text{SNW}) = (9134 + 15028) = 24161 \text{ Kgf}$

Failure Path [Path 2-2]:
 $= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$
 $= (9134 + 0 + 17932 + 0) = 27065 \text{ Kgf}$

Failure Path [Path 3-3]:
 $= (\text{Sonw} + \text{Tngw} + \text{Sinw})$
 $= (9134 + 17932 + 0) = 27065 \text{ Kgf}$

Summary of Failure Path Calculations:



Path 1-1 = 24161 Kgf, must exceed W = 6466 Kgf or W1 = 13790 Kgf
Path 2-2 = 27065 Kgf, must exceed W = 6466 Kgf or W2 = 18826 Kgf
Path 3-3 = 27065 Kgf, must exceed W = 6466 Kgf or W3 = 18826 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 10.8 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 0.7258 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 176.7258 mm.

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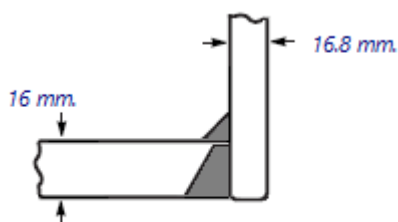
**Input, Nozzle Desc: K4-2"****From: 40**

Pressure for Reinforcement Calculations	P	10.915	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2450.00	mm.
Shell Finished (Minimum) Thickness	t	16.0000	mm.
Shell Internal Corrosion Allowance	cas	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		325.00	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm.
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	160.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	14.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Overall MAWP.

**Reinforcement CALCULATION, Description: K4-2"**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.660	in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 10.9 \cdot 1228 / (138 \cdot 1 - 0.6 \cdot 10.9)$
 $= 9.7600$ mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 10.9 \cdot 28.4 / (138 \cdot 1 - 0.6 \cdot 10.9)$
 $= 0.2257$ mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 113.6000 mm.
 Parallel to Vessel Wall, opening length d 56.8000 mm.
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 32.5000 mm.

Weld Strength Reduction Factor [fr1]:

$= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:

$= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:

$= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$

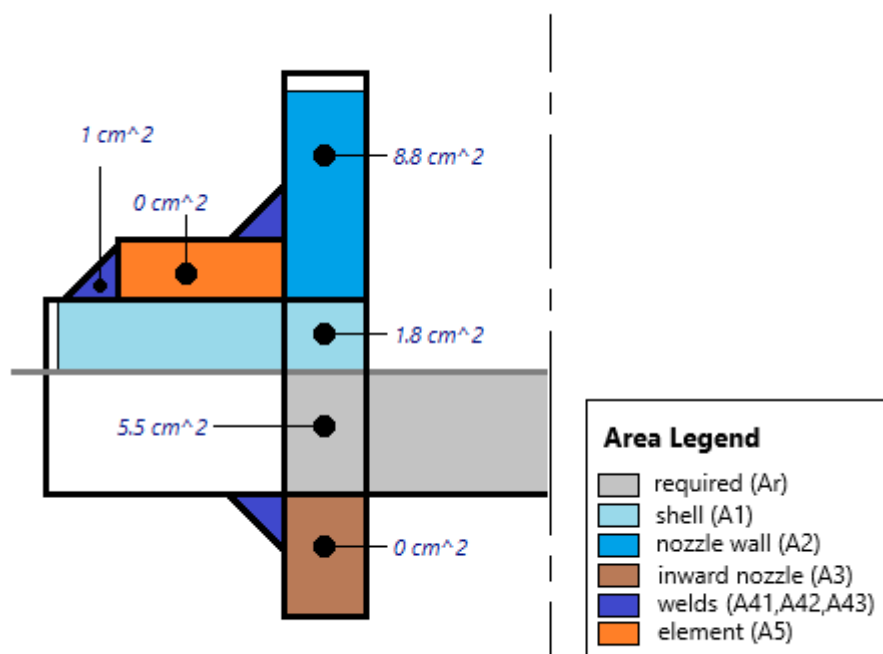


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	5.544	NA	NA
Area in Shell	A1	1.840	NA	NA
Area in Nozzle Wall	A2	8.800	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA



Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
TOTAL AREA AVAILABLE	Atot	11.640	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (56.8 \cdot 9.76 \cdot 1 + 2 \cdot 13.8 \cdot 9.76 \cdot 1 \cdot (1 - 1))$
 $= 5.544 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 56.8(1 \cdot 13 - 1 \cdot 9.76) - 2 \cdot 13.8$
 $(1 \cdot 13 - 1 \cdot 9.76) \cdot (1 - 1)$
 $= 1.840 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2$
 $= (2 \cdot 32.5)(13.8 - 0.23)1$
 $= 8.800 \text{ cm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10^2 \cdot 1 + (0)^2 \cdot 1$
 $= 1.000 \text{ cm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 3.2257 \text{ mm.}$
 Wall Thickness per UG16(b), $tr16b = 4.5000 \text{ mm.}$
 Wall Thickness, shell/head, internal pressure $trb1 = 12.7600 \text{ mm.}$
 Wall Thickness $tb1 = \max(trb1, tr16b) = 12.7600 \text{ mm.}$
 Wall Thickness $tb2 = \max(trb2, tr16b) = 4.5000 \text{ mm.}$
 Wall Thickness per table UG-45 $tb3 = 7.8000 \text{ mm.}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[7.8, \max(12.8, 4.5)]$
 $= 7.8000 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.23, 7.8)$
 $= 7.8000 \text{ mm.}$

Available Nozzle Neck Thickness = 16.7640 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:B, Curve:D)

Govrn. thk, $tg = 16$, $tr = 9.76$, $c = 3 \text{ mm.}$, $E* = 1$
 Thickness Ratio $= tr \cdot E* / (tg - c) = 0.75$, Temp. Reduction = 14 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -14 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -28 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -28 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $10.92/51.00 = 0.214$

Weld Size Calculations, Description: K4-2"

Intermediate Calc. for nozzle/shell Welds Tmin 13.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 * W_o \text{ mm.}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (5.54 - 1.84 + 2 \cdot 13.8 \cdot 1 \cdot (1 \cdot 13 - 9.76)) \cdot 138)$
 $= 6466.43 \text{ Kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (8.8 + 0 + 1 - 0 \cdot 1) \cdot 138$
 $= 13790.22 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (8.8 + 0 + 1 + (3.58)) \cdot 138$
 $= 18826.02 \text{ Kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (8.8 + 0 + 1 + 0 + (3.58)) \cdot 138$
 $= 18826.02 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.14/2.0) \cdot 84.3 \cdot 10 \cdot 0.49 \cdot 138$
 $= 9134. \text{ Kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.14 \cdot 35.3) \cdot (16.8 - 3) \cdot 0.7 \cdot 138$
 $= 15028. \text{ Kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng}$
 $= (3.14/2.0) \cdot 84.3 \cdot (16 - 3) \cdot 0.74 \cdot 138$
 $= 17932. \text{ Kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (\text{SONW} + \text{SNW}) = (9134 + 15028) = 24161 \text{ Kgf}$

Failure Path [Path 2-2]:
 $= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$
 $= (9134 + 0 + 17932 + 0) = 27065 \text{ Kgf}$

Failure Path [Path 3-3]:
 $= (\text{Sonw} + \text{Tngw} + \text{Sinw})$
 $= (9134 + 17932 + 0) = 27065 \text{ Kgf}$

Summary of Failure Path Calculations:



Path 1-1 = 24161 Kgf, must exceed W = 6466 Kgf or W1 = 13790 Kgf
Path 2-2 = 27065 Kgf, must exceed W = 6466 Kgf or W2 = 18826 Kgf
Path 3-3 = 27065 Kgf, must exceed W = 6466 Kgf or W3 = 18826 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 10.8 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 0.7258 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 176.7258 mm.

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Input, Nozzle Desc: M1-24"

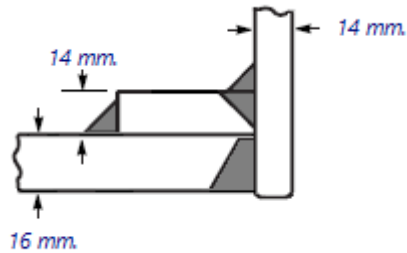
From: 40

Pressure for Reinforcement Calculations	P	10.915	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	2450.00	mm.
Shell Finished (Minimum) Thickness	t	16.0000	mm.
Shell Internal Corrosion Allowance	cas	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		800.00	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm2
Allowable Stress At Ambient	Sna	118.00	N./mm2
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	14.0000	mm.
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	350.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	14.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm2
Pad Allowable Stress At Ambient	Spa	138.00	N./mm2
Diameter of Pad along vessel surface	Dp	1050.0000	mm.
Thickness of Pad	te	14.0000	mm.
Weld leg size between Pad and Shell	Wp	12.0000	mm.
Groove weld depth between Pad and Nozzle	Wgpn	14.0000	mm.
Reinforcing Pad Width		220.2000	mm.
This is a Manway, Coupling or Access Opening.			
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was Overall MAWP.

**Reinforcement CALCULATION, Description: M1-24"**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	24.000 in.
Actual Thickness Used in Calculation	0.551 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 10.9 \cdot 1228 / (138 \cdot 1 - 0.6 \cdot 10.9)$
 $= 9.7600 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 10.9 \cdot 305 / (118 \cdot 1 + 0.4 \cdot 10.9)$
 $= 2.8093 \text{ mm.}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1175.2001 mm.
Parallel to Vessel Wall, opening length	d	587.6000 mm.
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		32.5000 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118 / 138)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118 / 138)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.86, 1)$
 $= 0.855$

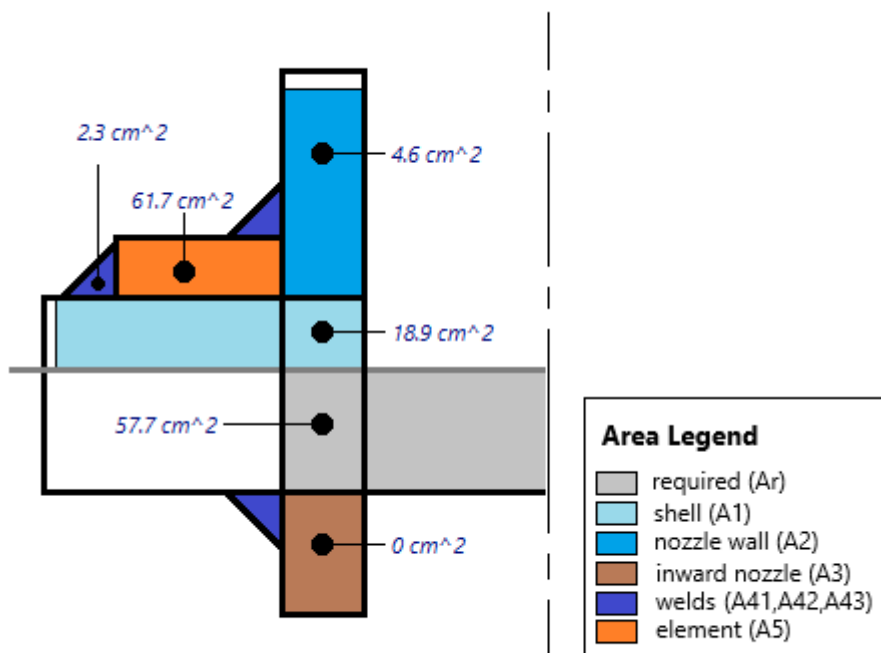


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	57.661	NA	NA
Area in Shell	A1	18.935	NA	NA
Area in Nozzle Wall	A2	4.552	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	2.295	NA	NA
Area in Element	A5	61.655	NA	NA
TOTAL AREA AVAILABLE	Atot	87.437	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 837.3127 14.0000 mm.
 Based on given Pad Diameter: 1050.0000 7.2388 mm.
 Based on Shell or Nozzle Thickness: 837.3127 14.0000 mm.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (588 \cdot 9.76 \cdot 1 + 2 \cdot 11 \cdot 9.76 \cdot 1 \cdot (1 - 0.86))$
 $= 57.661 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 588(1 \cdot 13 - 1 \cdot 9.76) - 2 \cdot 11$
 $(1 \cdot 13 - 1 \cdot 9.76) \cdot (1 - 0.86)$
 $= 18.935 \text{ cm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 32.5) \cdot (11 - 2.81) \cdot 0.86$

$$= 4.552 \text{ cm}^2$$

Area Available in Welds [A41 + A42 + A43]:
 $= W_o^2 \cdot fr3 + (W_i - can/0.707)^2 \cdot fr2 + W_p^2 \cdot fr4$
 $= 10^2 \cdot 0.86 + (0)^2 \cdot 0.86 + 12^2 \cdot 1$
 $= 2.295 \text{ cm}^2$

Area Available in Element [A5]:
 $= (\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, Tlwp, te)) \cdot fr4$
 $= (1050 - 610) \cdot 14 \cdot 1$
 $= 61.655 \text{ cm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures $t_a = 5.8093 \text{ mm}$.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= t_a$$

$$= 5.8093 \text{ mm}.$$

Available Nozzle Neck Thickness = 14.0000 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, $t_g = 14$, $t_r = 2.81$, $c = 3 \text{ mm.}$, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.26$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	2 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, $t_g = 14$, $t_r = 2.81$, $c = 3 \text{ mm.}$, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.26$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-18 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, $t_g = 14$, $t_r = 2.81$, $c = 3 \text{ mm.}$, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.26$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-18 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Shell to Pad Weld Junction at Pad OD, min(Curve:B, Curve:D)

Govrn. thk, $t_g = 14$, $c = 3 \text{ mm.}$, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.75$, Temp. Reduction = 14 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-18 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-32 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:B, Curve:D)

Govrn. thk, $t_g = 14$, $t_r = 2.81$, $c = 3 \text{ mm.}$, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.26$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-18 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the Nozzle	: -104 °C
Gov. MDMT of the Reinforcement Pad	: -32 °C
Gov. MDMT of the nozzle to shell joint welded assembly	: -32 °C

Flange MDMT including Temperature reduction per UCS-66.1:



Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -44 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 10.92/19.60 = 0.557

Weld Size Calculations, Description: M1-24"

Intermediate Calc. for nozzle/shell Welds Tmin 11.0000 mm.
 Intermediate Calc. for pad/shell Welds TminPad 13.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.
Pad Weld	6.5000 = 0.5*TminPad	8.4840 = 0.7 * Wp mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (57.7 - 18.9 + 2 \cdot 11 \cdot 0.86 \cdot (1 \cdot 13 - 9.76)) \cdot 138)$
 $= 55352.91 \text{ Kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (4.55 + 61.7 + 2.3 - 0 \cdot 0.86) \cdot 138$
 $= 96395.08 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (4.55 + 0 + 0.86 + (2.45)) \cdot 138$
 $= 11050.31 \text{ Kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (4.55 + 0 + 2.3 + 61.7 + (2.45)) \cdot 138$
 $= 99836.29 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.14/2.0) \cdot 610 \cdot 10 \cdot 0.49 \cdot 118$
 $= 56456. \text{ Kgf}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$
 $= (3.14/2.0) \cdot 1050 \cdot 12 \cdot 0.49 \cdot 138$
 $= 136467. \text{ Kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.14 \cdot 299) \cdot (14 - 3) \cdot 0.7 \cdot 118$
 $= 87115. \text{ Kgf}$

Tension, Pad Groove Weld [Tpgw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$
 $= (3.14/2) \cdot 610 \cdot 14 \cdot 0.74 \cdot 138$
 $= 139594. \text{ Kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng}$
 $= (3.14/2.0) \cdot 610 \cdot (16 - 3) \cdot 0.74 \cdot 138$
 $= 129625. \text{ Kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.



Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SPEW} + \text{SNW}) = (136467 + 87115) = 223582 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw}) \\ = (56456 + 139594 + 129625 + 0) = 325675 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (\text{Spew} + \text{Tngw} + \text{Sinw}) \\ = (136467 + 129625 + 0) = 266093 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 223582 Kgf, must exceed W = 55352 Kgf	or W1 = 96395 Kgf
Path 2-2 = 325675 Kgf, must exceed W = 55352 Kgf	or W2 = 11050 Kgf
Path 3-3 = 266092 Kgf, must exceed W = 55352 Kgf	or W3 = 99836 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 10.8 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 38.5254 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 404.5254 mm.

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm., while the distance from the center of the nozzle to the edge of the diameter limit is: 587.60 mm.

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Input, Nozzle Desc: N2-14"

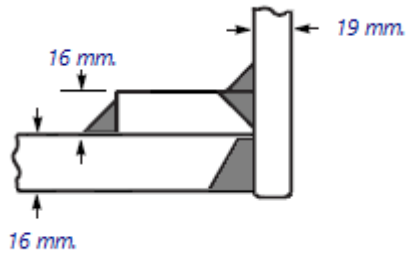
From: 50

Pressure for Reinforcement Calculations	P	10.915	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Elliptical Head	D	2450.00	mm.
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	16.0000	mm.
Head Internal Corrosion Allowance	cas	3.0000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	0.0000	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm2
Allowable Stress At Ambient	Sna	118.00	N./mm2
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		14.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		80	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	250.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	14.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm2
Pad Allowable Stress At Ambient	Spa	138.00	N./mm2
Diameter of Pad along vessel surface	Dp	550.0000	mm.
Thickness of Pad	te	16.0000	mm.
Weld leg size between Pad and Shell	Wp	12.0000	mm.
Groove weld depth between Pad and Nozzle	Wgpn	16.0000	mm.
Reinforcing Pad Width		97.2000	mm.
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was Overall MAWP.

**Reinforcement CALCULATION, Description: N2-14"**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	14.000 in.
Actual Thickness Used in Calculation	0.750 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 10.9 \cdot 2456 \cdot 0.9 / (2 \cdot 138 \cdot 1 - 0.2 \cdot 10.9)$
 $= 8.7289 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot Ro / (Sn \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 10.9 \cdot 178 / (118 \cdot 1 + 0.4 \cdot 10.9)$
 $= 1.6388 \text{ mm.}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	647.0000 mm.
Parallel to Vessel Wall, opening length	d	323.5000 mm.
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		32.5000 mm.

Weld Strength Reduction Factor [fr1]:

$$= \min(1, Sn/Sv)$$

$$= \min(1, 118/138)$$

$$= 0.855$$

Weld Strength Reduction Factor [fr2]:

$$= \min(1, Sn/Sv)$$

$$= \min(1, 118/138)$$

$$= 0.855$$

Weld Strength Reduction Factor [fr4]:

$$= \min(1, Sp/Sv)$$

$$= \min(1, 138/138)$$

$$= 1.000$$

Weld Strength Reduction Factor [fr3]:

$$= \min(fr2, fr4)$$

$$= \min(0.86, 1)$$

$$= 0.855$$

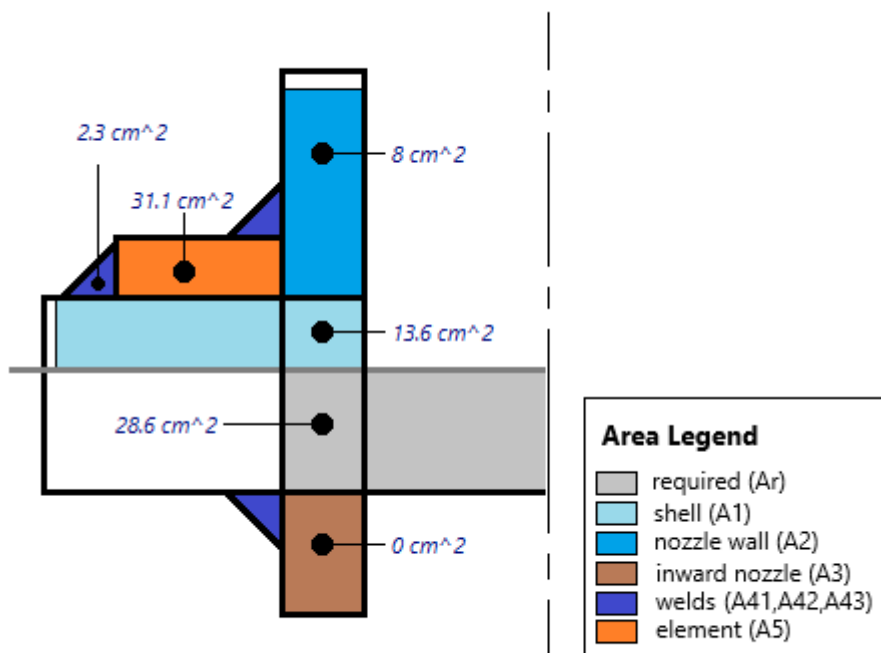


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	28.644	NA	NA
Area in Shell	A1	13.618	NA	NA
Area in Nozzle Wall	A2	8.010	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	2.295	NA	NA
Area in Element	A5	31.103	NA	NA
TOTAL AREA AVAILABLE	Atot	55.026	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 385.1081 16.0000 mm.
 Based on given Pad Diameter: 549.9999 2.4287 mm.
 Based on Shell or Nozzle Thickness: 385.1081 16.0000 mm.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (324 \cdot 8.73 \cdot 1 + 2 \cdot 16 \cdot 8.73 \cdot 1 \cdot (1 - 0.86))$
 $= 28.644 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 324(1 \cdot 13 - 1 \cdot 8.73) - 2 \cdot 16$
 $(1 \cdot 13 - 1 \cdot 8.73) \cdot (1 - 0.86)$
 $= 13.618 \text{ cm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 32.5) \cdot (16 - 1.64) \cdot 0.86$



$$= 8.010 \text{ cm}^2$$

Area Available in Welds [A41 + A42 + A43]:

$$= W_o^2 \cdot fr_3 + (W_i - c_{an}/0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$$

$$= 10^2 \cdot 0.86 + (0)^2 \cdot 0.86 + 12^2 \cdot 1$$

$$= 2.295 \text{ cm}^2$$

Area Available in Element [A5]:

$$= (\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, T_{lwp}, t_e)) \cdot fr_4$$

$$= (550 - 356) \cdot 16 \cdot 1$$

$$= 31.103 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	$t_a = 4.6388 \text{ mm.}$
Wall Thickness per UG16(b),	$tr_{16b} = 4.5000 \text{ mm.}$
Wall Thickness, shell/head, internal pressure	$trb_1 = 12.6899 \text{ mm.}$
Wall Thickness	$tb_1 = \max(trb_1, tr_{16b}) = 12.6899 \text{ mm.}$
Wall Thickness	$tb_2 = \max(trb_2, tr_{16b}) = 4.5000 \text{ mm.}$
Wall Thickness per table UG-45	$tb_3 = 11.3312 \text{ mm.}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb_3, \max(tb_1, tb_2)]$$

$$= \min[11.3, \max(12.7, 4.5)]$$

$$= 11.3312 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(t_a, t_b)$$

$$= \max(4.64, 11.3)$$

$$= 11.3312 \text{ mm.}$$

$$\text{Available Nozzle Neck Thickness} = 0.875 \cdot 19.050 = 16.669 \text{ mm.} \rightarrow \text{OK}$$

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 24.2,	Allowable	: 118.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 270.8 N./mm ²	Passed
Occasional	: 5.2,	Allowable	: 156.9 N./mm ²	Passed
Shear	: 12.3,	Allowable	: 82.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, $t_g = 16.7$, $tr = 1.64$, $c = 3 \text{ mm.}$, $E^* = 1$
 Thickness Ratio = $tr \cdot E^* / (t_g - c) = 0.12$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	7 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, $t_g = 16$, $c = 3 \text{ mm.}$, $E^* = 1$
 Thickness Ratio = $tr \cdot E^* / (t_g - c) = 0.67$, Temp. Reduction = 18 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-32 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, $t_g = 16$, $c = 3 \text{ mm.}$, $E^* = 1$
 Thickness Ratio = $tr \cdot E^* / (t_g - c) = 0.67$, Temp. Reduction = 18 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-32 °C

Shell to Pad Weld Junction at Pad OD, min(Curve:B, Curve:D)

Govrn. thk, tg = 16, tr = 8.73, c = 3 mm., E* = 1
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.67$, Temp. Reduction = 18 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-32 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:B, Curve:D)

Govrn. thk, tg = 16, tr = 8.73, c = 3 mm., E* = 1
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.67$, Temp. Reduction = 18 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-32 °C

Gov. MDMT of the Nozzle	: -32 °C
Gov. MDMT of the Reinforcement Pad	: -32 °C
Gov. MDMT of the nozzle to shell joint welded assembly	: -32 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-44 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $10.92/19.60 = 0.557$

Weld Size Calculations, Description: N2-14"

Intermediate Calc. for nozzle/shell Welds	Tmin	16.0000 mm.
Intermediate Calc. for pad/shell Welds	TminPad	16.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = $0.7 \cdot W_o$ mm.
Pad Weld	8.0000 = $0.5 \cdot T_{minPad}$	8.4840 = $0.7 \cdot W_p$ mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A1 + 2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - tr)) \cdot S_v)$
 $= \max(0, (28.6 - 13.6 + 2 \cdot 16 \cdot 0.86 \cdot (1 \cdot 13 - 8.73)) \cdot 138)$
 $= 22793.71 \text{ Kg}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2 + A5 + A4 - (W_i - Can / .707)^2 \cdot f_{r2}) \cdot S_v$
 $= (8.01 + 31.1 + 2.3 - 0 \cdot 0.86) \cdot 138$
 $= 58268.65 \text{ Kg}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (8.01 + 0 + 0.86 + (3.57)) \cdot 138$
 $= 17495.19 \text{ Kg}$

Weld Load [W3]:
 $= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (8.01 + 0 + 2.3 + 31.1 + (3.57)) \cdot 138$
 $= 63289.70 \text{ Kg}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.14/2.0) \cdot 356 \cdot 10 \cdot 0.49 \cdot 118$
 $= 32932. \text{ Kg}$



Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$
 $= (3.14/2.0) \cdot 550 \cdot 12 \cdot 0.49 \cdot 138$
 $= 71483. \text{ Kg}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$
 $= (3.14 \cdot 170) \cdot (19 - 3) \cdot 0.7 \cdot 118$
 $= 72101. \text{ Kg}$

Tension, Pad Groove Weld [Tpgw]:
 $= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$
 $= (3.14/2) \cdot 356 \cdot 16 \cdot 0.74 \cdot 138$
 $= 93063. \text{ Kg}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$
 $= (3.14/2.0) \cdot 356 \cdot (16 - 3) \cdot 0.74 \cdot 138$
 $= 75615. \text{ Kg}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (SPEW + SNW) = (71483 + 72101) = 143584 \text{ Kg}$

Failure Path [Path 2-2]:
 $= (Sonw + Tpgw + Tngw + Sinw)$
 $= (32932 + 93063 + 75615 + 0) = 201610 \text{ Kg}$

Failure Path [Path 3-3]:
 $= (Spew + Tngw + Sinw)$
 $= (71483 + 75615 + 0) = 147098 \text{ Kg}$

Summary of Failure Path Calculations:

Path 1-1 = 143584 Kg, must exceed W = 22793 Kg or W1 = 58268 Kg
 Path 2-2 = 201610 Kg, must exceed W = 22793 Kg or W2 = 17495 Kg
 Path 3-3 = 147097 Kg, must exceed W = 22793 Kg or W3 = 63289 Kg

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 10.8 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 7.1385 mm.
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 273.1385 mm.

Input Echo, WRC107/537 Item 1, Description: N2-14" :

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	3.0000	mm.
Vessel Diameter	Dv	4410.000	mm.
Vessel Thickness	Tv	16.000	mm.
Design Temperature	T1	82.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh	138.00	N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Attachment Type	Type	Round
WRC107 Attachment Classification	Holsol	Hollow
Diameter Basis for Nozzle	Nbasis	OD



Corrosion Allowance for Nozzle	Can	3.0000	mm.
Nozzle Diameter	Dn	355.600	mm.
Nozzle Thickness	Tn	19.050	mm.
Nozzle Material		SA-106 B	
Nozzle UNS Number		K03006	
Nozzle Cold S.I. Allowable	SNmc	118.00	N./mm2
Nozzle Hot S.I. Allowable	SNmh	118.00	N./mm2
Thickness of Reinforcing Pad	Tpad	16.000	mm.
Diameter of Reinforcing Pad	Dpad	550.000	mm.
Design Internal Pressure	Dp	10.915	bars
Include Pressure Thrust		No	

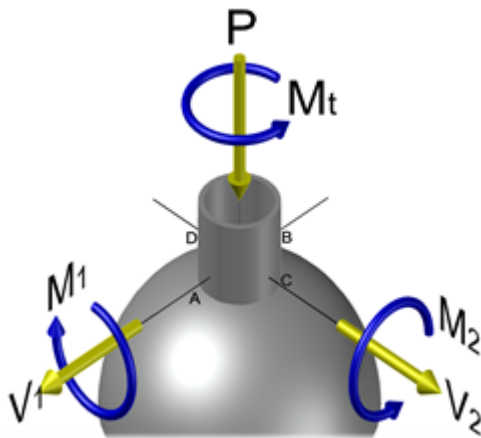
External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)		P	2676.9	Kgf
Longitudinal Shear (SUS)	(V1)	V1	1784.6	Kgf
Circumferential Shear (SUS)	(Vc)	V2	1784.6	Kgf
Circumferential Moment (SUS)	(Mc)	M1	1748.3	Kg-m.
Longitudinal Moment (SUS)	(Ml)	M2	1748.3	Kg-m.
Torsional Moment (SUS)		Mt	2497.5	Kg-m.

Use Interactive Control No
WRC107 Version Version March 1979

Include Pressure Stress Indices per Div. 2 No
Compute Pressure Stress per WRC-368 No
Local Loads applied at end of Nozzle/Attachment No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$$

$$= 355.6 + 2 \cdot 1.65 \cdot \sqrt{2214.5 (16.0 - 3.0)}$$

$$= 915.517 \text{ mm.}$$

WRC 107 Stress Calculation for SUStained loads:

Radial Load		P	2676.9	Kgf
Circumferential Shear	(VC)	V2	1784.6	Kgf
Longitudinal Shear	(VL)	V1	1784.6	Kgf
Circumferential Moment	(MC)	M1	1748.3	Kg-m.
Longitudinal Moment	(ML)	M2	1748.3	Kg-m.
Torsional Moment		MT	2497.5	Kg-m.

Dimensionless Parameters: U = 0.70 TAU = 10.58 RHO = 1.81

**Dimensionless Loads for Spherical Shells at Attachment Junction:**

Curves read for 1979	Figure	Value	Location
$N(x) * T / P$	SP 2	0.04860	(A,B,C,D)
$M(x) / P$	SP 2	0.04381	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / MC$	SM 2	0.09118	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / MC$	SM 2	0.09289	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / ML$	SM 2	0.09118	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / ML$	SM 2	0.09289	(A,B,C,D)
$N(y) * T / P$	SP 2	0.16139	(A,B,C,D)
$M(y) / P$	SP 2	0.03118	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / MC$	SM 2	0.16752	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / MC$	SM 2	0.08446	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / ML$	SM 2	0.16752	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / ML$	SM 2	0.08446	(A,B,C,D)

Stress Concentration Factors: $K_n = 1.00$, $K_b = 1.00$ **Stresses in the Vessel at the Attachment Junction (N./mm2)**

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Memb. P	-1.5	-1.5	-1.5	-1.5	-1.5	-1.5	-1.5	-1.5
Rad.	Bend. P	-8.2	8.2	-8.2	8.2	-8.2	8.2	-8.2	8.2
Rad.	Memb. MC	0.0	0.0	0.0	0.0	-7.3	7.3	7.3	7.3
Rad.	Bend. MC	0.0	0.0	0.0	0.0	-44.8	44.8	44.8	-44.8
Rad.	Memb. ML	-7.3	-7.3	7.3	7.3	0.0	0.0	0.0	0.0
Rad.	Bend. ML	-44.8	44.8	44.8	-44.8	0.0	0.0	0.0	0.0
Tot. Rad.	Str.	-61.8	44.1	42.4	-30.7	-61.8	44.1	42.4	-30.7
Tang.	Memb. P	-5.0	-5.0	-5.0	-5.0	-5.0	-5.0	-5.0	-5.0
Tang.	Bend. P	-5.8	5.8	-5.8	5.8	-5.8	5.8	-5.8	5.8
Tang.	Memb. MC	0.0	0.0	0.0	0.0	-13.5	13.5	13.5	13.5
Tang.	Bend. MC	0.0	0.0	0.0	0.0	-40.7	40.7	40.7	-40.7
Tang.	Memb. ML	-13.5	-13.5	13.5	13.5	0.0	0.0	0.0	0.0
Tang.	Bend. ML	-40.7	40.7	40.7	-40.7	0.0	0.0	0.0	0.0
Tot. Tang.	Str.	-65.0	28.0	43.3	-26.4	-65.0	28.0	43.3	-26.4
	Shear VC	1.1	1.1	-1.1	-1.1	0.0	0.0	0.0	0.0
	Shear VL	0.0	0.0	0.0	0.0	-1.1	-1.1	1.1	1.1
	Shear MT	4.3	4.3	4.3	4.3	4.3	4.3	4.3	4.3
Tot. Shear		5.3	5.3	3.2	3.2	3.2	3.2	5.3	5.3
Str. Int.		69.0	45.7	46.0	32.4	67.0	44.7	48.2	34.3

Unitless Prm: $U = 1.62$ TAU = 0.00 (16.63) RHO = 0.00 (0.81)**Dimensionless Loads for Spherical Shells at Pad edge:**

Curves read for 1979	Figure	Value	Location
$N(x) * T / P$	SR 2	0.03407	(A,B,C,D)
$M(x) / P$	SR 2	0.01650	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / MC$	SR 3	0.03286	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / MC$	SR 3	0.02333	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / ML$	SR 3	0.03286	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / ML$	SR 3	0.02333	(A,B,C,D)
$N(y) * T / P$	SR 2	0.01015	(A,B,C,D)
$M(y) / P$	SR 2	0.00494	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / MC$	SR 3	0.00984	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / MC$	SR 3	0.00690	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / ML$	SR 3	0.00984	(A,B,C,D)



$$M(y) * \text{SQRT}(R_m * T) / ML \quad SR \ 3 \quad 0.00690 \quad (A,B,C,D)$$
Stress Concentration Factors: $K_n = 1.00$, $K_b = 1.00$ **Stresses in the Vessel at the Edge of Reinforcing Pad (N./mm2)**

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb. P		-5.3	-5.3	-5.3	-5.3	-5.3	-5.3	-5.3	-5.3
Rad. Bend. P		-15.4	15.4	-15.4	15.4	-15.4	15.4	-15.4	15.4
Rad. Memb. MC		0.0	0.0	0.0	0.0	-19.6	-19.6	19.6	19.6
Rad. Bend. MC		0.0	0.0	0.0	0.0	-83.7	83.7	83.7	-83.7
Rad. Memb. ML		-19.6	-19.6	19.6	19.6	0.0	0.0	0.0	0.0
Rad. Bend. ML		-83.7	83.7	83.7	-83.7	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-124.0	74.1	82.7	-53.9	-124.0	74.1	82.7	-53.9
Tang. Memb. P		-1.6	-1.6	-1.6	-1.6	-1.6	-1.6	-1.6	-1.6
Tang. Bend. P		-4.6	4.6	-4.6	4.6	-4.6	4.6	-4.6	4.6
Tang. Memb. MC		0.0	0.0	0.0	0.0	-5.9	-5.9	5.9	5.9
Tang. Bend. MC		0.0	0.0	0.0	0.0	-24.8	24.8	24.8	-24.8
Tang. Memb. ML		-5.9	-5.9	5.9	5.9	0.0	0.0	0.0	0.0
Tang. Bend. ML		-24.8	24.8	24.8	-24.8	0.0	0.0	0.0	0.0
Tot. Tang. Str.		-36.8	21.9	24.5	-15.8	-36.8	21.9	24.5	-15.8
Shear VC		1.6	1.6	-1.6	-1.6	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-1.6	-1.6	1.6	1.6
Shear MT		4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0
Tot. Shear		5.5	5.5	2.4	2.4	2.4	2.4	5.5	5.5
Str. Int.		124.3	74.7	82.8	54.1	124.1	74.2	83.2	54.7

WRC 107/537 Stress Summations:**Vessel Stress Summation at Attachment Junction (N./mm2)**

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		41.6	41.6	41.6	41.6	41.6	41.6	41.6	41.6
Rad. Pl (SUS)		-8.8	-8.8	5.8	5.8	-8.8	-8.8	5.8	5.8
Rad. Q (SUS)		-53.0	53.0	36.5	-36.5	-53.0	53.0	36.5	-36.5
Long. Pm (SUS)		41.6	41.6	41.6	41.6	41.6	41.6	41.6	41.6
Long. Pl (SUS)		-18.5	-18.5	8.4	8.4	-18.5	-18.5	8.4	8.4
Long. Q (SUS)		-46.5	46.5	34.9	-34.9	-46.5	46.5	34.9	-34.9
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		1.1	1.1	-1.1	-1.1	-1.1	-1.1	1.1	1.1
Shear Q (SUS)		4.3	4.3	4.3	4.3	4.3	4.3	4.3	4.3
Pm (SUS)		41.6	41.6	41.6	41.6	41.6	41.6	41.6	41.6
Pm+Pl (SUS)		32.8	32.8	50.4	50.4	32.8	32.8	50.4	50.4
Pm+Pl+Q (Total)		27.4	87.3	87.6	16.8	25.4	86.3	89.7	18.7

Vessel Stress Summation Comparison (N./mm2):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	41.56	138.00	Passed



Pm+Pl (SUS)	50.36	207.00	Passed
Pm+Pl+Q (TOTAL)	89.72	414.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 107/537 Stress Summations:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Pm (SUS)	92.7	92.7	92.7	92.7	92.7	92.7	92.7	92.7
Rad.	Pl (SUS)	-24.9	-24.9	14.4	14.4	-24.9	-24.9	14.4	14.4
Rad.	Q (SUS)	-99.1	99.1	68.3	-68.3	-99.1	99.1	68.3	-68.3
Long.	Pm (SUS)	92.7	92.7	92.7	92.7	92.7	92.7	92.7	92.7
Long.	Pl (SUS)	-7.5	-7.5	4.3	4.3	-7.5	-7.5	4.3	4.3
Long.	Q (SUS)	-29.4	29.4	20.1	-20.1	-29.4	29.4	20.1	-20.1
Shear	Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear	Pl (SUS)	1.6	1.6	-1.6	-1.6	-1.6	-1.6	1.6	1.6
Shear	Q (SUS)	4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0
Pm (SUS)		92.7	92.7	92.7	92.7	92.7	92.7	92.7	92.7
Pm+Pl (SUS)		85.4	85.4	107.3	107.3	85.4	85.4	107.3	107.3
Pm+Pl+Q (Total)		87.9	167.4	175.5	77.0	87.3	166.9	175.9	77.6

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	92.70	138.00	Passed
Pm+Pl (SUS)	107.30	207.00	Passed
Pm+Pl+Q (TOTAL)	175.88	414.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: K5-2"

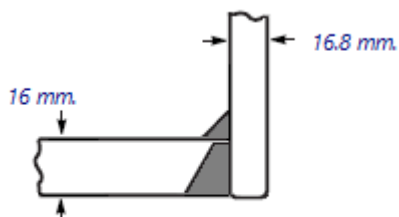
From: 50

Pressure for Reinforcement Calculations	P	10.915	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Elliptical Head	D	2450.00	mm.
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	16.0000	mm.
Head Internal Corrosion Allowance	cas	3.0000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	650.0000	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm.
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	480.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	14.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Overall MAWP.



Note : Checking Nozzle in the Meridional direction.

Reinforcement CALCULATION, Description: K5-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.660 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 10.9 \cdot 2456 \cdot 0.9 / (2 \cdot 138 \cdot 1 - 0.2 \cdot 10.9)$
 $= 8.7289 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (Sn \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 10.9 \cdot 28.4 / (138 \cdot 1 - 0.6 \cdot 10.9)$
 $= 0.2257 \text{ mm.}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 119.0078 mm.
 Parallel to Vessel Wall, opening length d 59.5039 mm.
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 32.5000 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 138/138)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 138/138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$

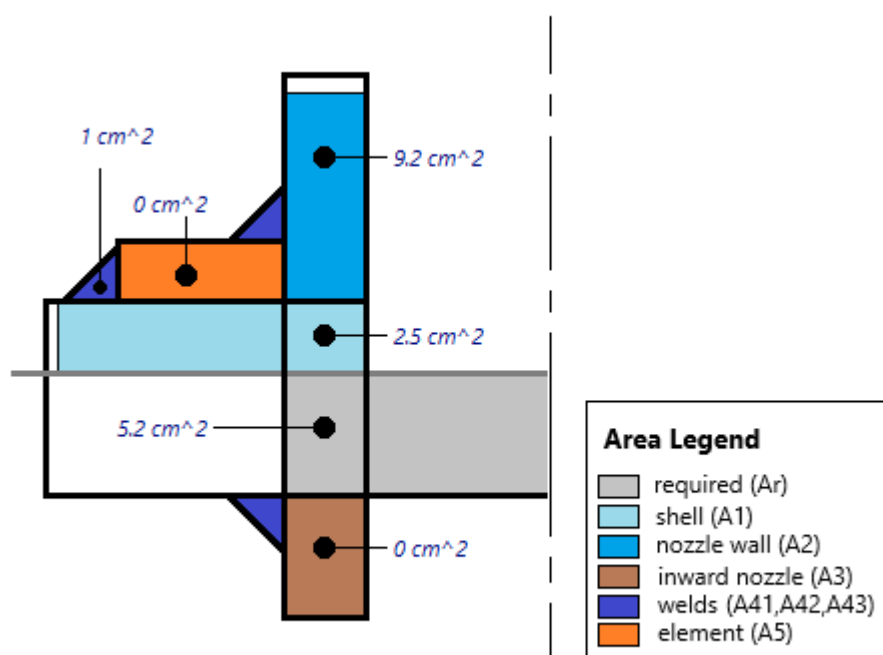


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	5.194	NA	NA



Area in Shell	A1	2.541	NA	NA
Area in Nozzle Wall	A2	9.161	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
TOTAL AREA AVAILABLE	Atot	12.702	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 72.66 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (59.5 \cdot 8.73 \cdot 1 + 2 \cdot 13.8 \cdot 8.73 \cdot 1 \cdot (1 - 1))$
 $= 5.194 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 59.5(1 \cdot 13 - 1 \cdot 8.73) - 2 \cdot 13.8$
 $(1 \cdot 13 - 1 \cdot 8.73) \cdot (1 - 1)$
 $= 2.541 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tnp)(tn - trn)fr2/\sin(\alpha3)$
 $= (2 \cdot 32.5)(13.8 - 0.23)/\sin(73.9)$
 $= 9.161 \text{ cm}^2$

See ASME VIII-1 2011(a) Appendix L, L-7.7.7(b) for more information.

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10^2 \cdot 1 + (0)^2 \cdot 1$
 $= 1.000 \text{ cm}^2$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(b)), min(Curve:B, Curve:D)

Govrn. thk, $tg = 16$, $tr = 8.73$, $c = 3 \text{ mm.}$, $E* = 1$
 Thickness Ratio $= tr \cdot E*/(tg - c) = 0.67$, Temp. Reduction = 18 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -14 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -32 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -32 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $10.92/51.00 = 0.214$

Weld Size Calculations, Description: K5-2"

Intermediate Calc. for nozzle/shell Welds Tmin 13.7640 mm.

Results Per UW-16.1:

Required Thickness Actual Thickness
 Nozzle Weld 6.0000 = Min per Code 7.0700 = $0.7 \cdot Wo \text{ mm.}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:



$$\begin{aligned}
 &= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v) \\
 &= \max(0, (5.19 - 2.54 + 2 \cdot 13.8 \cdot 1 \cdot (1 \cdot 13 - 8.73)) \cdot 138) \\
 &= 5387.13 \text{ Kgf}
 \end{aligned}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$\begin{aligned}
 &= (A2+A5+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v \\
 &= (9.16 + 0 + 1 - 0 \cdot 1) \cdot 138 \\
 &= 14298.20 \text{ Kgf}
 \end{aligned}$$

Weld Load [W2]:

$$\begin{aligned}
 &= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v \\
 &= (9.16 + 0 + 1 + (3.58)) \cdot 138 \\
 &= 19334.00 \text{ Kgf}
 \end{aligned}$$

Weld Load [W3]:

$$\begin{aligned}
 &= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S \\
 &= (9.16 + 0 + 1 + 0 + (3.58)) \cdot 138 \\
 &= 19334.00 \text{ Kgf}
 \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\
 &= (3.14/2.0) \cdot 88.3 \cdot 10 \cdot 0.49 \cdot 138 \\
 &= 9568. \text{ Kgf}
 \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned}
 &= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n \\
 &= (3.14 \cdot 37) \cdot (16.8 - 3) \cdot 0.7 \cdot 138 \\
 &= 15743. \text{ Kgf}
 \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng} \\
 &= (3.14/2.0) \cdot 88.3 \cdot (16 - 3) \cdot 0.74 \cdot 138 \\
 &= 18785. \text{ Kgf}
 \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SONW} + \text{SNW}) = (9568 + 15743) = 25312 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$\begin{aligned}
 &= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw}) \\
 &= (9568 + 0 + 18785 + 0) = 28353 \text{ Kgf}
 \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned}
 &= (\text{Sonw} + \text{Tngw} + \text{Sinw}) \\
 &= (9568 + 18785 + 0) = 28353 \text{ Kgf}
 \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 25311	Kgf, must exceed W = 5387 Kgf	or W1 = 14298 Kgf
Path 2-2 = 28353	Kgf, must exceed W = 5387 Kgf	or W2 = 19334 Kgf
Path 3-3 = 28353	Kgf, must exceed W = 5387 Kgf	or W3 = 19334 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 10.8 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

Note : Checking Nozzle in the Latitudinal direction.

Reinforcement CALCULATION, Description: K5-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.660 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 10.9 \cdot 2456 \cdot 0.9 / (2 \cdot 138 \cdot 1 - 0.2 \cdot 10.9)$
 $= 8.7289 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (Sn \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 10.9 \cdot 28.4 / (138 \cdot 1 - 0.6 \cdot 10.9)$
 $= 0.2257 \text{ mm.}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 113.6000 mm.
 Parallel to Vessel Wall, opening length d 56.8000 mm.
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 32.5000 mm.

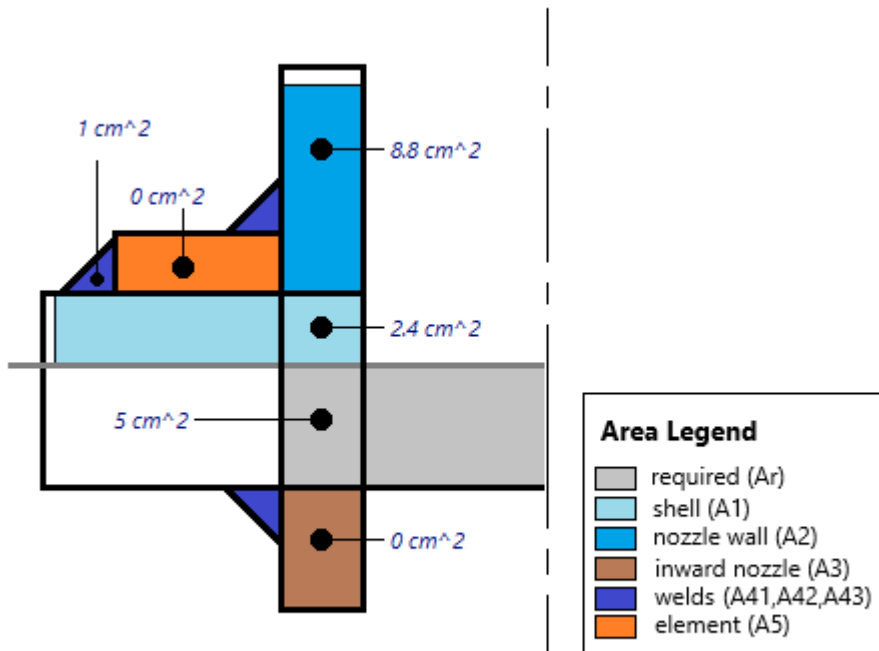


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	4.958	NA	NA
Area in Shell	A1	2.426	NA	NA
Area in Nozzle Wall	A2	8.800	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
TOTAL AREA AVAILABLE	Atot	12.226	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)

$$= (56.8 \cdot 8.73 \cdot 1 + 2 \cdot 13.8 \cdot 8.73 \cdot 1 \cdot (1-1))$$

$$= 4.958 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 56.8 (1 \cdot 13 - 1 \cdot 8.73) - 2 \cdot 13.8$$

$$(1 \cdot 13 - 1 \cdot 8.73) \cdot (1 - 1)$$

$$= 2.426 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp) (tn - trn) fr2$$

$$= (2 \cdot 32.5) (13.8 - 0.23) \cdot 1$$

$$= 8.800 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi - can / 0.707)^2 \cdot fr2$$

$$= 10^2 \cdot 1 + (0)^2 \cdot 1$$

$$= 1.000 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.2257 mm.
Wall Thickness per UG16(b),	tr16b = 4.5000 mm.
Wall Thickness, shell/head, internal pressure	trb1 = 12.6899 mm.
Wall Thickness	tb1 = max(trb1, tr16b) = 12.6899 mm.
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm.
Wall Thickness per table UG-45	tb3 = 7.8000 mm.

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[7.8, \max(12.7, 4.5)]$$

$$= 7.8000 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(3.23, 7.8)$$

$$= 7.8000 \text{ mm.}$$

Available Nozzle Neck Thickness = 16.7640 mm. -> OK

Weld Size Calculations, Description: K5-2"

Intermediate Calc. for nozzle/shell Welds Tmin 13.7640 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A1 + 2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$$

$$= \max(0, (4.96 - 2.43 + 2 \cdot 13.8 \cdot 1 \cdot$$

$$(1 \cdot 13 - 8.73) \cdot 138)$$

$$= 5217.52 \text{ Kg}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2 + A5 + A4 - (Wi - Can / 0.707)^2 \cdot fr2) \cdot Sv$$

$$= (8.8 + 0 + 1 - 0 \cdot 1) \cdot 138$$

$$= 13790.22 \text{ Kg}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$$

$$= (8.8 + 0 + 1 + (3.58)) \cdot 138$$

$$= 18826.02 \text{ Kg}$$

Weld Load [W3]:

$$\begin{aligned}
 &= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f r 1)) \cdot S \\
 &= (8.8 + 0 + 1 + 0 + (3.58)) \cdot 138 \\
 &= 18826.02 \text{ Kg}
 \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\
 &= (3.14/2.0) \cdot 84.3 \cdot 10 \cdot 0.49 \cdot 138 \\
 &= 9134. \text{ Kg}
 \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned}
 &= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n \\
 &= (3.14 \cdot 35.3) \cdot (16.8 - 3) \cdot 0.7 \cdot 138 \\
 &= 15028. \text{ Kg}
 \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng} \\
 &= (3.14/2.0) \cdot 84.3 \cdot (16 - 3) \cdot 0.74 \cdot 138 \\
 &= 17932. \text{ Kg}
 \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (9134 + 15028) = 24161 \text{ Kg}$$

Failure Path [Path 2-2]:

$$\begin{aligned}
 &= (Sonw + Tpgw + Tngw + Sinw) \\
 &= (9134 + 0 + 17932 + 0) = 27065 \text{ Kg}
 \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned}
 &= (Sonw + Tngw + Sinw) \\
 &= (9134 + 17932 + 0) = 27065 \text{ Kg}
 \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 24161	Kgf, must exceed W = 5217 Kg	or W1 = 13790 Kg
Path 2-2 = 27065	Kgf, must exceed W = 5217 Kg	or W2 = 18826 Kg
Path 3-3 = 27065	Kgf, must exceed W = 5217 Kg	or W3 = 18826 Kg

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 10.8 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 13.3866 mm.
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 510.1158 mm.

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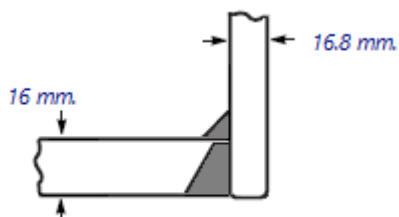
**Input, Nozzle Desc: K6-2"****From: 50**

Pressure for Reinforcement Calculations	P	10.915	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Elliptical Head	D	2450.00	mm.
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	16.0000	mm.
Head Internal Corrosion Allowance	cas	3.0000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	650.0000	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm.
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	480.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	14.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Overall MAWP.

Note : Checking Nozzle in the Meridional direction.**Reinforcement CALCULATION, Description: K6-2"**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 2.000 in.
Actual Thickness Used in Calculation 0.660 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
= $P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
= $10.9 \cdot 2456 \cdot 0.9 / (2 \cdot 138 \cdot 1 - 0.2 \cdot 10.9)$
= 8.7289 mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (Sn \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $10.9 \cdot 28.4 / (138 \cdot 1 - 0.6 \cdot 10.9)$
= 0.2257 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 119.0078 mm.
Parallel to Vessel Wall, opening length d 59.5039 mm.
Normal to Vessel Wall (Thickness Limit), no pad Tlnp 32.5000 mm.

Weld Strength Reduction Factor [fr1]:
= min(1, Sn/Sv)
= min(1, 138/138)
= 1.000

Weld Strength Reduction Factor [fr2]:
= min(1, Sn/Sv)
= min(1, 138/138)
= 1.000

Weld Strength Reduction Factor [fr3]:
= min(fr2, fr4)
= min(1, 1)
= 1.000

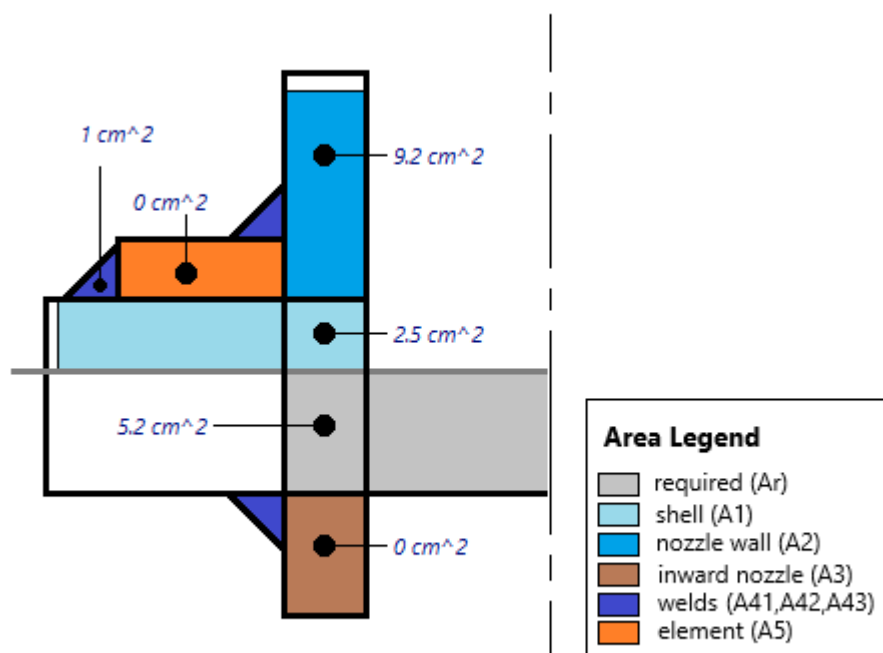


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	5.194	NA	NA



Area in Shell	A1	2.541	NA	NA
Area in Nozzle Wall	A2	9.161	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
TOTAL AREA AVAILABLE	Atot	12.702	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 72.66 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (59.5 \cdot 8.73 \cdot 1 + 2 \cdot 13.8 \cdot 8.73 \cdot 1 \cdot (1 - 1))$
 $= 5.194 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 59.5(1 \cdot 13 - 1 \cdot 8.73) - 2 \cdot 13.8$
 $(1 \cdot 13 - 1 \cdot 8.73) \cdot (1 - 1)$
 $= 2.541 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tnp)(tn - trn)fr2/\sin(\alpha3)$
 $= (2 \cdot 32.5)(13.8 - 0.23)/\sin(73.9)$
 $= 9.161 \text{ cm}^2$

See ASME VIII-1 2011(a) Appendix L, L-7.7.7(b) for more information.

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10^2 \cdot 1 + (0)^2 \cdot 1$
 $= 1.000 \text{ cm}^2$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(b)), min(Curve:B, Curve:D)

Govrn. thk, $tg = 16$, $tr = 8.73$, $c = 3 \text{ mm.}$, $E* = 1$
 Thickness Ratio $= tr \cdot E*/(tg - c) = 0.67$, Temp. Reduction = 18 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -14 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -32 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -32 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $10.92/51.00 = 0.214$

Weld Size Calculations, Description: K6-2"

Intermediate Calc. for nozzle/shell Welds Tmin 13.7640 mm.

Results Per UW-16.1:

Required Thickness Actual Thickness
 Nozzle Weld 6.0000 = Min per Code 7.0700 = $0.7 \cdot Wo \text{ mm.}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:



$$\begin{aligned}
 &= \max(0, (A-A1+2 \cdot t_n \cdot f r_1 \cdot (E1 \cdot t - t_r)) \cdot S_v) \\
 &= \max(0, (5.19 - 2.54 + 2 \cdot 13.8 \cdot 1 \cdot \\
 &\quad (1 \cdot 13 - 8.73)) \cdot 138) \\
 &= 5387.13 \text{ Kgf}
 \end{aligned}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$\begin{aligned}
 &= (A2+A5+A4-(W_i-Can/.707)^2 \cdot f r_2) \cdot S_v \\
 &= (9.16 + 0 + 1 - 0 \cdot 1) \cdot 138 \\
 &= 14298.20 \text{ Kgf}
 \end{aligned}$$

Weld Load [W2]:

$$\begin{aligned}
 &= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f r_1)) \cdot S_v \\
 &= (9.16 + 0 + 1 + (3.58)) \cdot 138 \\
 &= 19334.00 \text{ Kgf}
 \end{aligned}$$

Weld Load [W3]:

$$\begin{aligned}
 &= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f r_1)) \cdot S \\
 &= (9.16 + 0 + 1 + 0 + (3.58)) \cdot 138 \\
 &= 19334.00 \text{ Kgf}
 \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\
 &= (3.14/2.0) \cdot 88.3 \cdot 10 \cdot 0.49 \cdot 138 \\
 &= 9568. \text{ Kgf}
 \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned}
 &= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n \\
 &= (3.14 \cdot 37) \cdot (16.8 - 3) \cdot 0.7 \cdot 138 \\
 &= 15743. \text{ Kgf}
 \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng} \\
 &= (3.14/2.0) \cdot 88.3 \cdot (16 - 3) \cdot 0.74 \cdot 138 \\
 &= 18785. \text{ Kgf}
 \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (9568 + 15743) = 25312 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$\begin{aligned}
 &= (Sonw + Tpgw + Tngw + Sinw) \\
 &= (9568 + 0 + 18785 + 0) = 28353 \text{ Kgf}
 \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned}
 &= (Sonw + Tngw + Sinw) \\
 &= (9568 + 18785 + 0) = 28353 \text{ Kgf}
 \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 25311	Kgf, must exceed W = 5387 Kgf	or W1 = 14298 Kgf
Path 2-2 = 28353	Kgf, must exceed W = 5387 Kgf	or W2 = 19334 Kgf
Path 3-3 = 28353	Kgf, must exceed W = 5387 Kgf	or W3 = 19334 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 10.8 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

Note : Checking Nozzle in the Latitudinal direction.

Reinforcement CALCULATION, Description: K6-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.660 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 10.9 \cdot 2456 \cdot 0.9 / (2 \cdot 138 \cdot 1 - 0.2 \cdot 10.9)$
 $= 8.7289 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (Sn \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 10.9 \cdot 28.4 / (138 \cdot 1 - 0.6 \cdot 10.9)$
 $= 0.2257 \text{ mm.}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 113.6000 mm.
 Parallel to Vessel Wall, opening length d 56.8000 mm.
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 32.5000 mm.

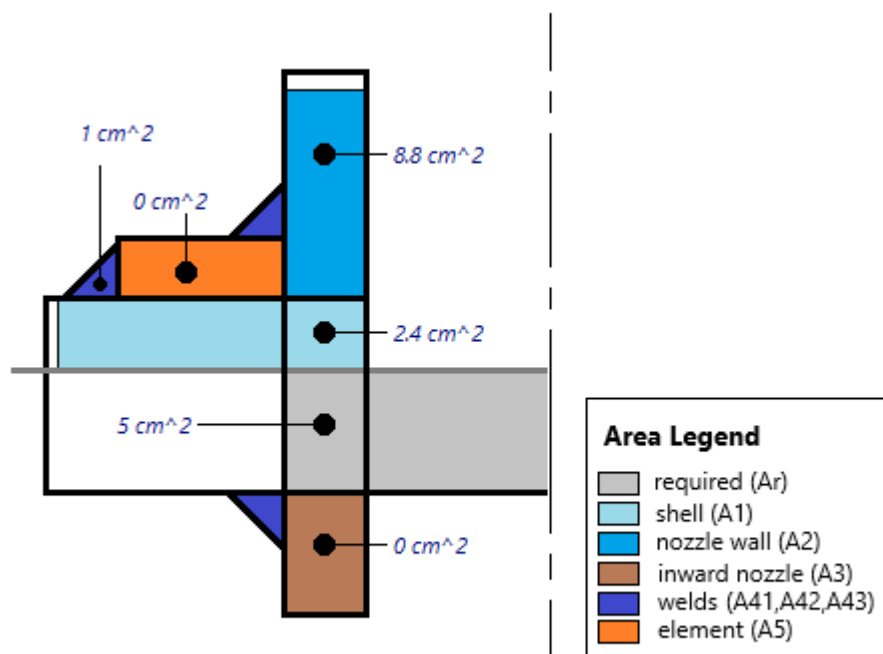


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	4.958	NA	NA
Area in Shell	A1	2.426	NA	NA
Area in Nozzle Wall	A2	8.800	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
TOTAL AREA AVAILABLE	Atot	12.226	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)

$$= (56.8 \cdot 8.73 \cdot 1 + 2 \cdot 13.8 \cdot 8.73 \cdot 1 \cdot (1-1))$$

$$= 4.958 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 56.8 (1 \cdot 13 - 1 \cdot 8.73) - 2 \cdot 13.8$$

$$(1 \cdot 13 - 1 \cdot 8.73) \cdot (1 - 1)$$

$$= 2.426 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp) (tn - trn) fr2$$

$$= (2 \cdot 32.5) (13.8 - 0.23) \cdot 1$$

$$= 8.800 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi \cdot can / 0.707)^2 \cdot fr2$$

$$= 10^2 \cdot 1 + (0)^2 \cdot 1$$

$$= 1.000 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.2257 mm.
Wall Thickness per UG16(b),	tr16b = 4.5000 mm.
Wall Thickness, shell/head, internal pressure	trb1 = 12.6899 mm.
Wall Thickness	tb1 = max(trb1, tr16b) = 12.6899 mm.
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm.
Wall Thickness per table UG-45	tb3 = 7.8000 mm.

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[7.8, \max(12.7, 4.5)]$$

$$= 7.8000 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(3.23, 7.8)$$

$$= 7.8000 \text{ mm.}$$

Available Nozzle Neck Thickness = 16.7640 mm. -> OK

Weld Size Calculations, Description: K6-2"

Intermediate Calc. for nozzle/shell Welds Tmin 13.7640 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A1 + 2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$$

$$= \max(0, (4.96 - 2.43 + 2 \cdot 13.8 \cdot 1 \cdot$$

$$(1 \cdot 13 - 8.73) \cdot 138)$$

$$= 5217.52 \text{ Kg}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2 + A5 + A4 - (Wi \cdot Can / 0.707)^2 \cdot fr2) \cdot Sv$$

$$= (8.8 + 0 + 1 - 0 \cdot 1) \cdot 138$$

$$= 13790.22 \text{ Kg}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$$

$$= (8.8 + 0 + 1 + (3.58)) \cdot 138$$

$$= 18826.02 \text{ Kg}$$

Weld Load [W3]:

$$\begin{aligned}
 &= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f r 1)) \cdot S \\
 &= (8.8 + 0 + 1 + 0 + (3.58)) \cdot 138 \\
 &= 18826.02 \text{ Kgf}
 \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\
 &= (3.14/2.0) \cdot 84.3 \cdot 10 \cdot 0.49 \cdot 138 \\
 &= 9134. \text{ Kgf}
 \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned}
 &= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n \\
 &= (3.14 \cdot 35.3) \cdot (16.8 - 3) \cdot 0.7 \cdot 138 \\
 &= 15028. \text{ Kgf}
 \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng} \\
 &= (3.14/2.0) \cdot 84.3 \cdot (16 - 3) \cdot 0.74 \cdot 138 \\
 &= 17932. \text{ Kgf}
 \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (9134 + 15028) = 24161 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$\begin{aligned}
 &= (Sonw + Tpgw + Tngw + Sinw) \\
 &= (9134 + 0 + 17932 + 0) = 27065 \text{ Kgf}
 \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned}
 &= (Sonw + Tngw + Sinw) \\
 &= (9134 + 17932 + 0) = 27065 \text{ Kgf}
 \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 24161	Kgf, must exceed W = 5217 Kgf	or W1 = 13790 Kgf
Path 2-2 = 27065	Kgf, must exceed W = 5217 Kgf	or W2 = 18826 Kgf
Path 3-3 = 27065	Kgf, must exceed W = 5217 Kgf	or W3 = 18826 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 10.8 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 13.3866 mm.
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 510.1158 mm.

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PV Elite[®] Pressure Vessel Design Notes

Project Information	
Project Number	
Client	
Date	
Designer	
Design Code	
Additional Standards	

Design Conditions		
Internal Design Pressure:		
External Design Pressure:		
Design Int. Temperature:		
Design Ext. Temperature:		
Corrosion Allowance:		

Vessel Geometry		
Vessel Type		
Diameter		
Tangent-to-Tangent Length		
Head Type		
Number of Segments		

Material Specifications	
Shell:	
Head:	
Nozzle:	

Nozzle/Opening Details	ID/OD	Size	Rating	Matl.	Reinf.

Miscellaneous		
Support Type		
Number of Supports		
Support Material		
Seismic Code		
Wind Code		
Insulation		
Pressure Test		
Radiography		
Fatigue Analysis		

Analysis Notes: